PYTHTB-SKILL · UNDERGRADUATE COURSE

Tight-Binding Physics with PythTB

From atoms to topological matter in ten lectures — every figure computed live in the companion notebook

Why tight binding?

A crystal is 10^{23} atoms. Tight binding keeps **one number per atom** (an energy) and **one number per bond** (a hopping) — and still predicts metals, insulators, and topological matter.

- Electrons sit on atomic orbitals and hop to neighbours: the Hamiltonian is a sparse matrix of hoppings.
- Periodicity turns that infinite matrix into a small matrix $H(k)$ for each crystal momentum k.
- Its eigenvalues are the **band structure**; its eigenvectors carry the geometry — Berry phases, polarization, topology.
- PythTB does exactly this and nothing else: build a lattice model, solve it on paths and meshes, read out the geometry.

Ten lectures

Each lecture is one vertical stack of slides: a picture first, the physics next, the equations last. Notebook sections are given for every lecture.

How each lecture is built

The deck is a two-dimensional grid: → jumps between lectures, ↓ descends through one lecture from the simplest slide to the most mathematical.

- intro slides: one figure, plain language, no symbols beyond E and k.
- core slides: the physics and how PythTB computes it (which method, which object).
- math slides: the formulas behind the figures, with the derivation sketched in the lecturer notes.
- To run the notebooks: `install_pythtb_windows.ps1` (or `pip install -r requirements.txt`), then open `chapters/README.md` and start with chapter 0; exercises with solutions in `chapters/PythTB_Exercises_I-II.ipynb` and `…_III-IV.ipynb`.
- Speaker view: press `S`. Section jump: `Shift+←/→`. Overview: `Esc`.

MATH

The one Hamiltonian of the course

Everything that follows is a special case of a second-quantized hopping Hamiltonian and its Bloch transform.

REAL SPACE

$$H = \Sigma_i\, \varepsilon_i\, c^{\dagger}{}_i c_i + \Sigma_{\langle ij \rangle} (\, t_{ij}\, c^{\dagger}{}_i c_j + \mathrm{h.c.}\,)$$

BLOCH HAMILTONIAN

$$H_{ab}(k) = \Sigma_R\, e^{\, i\, k \cdot (R \,+\, \tau_b \,-\, \tau_a)}\, H_{ab}(R)$$

EIGENPROBLEM

$$H(k)\, u_n(k) = E_n(k)\, u_n(k)$$

- ε_i: onsite energies; t_{ij}: hoppings (complex in general — hermiticity fixes the reverse hop).
- R: lattice vector, τ_a: orbital position in the cell. PythTB uses this atomic gauge: the phases carry the orbital positions, so Berry phases have the right meaning.
- `TBModel.set_onsite`, `set_hop`, `solve_path` / `solve_mesh` are these three lines.

From atoms to bands

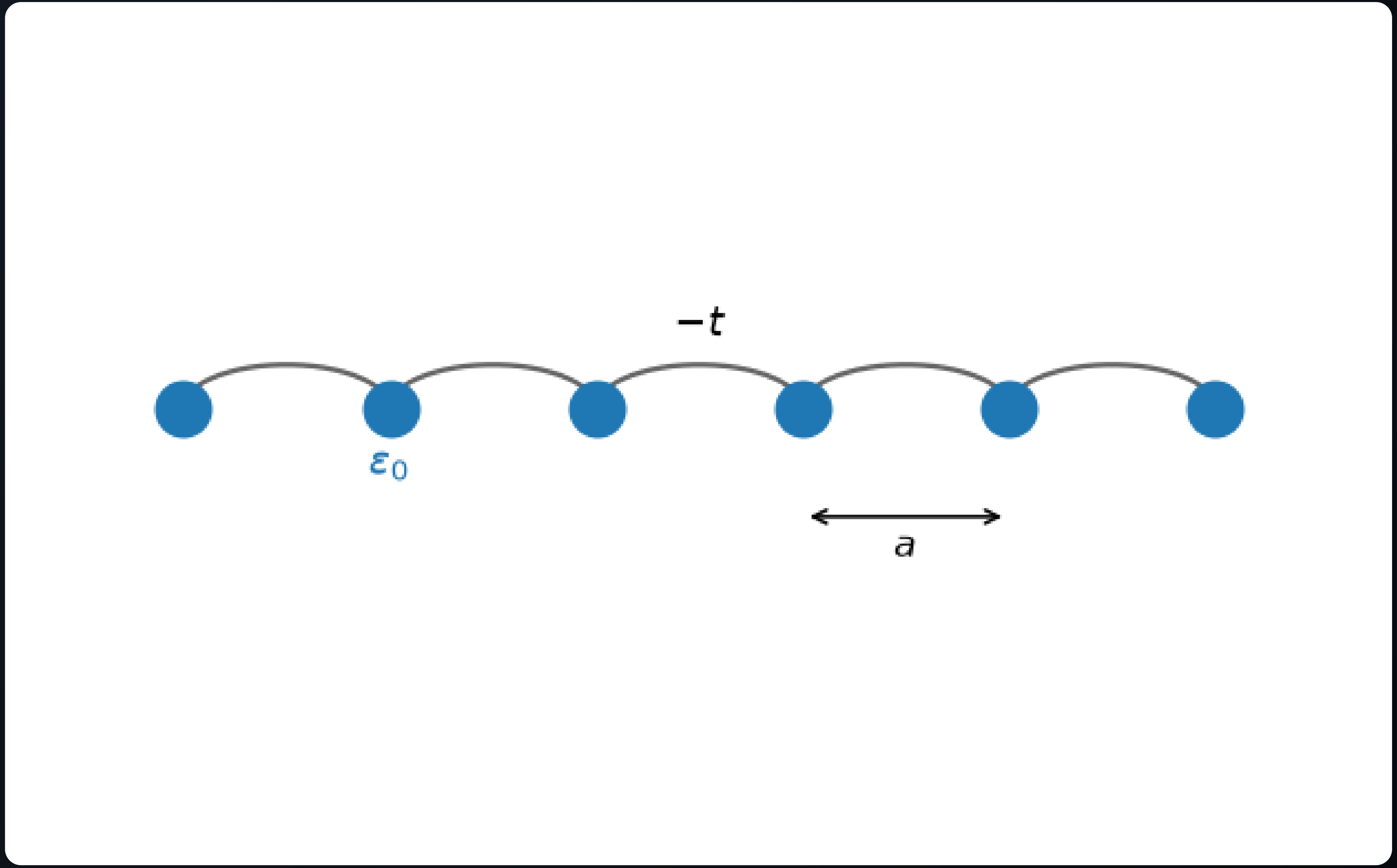
A single orbital per atom, one hopping, and Bloch's theorem give the first band structure — and the first PythTB model.

1. One orbital per atom, one hop per bond
2. The object model: Lattice → TBModel
3. The first band structure
4. Bloch's theorem on one line

One orbital per atom, one hop per bond

The simplest crystal: a row of identical atoms, spacing a. Each has one orbital at energy ε_o; an electron hops to the neighbour with amplitude t.

- The figure is what PythTB stores: the sites, their onsite energy, the bond.
- Two numbers, ε_o and t, will produce a whole band of allowed energies.
- The sign of t is set by the orbital overlap: for s-orbitals it is negative.



The monatomic chain as PythTB stores it: one orbital per cell at onsite energy ε_0 (blue), nearest-neighbour hopping −t (arcs), lattice constant a. Every model in this notebook reduces to exactly this data: positions, onsite energies, and a list of hoppings. — Figure 1 · notebook §2 · chapter 1, cell 5

CORE

The object model: `Lattice` → `TBModel`

A PythTB model is a lattice (vectors + orbital positions) plus a table of onsite energies and hoppings. Nothing is solved until you ask.

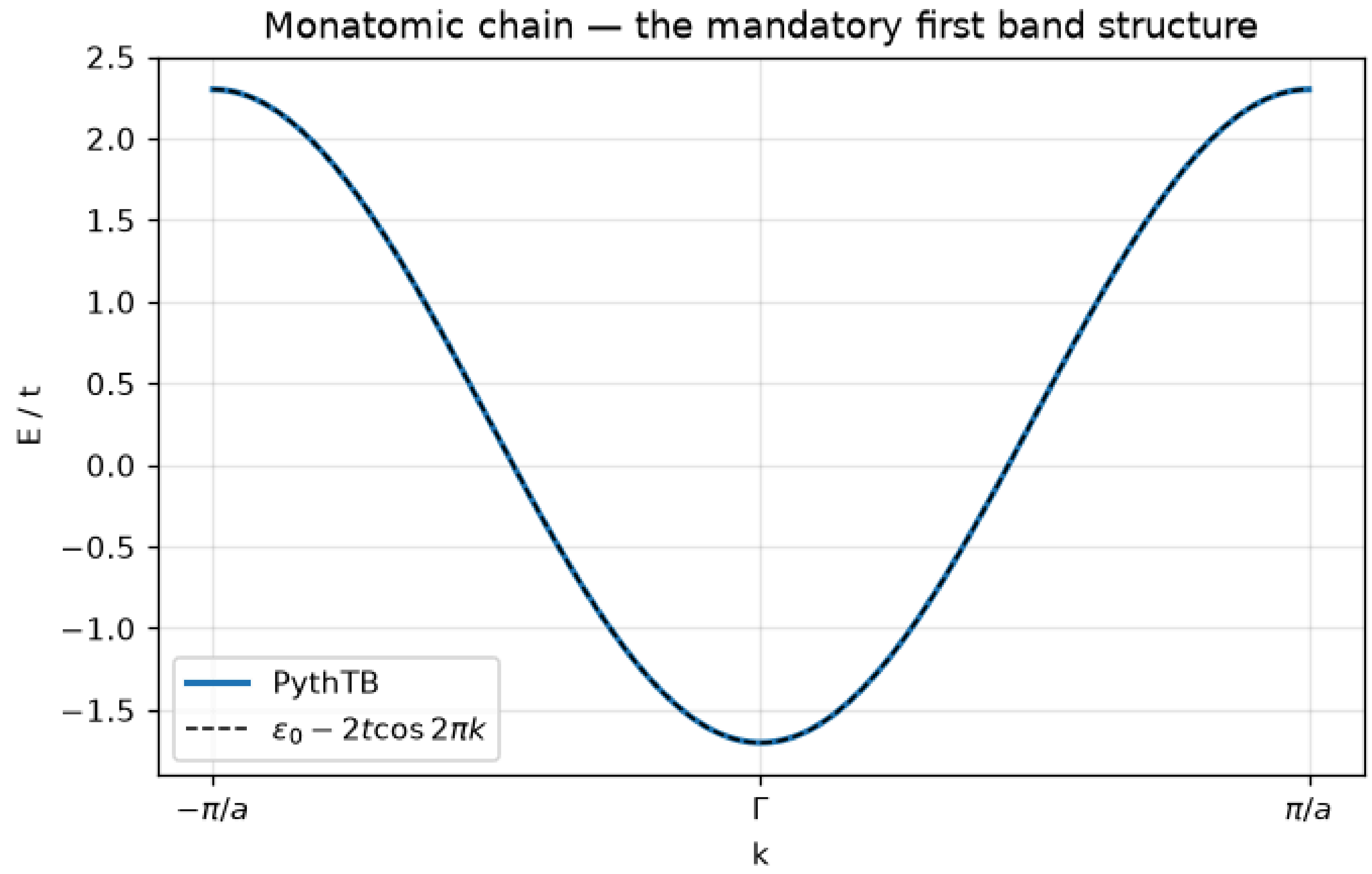
```
from pythtb import Lattice, TBModel

lat = Lattice(lat_vecs=[[1.0]], orb_vecs=[[0.0]])     # a = 1, one orbital at τ = 0
chain = TBModel(dim_k=1, dim_r=1, lattice=lat)
chain.set_onsite([0.0])                               # ε0
chain.set_hop(-1.0, 0, 0, [1])                        # t = -1 from orbital 0 to its image in cell R = +1

k_path, k_dist, k_nodes = chain.k_path([[-0.5], [0.0], [0.5]], nk=201)
energies = chain.solve_path(k_path)                   # shape (n_bands, nk)
```

- `dim_k` and `dim_r` can differ: a ribbon is `dim_k=1` in `dim_r=2`.
- `set_hop(t, i, j, R)` connects orbital i in the home cell to orbital j in cell R; the reverse hop is implied.
- Reduced coordinates everywhere: k in units of the reciprocal vectors, positions in units of the lattice vectors.

CORE

The first band structure

Numerical eigenvalues along the 1D Brillouin zone lie exactly on the analytic cosine. Bandwidth $4|t|$, minimum at $k = 0$.



Band structure of the chain above: the numerical eigenvalues (solid) lie exactly on the analytic tight-binding cosine band (dashed). The bandwidth 4t and the band minimum at Γ (for t > 0 in our sign convention) are the two numbers to read off any 1D band plot. — `Figure 2 · notebook §2 · chapter 1, cell 6`

MATH

Bloch's theorem on one line

BLOCH STATE

$$|\psi_k\rangle = \Sigma_n e^{ikna} |n\rangle$$

DISPERSION

$$E(k) = \varepsilon_0 + 2t\cos(ka)$$

NEAR THE BOTTOM

$$E \approx \varepsilon_0 + 2t + |t|\, a^2 k^2 \Rightarrow m^* = \hbar^2 / (2|t|a^2)$$

- Translation by a commutes with H; its eigenvalues e^{ika} label the eigenstates.
- k and $k + 2\pi/a$ give the same state: the Brillouin zone is $(-\pi/a, \pi/a]$.
- A stronger hop means a wider band and a lighter electron — the effective mass is set by t.

The SSH chain

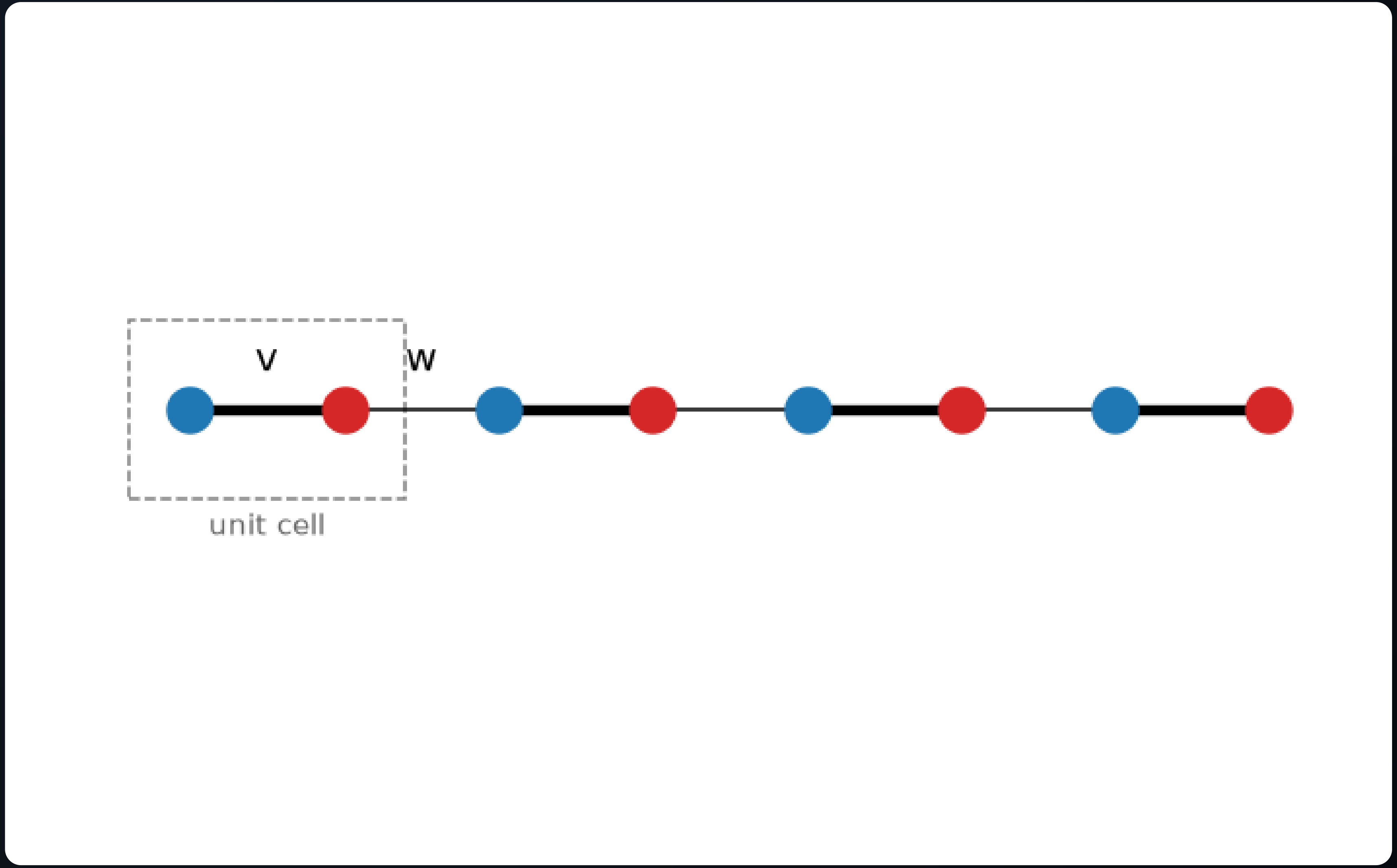
Two sites per cell teach dimerization, gap closing, end states and the winding number — the whole of topology in one dimension.

1. Two sites per cell, alternating bonds
2. One symbolic model, three insulators
3. The gap closes at exactly one point
4. Cut the chain: end modes appear
5. Where the electron actually sits: the Wannier function
6. The winding number

Two sites per cell, alternating bonds

Polyacetylene-style: bonds alternate strong, weak, strong, weak. The unit cell holds two orbitals, at $\tau = 0$ and $\tau = ½$, joined by the intracell hop v; the intercell hop w joins cells.

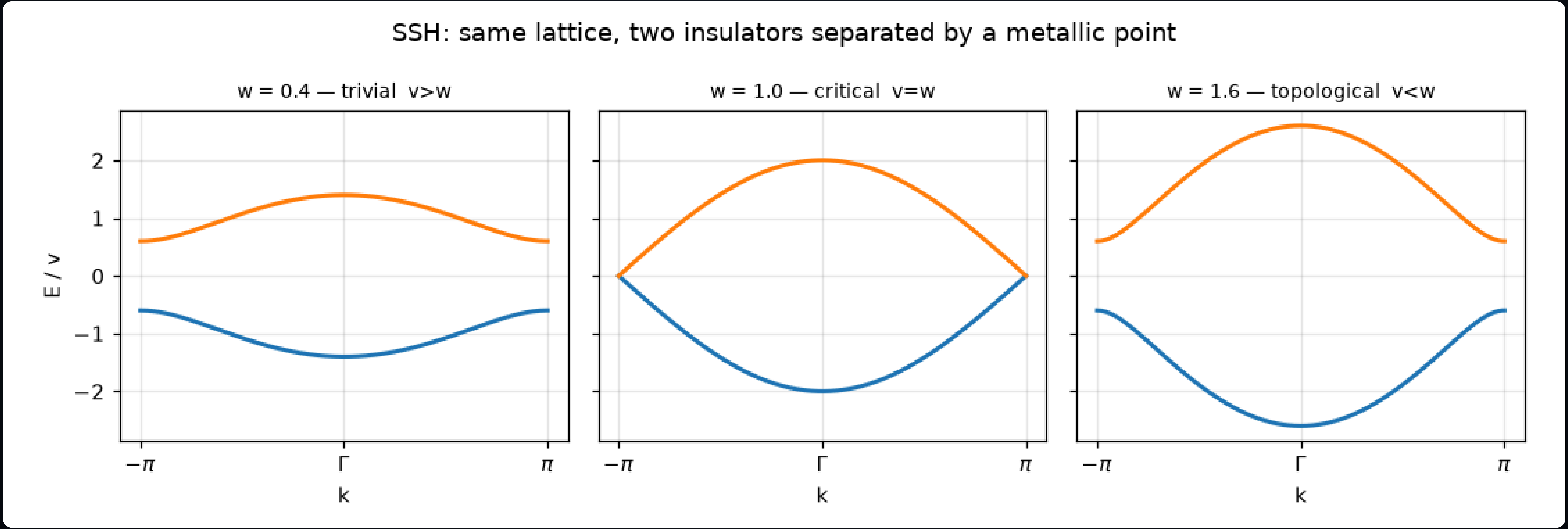
- Same atoms, same spacing as the chain — only the hoppings alternate.
- One question decides everything to come: **which bond is inside the unit cell?**
- The choice looks arbitrary. It is not: it becomes physical the moment the chain ends.



The SSH chain: two orbitals per cell (blue at τ = 0, red at τ = ½) with alternating intracell hopping v (thick) and intercell hopping w (thin). Which bond is 'inside' the cell is a choice — but once the chain ends somewhere, the choice of where to cut becomes physical, and that is the entire topological story of this model. — Figure 3 · notebook §3 · chapter 1, cell 9

One symbolic model, three insulators

PythTB lets a hopping be a parameter. The same model evaluated at $w < v$, $w = v$, $w > v$ gives two gapped band structures and one metal in between.



SSH bands for three dimerizations, computed from ONE symbolic model by broadcasting over w. The spectra of w = 0.4 and w = 1.6 look identical — band energies cannot distinguish the two insulators — yet they are topologically distinct: no smooth deformation connects them without passing through the gapless point w = v (middle panel).

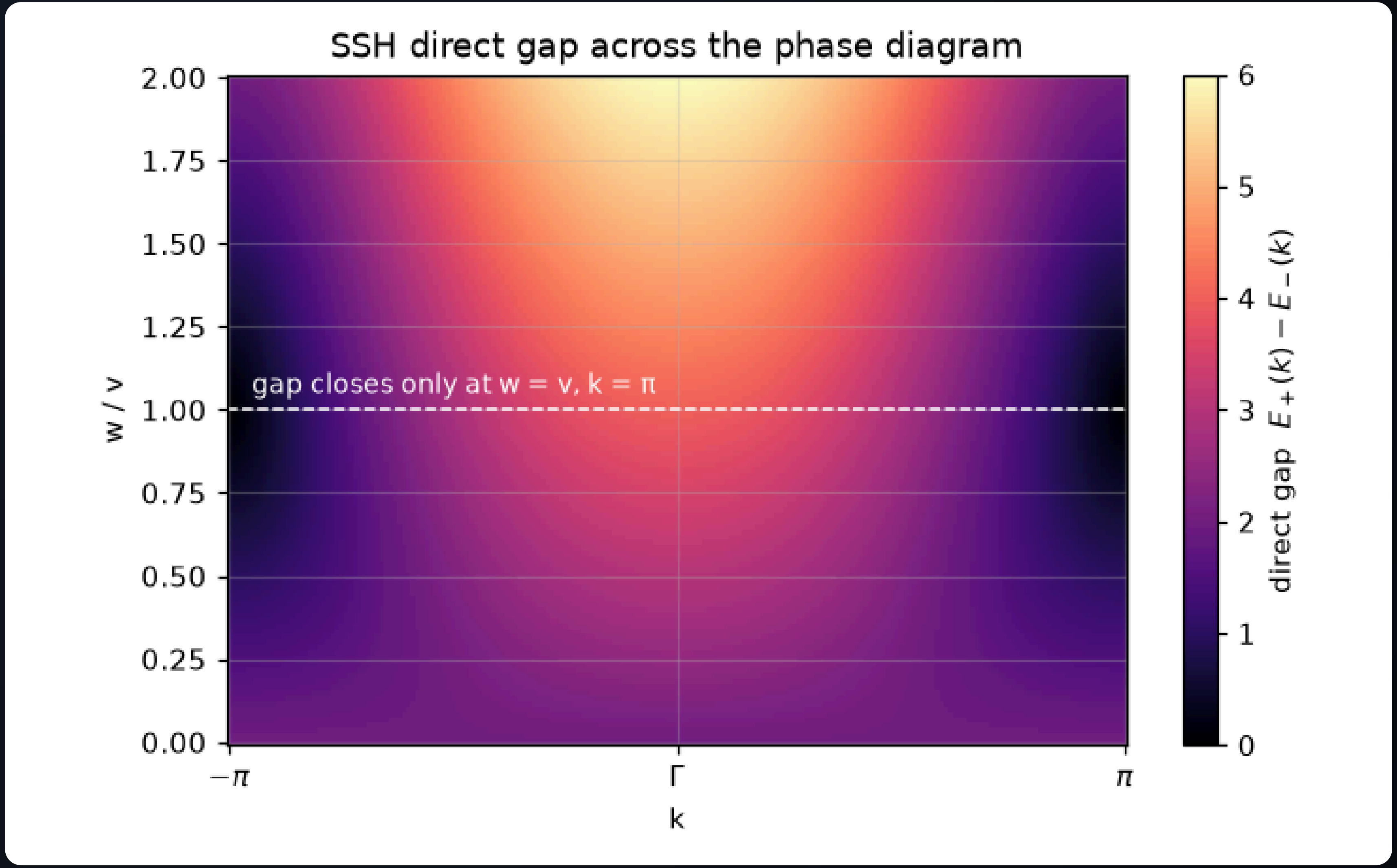
— Figure 4 · notebook §3 · chapter 1, cell 10

CORE

The gap closes at exactly one point

The direct gap over the whole (k, w) plane vanishes only at $w = v$, $k = \pi$.

- To go from one insulator to the other you must close the gap: they are not adiabatically connected.
- That is the definition of distinct **topological phases** — inside a symmetry class, phases separated by a gap closing.
- The symmetry here: chiral (sublattice) symmetry — no hopping within a sublattice.

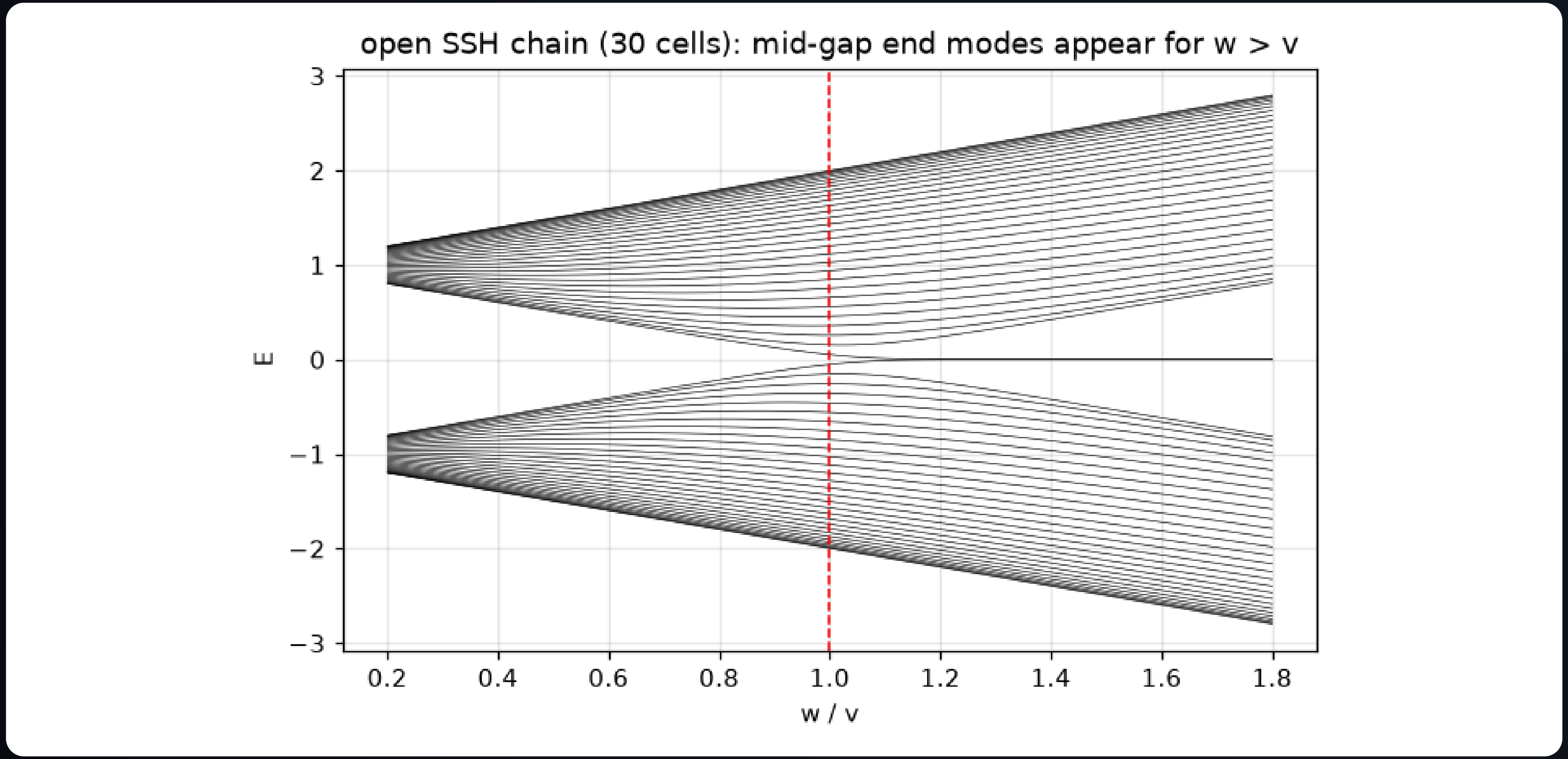


The direct gap over the whole (k, w) plane: it vanishes at exactly one point, (k, w) = (π, v). Any path from the trivial (bottom) to the topological (top) region must cross that point — a gap closing is the unavoidable toll of a topological transition. — Figure 5 · notebook §3 · chapter 1, cell 11

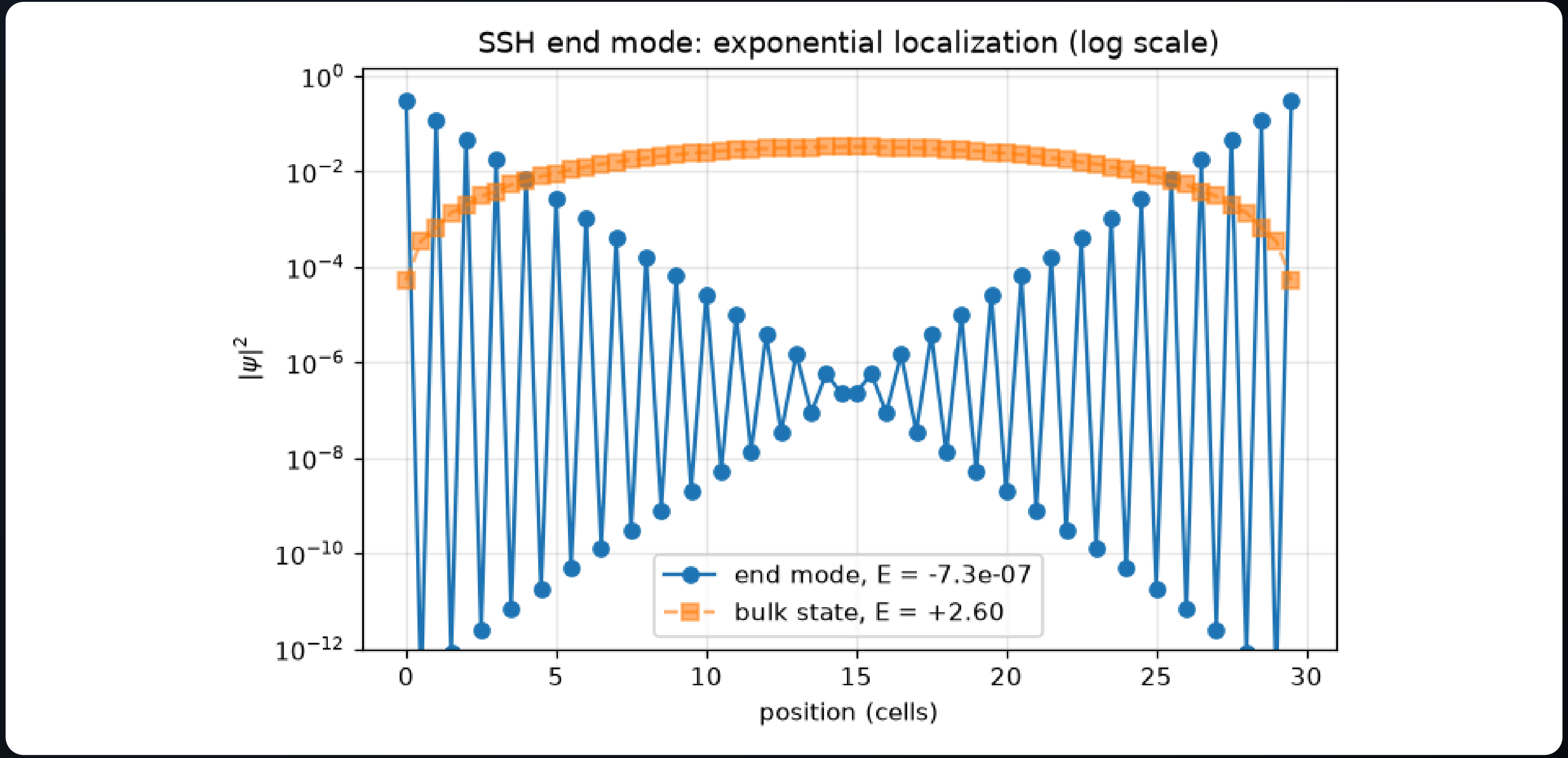
CORE

Cut the chain: end modes appear

`cut_piece` makes a finite chain. For $w > v$ two states sit in the middle of the gap, one per end, exponentially localized.



Spectrum of an open SSH chain versus dimerization. For w < v the finite chain simply shows the bulk bands with a clean gap; the moment w exceeds v the chain terminates on weak bonds, and two states peel off into the gap and pin exponentially to E = 0: the bulk-boundary correspondence in its simplest incarnation — the bulk invariant (Berry phase π, computed in §9) forces boundary states to exist, whatever the microscopic details of the cut. — Figure 15 · notebook §6 · chapter 2, cell 4



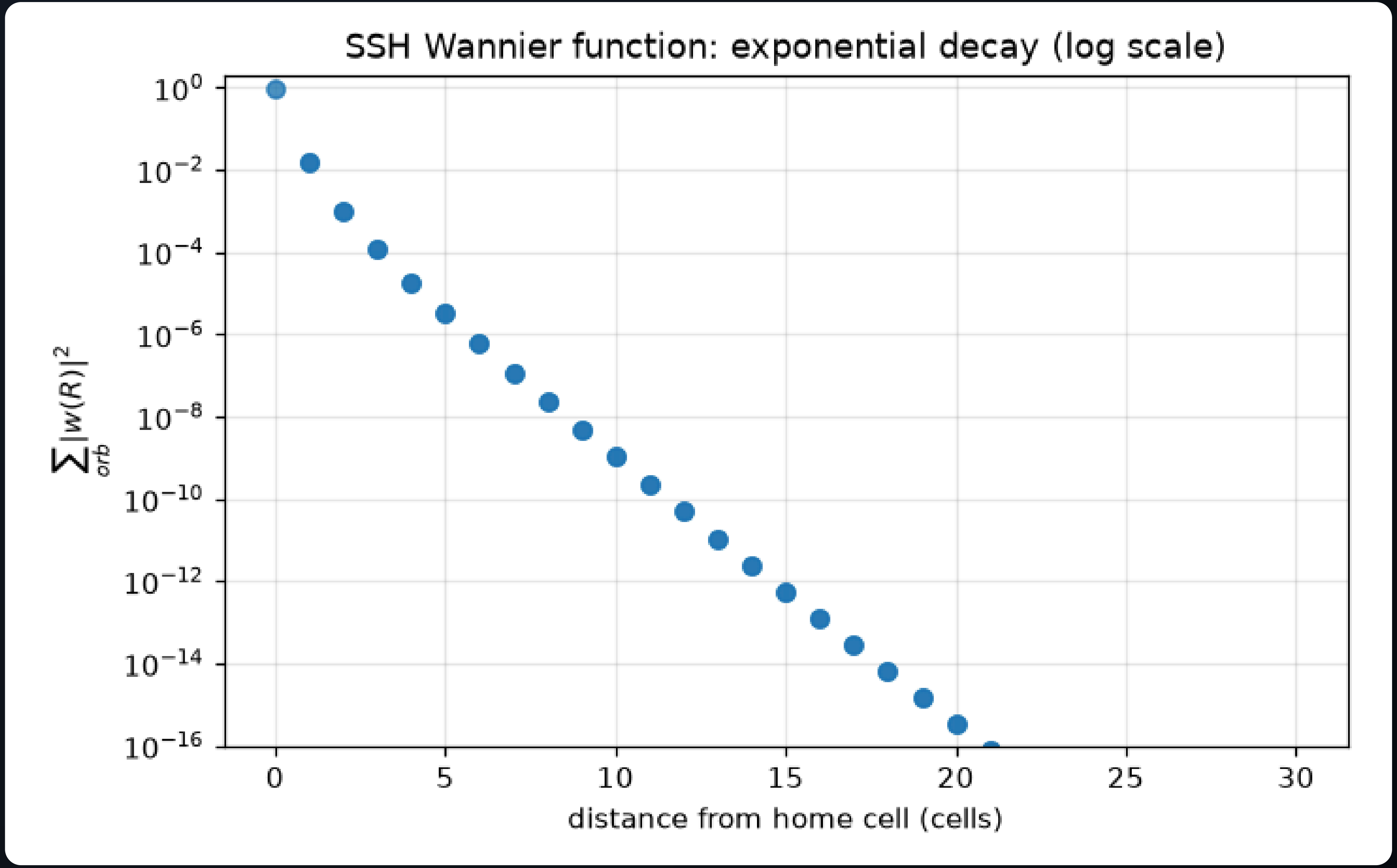
Where the end mode lives: $|\psi|^2$ of one E ≈ 0 state (circles) against a generic bulk state (squares), on a log scale. The end mode decays exponentially with decay factor $(v/w)^2$ per cell and — a chiral-symmetry fingerprint — has weight on only ONE sublattice: the even-numbered points interlace the odd ones because alternate sites carry strictly zero amplitude. — Figure 16 · notebook §6 · chapter 2, cell 5

- The open spectrum versus dimerization (left): mid-gap states exist only on one side of $w = v$.
- Their profile (right): $|\psi|^2$ falls off as $(v/w)^{2n}$, on one sublattice only.
- **Bulk–boundary correspondence**: a bulk invariant (next slides) counts the protected end states.

Where the electron actually sits: the Wannier function

Fourier-transform the occupied Bloch band back to real space and you get a localized orbital — the **Wannier function**. Its centre is the position of the electron's charge.

- Built here with PythTB's `Wannier` class (maximal localization inside PythTB, no external code).
- Exponential decay, like the end mode: the same length scale $\xi \propto 1/\ln(w/v)$ governs both.
- In the trivial phase the centre is on the atom ($\bar{x} = 0$); in the topological phase it sits on the bond ($\bar{x} = ½$).



Real-space profile of the maximally-localized Wannier function built by PythTB's own Wannier engine from the SSH Bloch states: exponential decay over ten orders of magnitude, centred on the strong bond. The decay rate is set by the gap — shrink the gap and the Wannier function fattens, which is the localization-topology trade-off behind everything in Part II. — `Figure 34 · notebook §13 · chapter 3, cell 14`

The winding number

BLOCH HAMILTONIAN

$$H(k) = d(k)\cdot\sigma,\ d(k) = (\, v + w\cos k,\ w\sin k,\ 0\,)$$

BANDS

$$E_{\pm}(k) = \pm\,|v + w\,e^{ik}| = \pm\,\sqrt{(\, v^2 + w^2 + 2vw\cos k\,)}$$

INVARIANT

$$\nu = (1/2\pi)\oint dk\,\partial_k \arg(\, v + w\,e^{ik}\,) \in \{0, 1\}$$

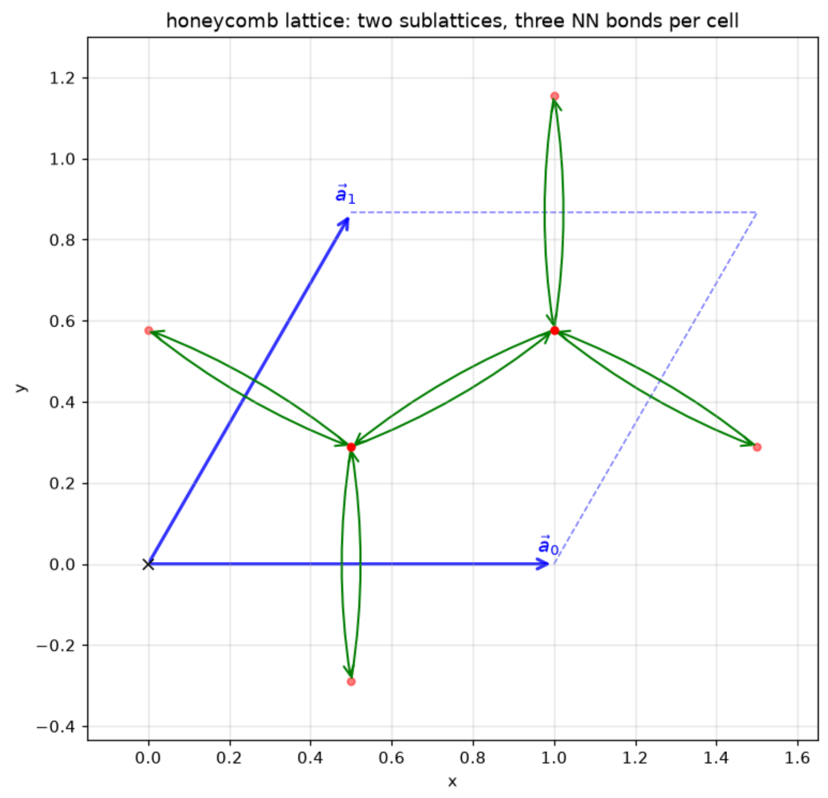
- $d(k)$ traces a circle of radius w centred at $(v, 0)$; it encloses the origin iff $w > v$.
- Chiral symmetry $\sigma_z H \sigma_z = -H$ keeps $d_z = 0$: the curve stays in a plane and the winding is well defined.
- Berry phase of the lower band $\varphi = \pi\nu$; Wannier centre $\bar{x} = \varphi/2\pi = \nu/2$.

Graphene & flat bands

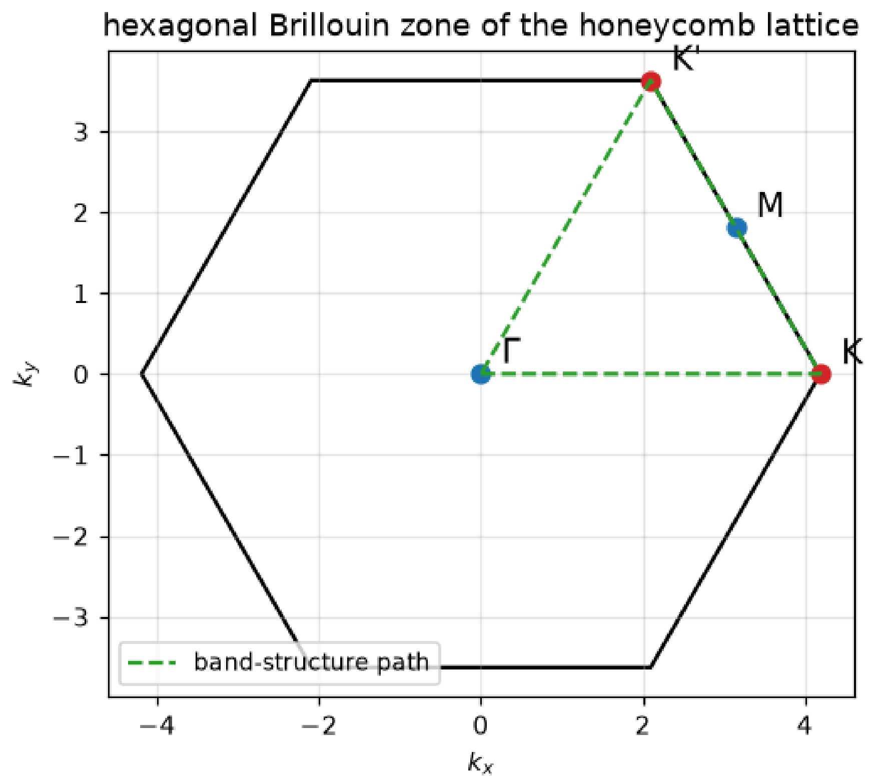
Honeycomb Dirac cones, how a sublattice potential gaps them, exactly flat bands, ribbons, supercells and defects.

The honeycomb: two sublattices, one hexagonal zone

Graphene is a triangular lattice with a two-atom basis (A and B). Every A neighbours three B's. Its Brillouin zone is a hexagon whose corners K, K' will carry all the physics.



The honeycomb lattice as PythTB sees it (built-in visualize): sublattice A and B orbitals, the two lattice vectors, and the three nearest-neighbour bonds each cell contributes. The two-site basis — not the hexagonal shape — is what produces two bands and makes a Dirac point possible. — `Figure 6 · notebook §4 · chapter 1, cell 13`



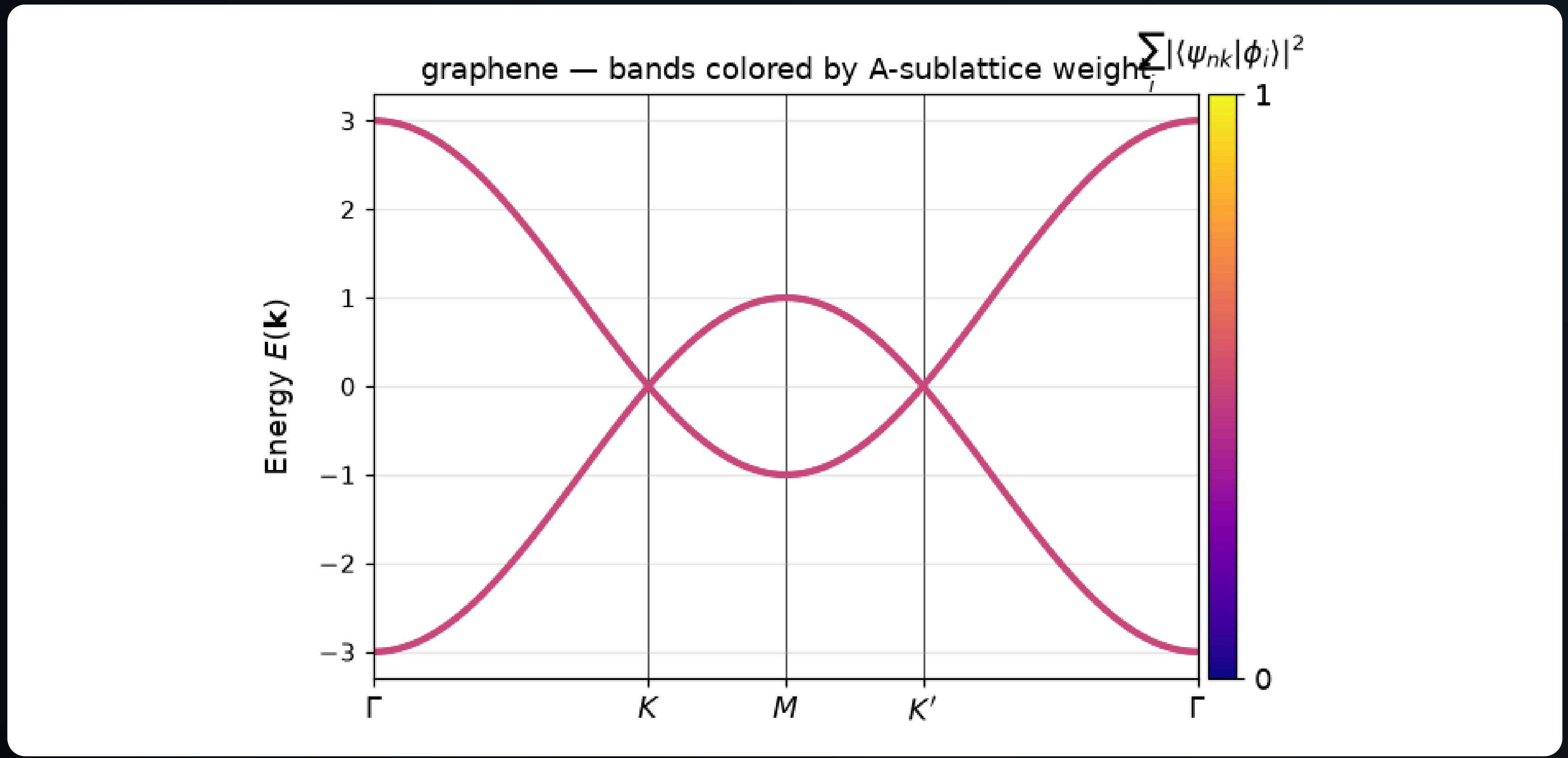
The Brillouin zone with the high-symmetry points and the Γ–K–M–K'–Γ path used in every honeycomb band plot below. K and K' (red) are the two inequivalent zone corners where the Dirac cones live; they are NOT related by a reciprocal lattice vector, which is why breaking the symmetry between them (as the Haldane model does in §14) has physical consequences. — `Figure 7 · notebook §4 · chapter 1, cell 14`

- Built-in `visualize` draws exactly what the model contains: positions and bonds.
- The band path $\Gamma–K–M–K'–\Gamma$ visits every high-symmetry point once.
- One p_z orbital per carbon, hopping $t \approx 2.7\,eV$ to nearest neighbours: the whole model.

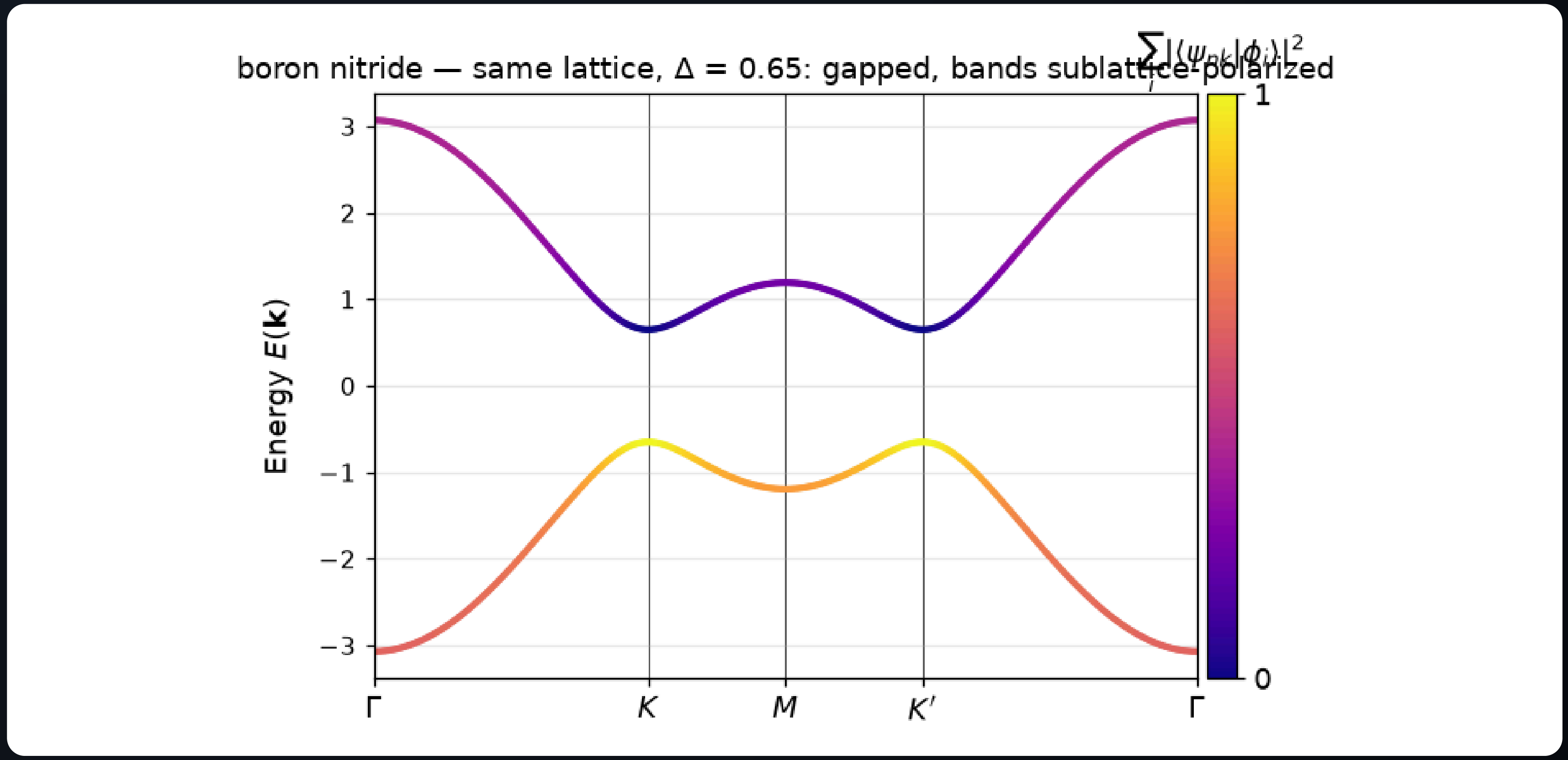
CORE

Dirac cones — and how to gap them

Left: graphene's bands touch at K and K' in linear crossings, coloured by sublattice weight. Right: boron nitride — same lattice, different atoms on A and B — opens a gap of 2Δ.



Graphene bands along the BZ path just shown, colored by the weight of each state on sublattice A. Away from the Dirac points the bands mix both sublattices equally (green); the linear crossings at K and K' are protected by the equivalence of the two sublattices — inversion symmetry. – `Figure 8 · notebook §4 · chapter 1, cell 15`



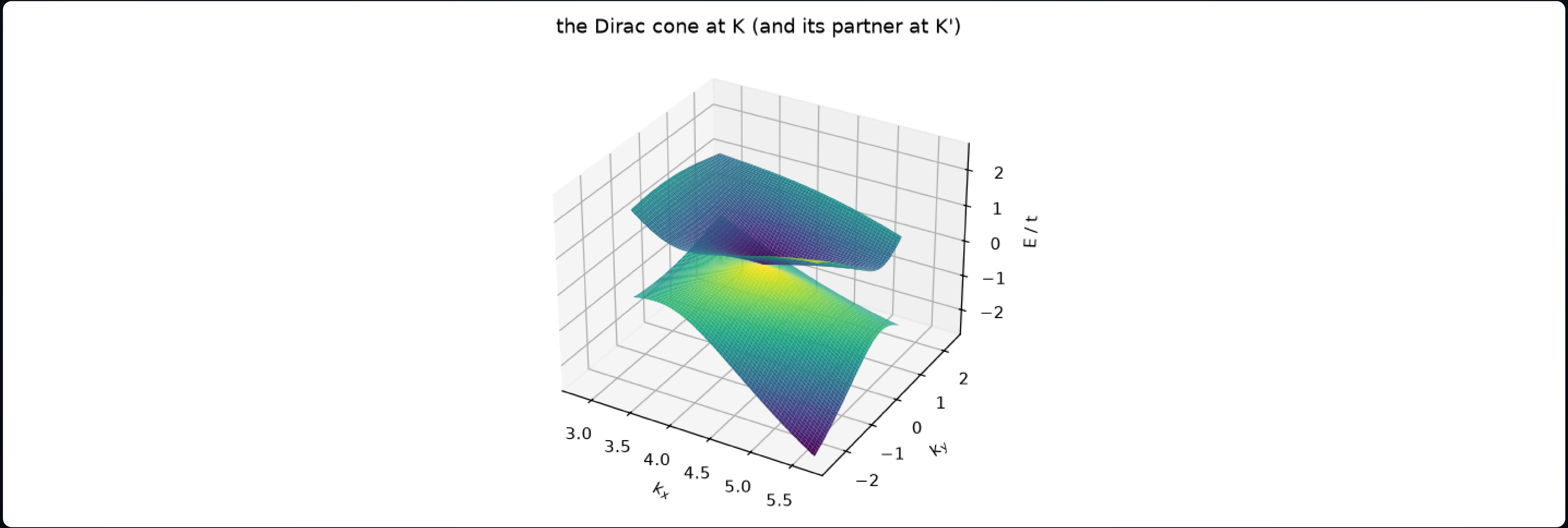
Boron nitride: chemically, B and N atoms make the two sublattices inequivalent — modeled by the staggered onsite ±Δ. The Dirac points gap out and the band coloring shows the physics: the valence band concentrates on the low-energy (N-like) sublattice, the conduction band on the other. Same lattice, same hoppings — a semimetal became an insulator by symmetry breaking. – `Figure 9 · notebook §4 · chapter 1, cell 15`

- At the touching point the electron has **zero effective mass**: the dispersion is a cone, not a parabola.
- The colour shows the eigenvector: away from K the bands mix A and B equally; the sublattice potential $\pm\Delta$ polarizes them.
- `set_onsite([Δ, -Δ])` is the entire difference between graphene and BN.

CORE

The cone as a surface

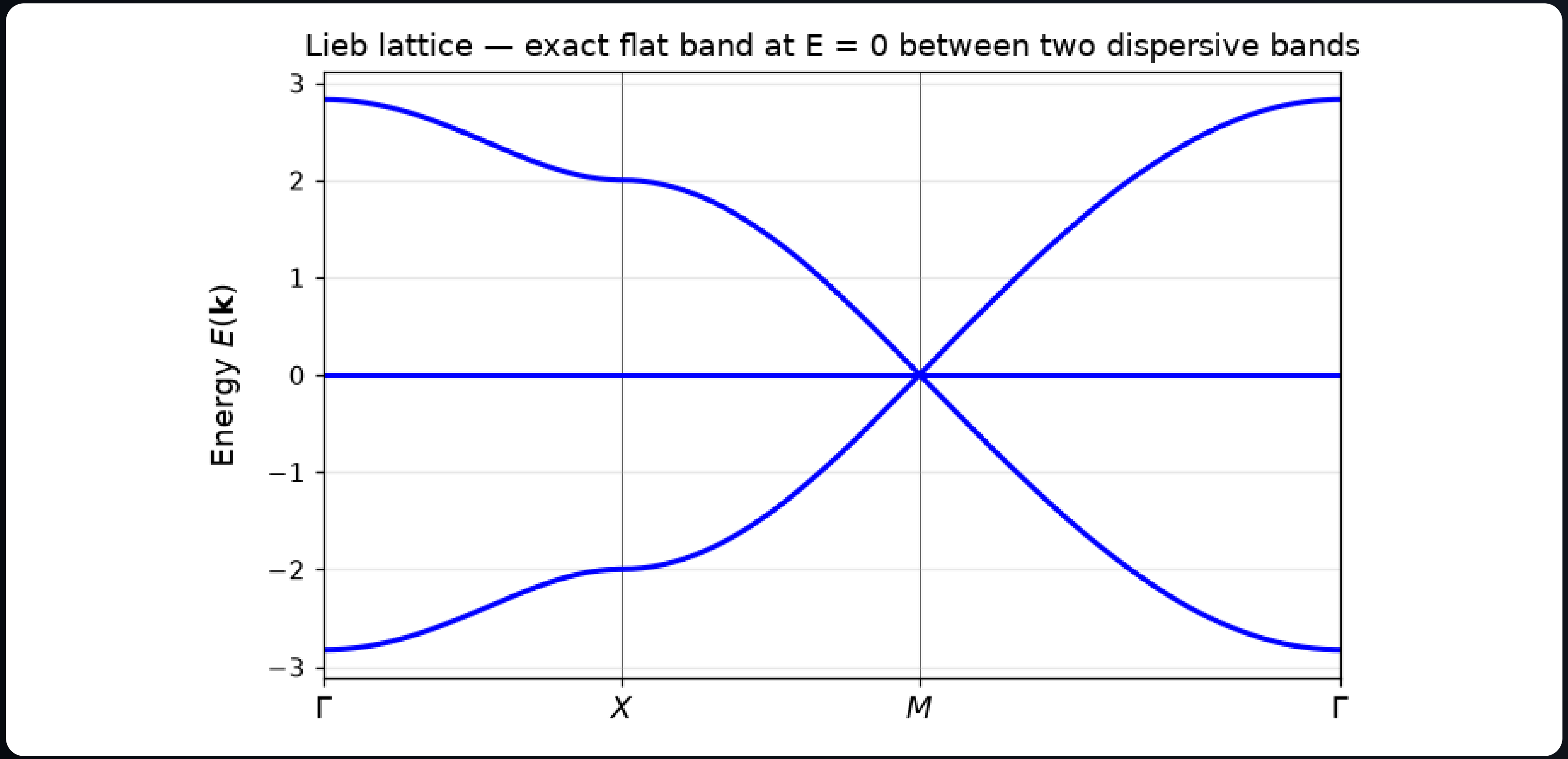
`solve_mesh` over a patch of the zone around K: the two bands form a double cone, isotropic to first order.



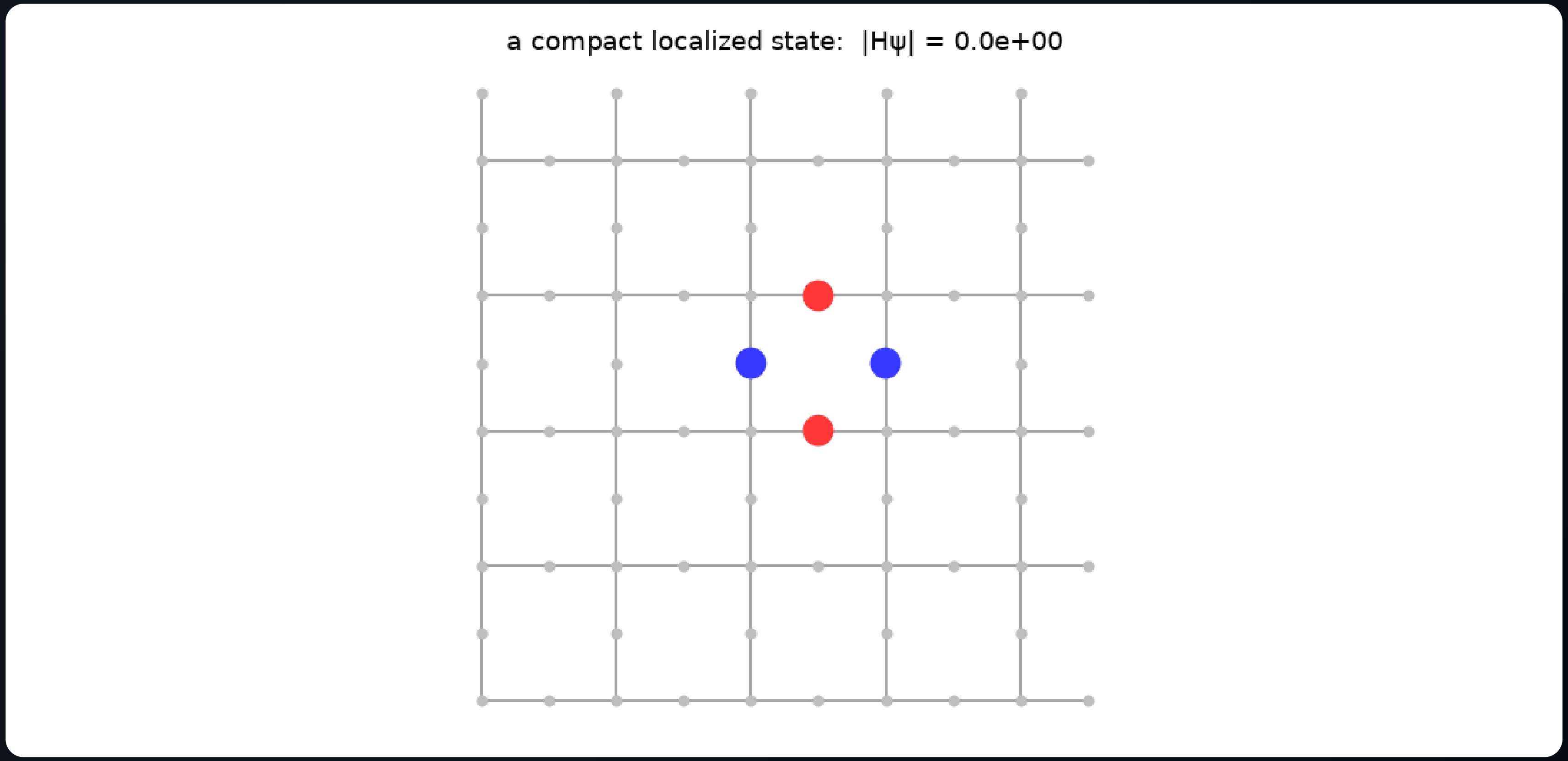
The energy surface E(kx, ky) near the zone corner: the two bands meet in a cone — dispersion linear in every direction, the hallmark of massless Dirac fermions. The slope is the Fermi velocity; in real graphene it is c/300, which is why relativistic phenomenology shows up at meV energies. — `Figure 10 · notebook §4 · chapter 1, cell 16`

Exactly flat bands: the Lieb lattice

Three sites per square cell (one corner, two edges). One band has no dispersion at all — and a compact localized state explains why in one picture.



Lieb-lattice bands: two dispersive bands touching a perfectly flat band at E = 0. Flat means infinite effective mass — electrons in this band cannot propagate at all, because every state is built from the interference-locked plaquette configurations shown in the next figure. — Figure 11 · notebook §5 · chapter 1, cell 18

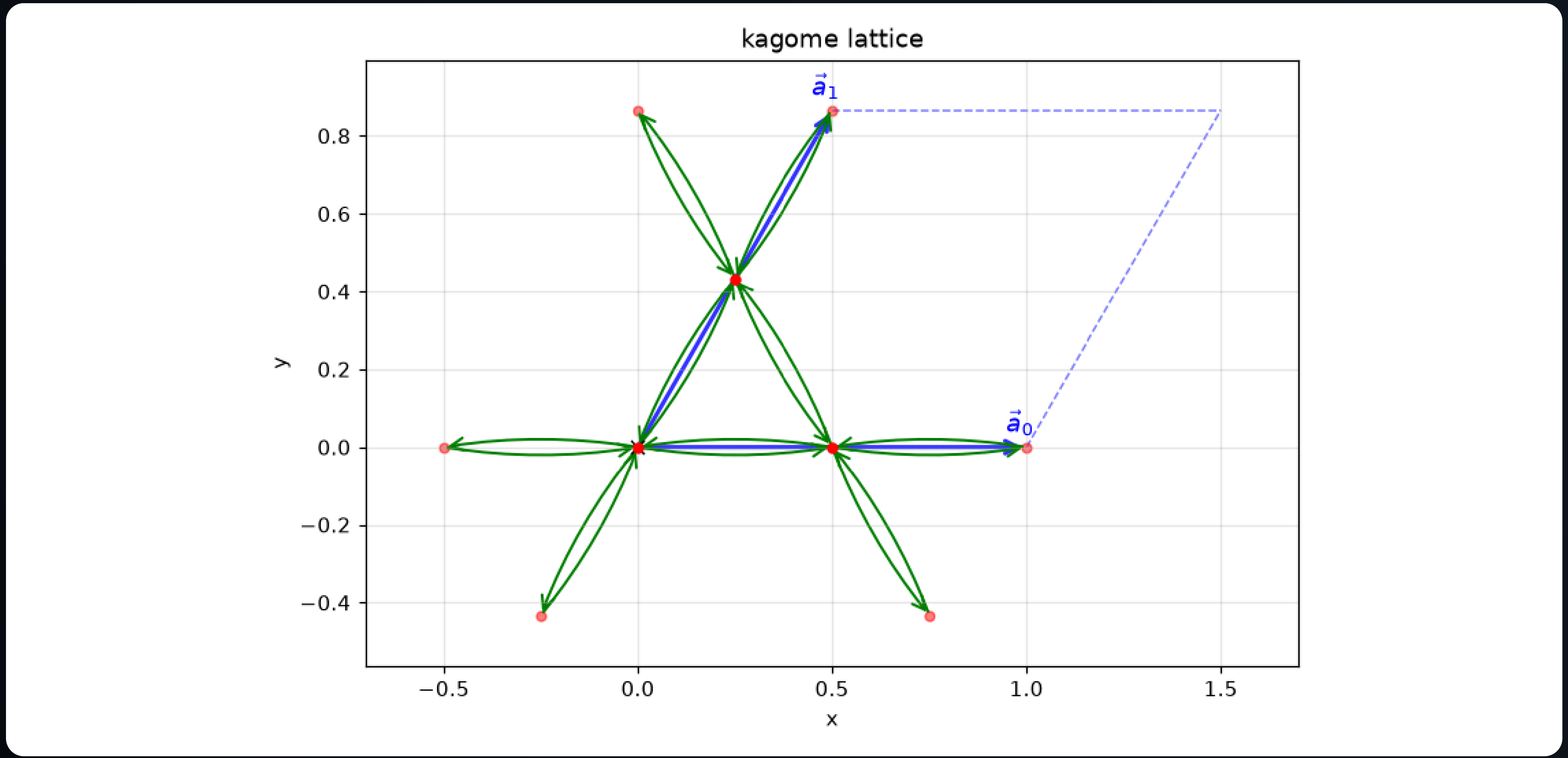


The flat band, explained in one picture: a state with alternating signs (red/blue) on the four edge sites of a single plaquette. Each neighbouring corner site couples to a +1 and a −1 with equal hopping, so the amplitudes cancel exactly and the state is a strict eigenstate at E = 0. One such state exists per plaquette — a macroscopically degenerate band of them. — Figure 12 · notebook §5 · chapter 1, cell 19

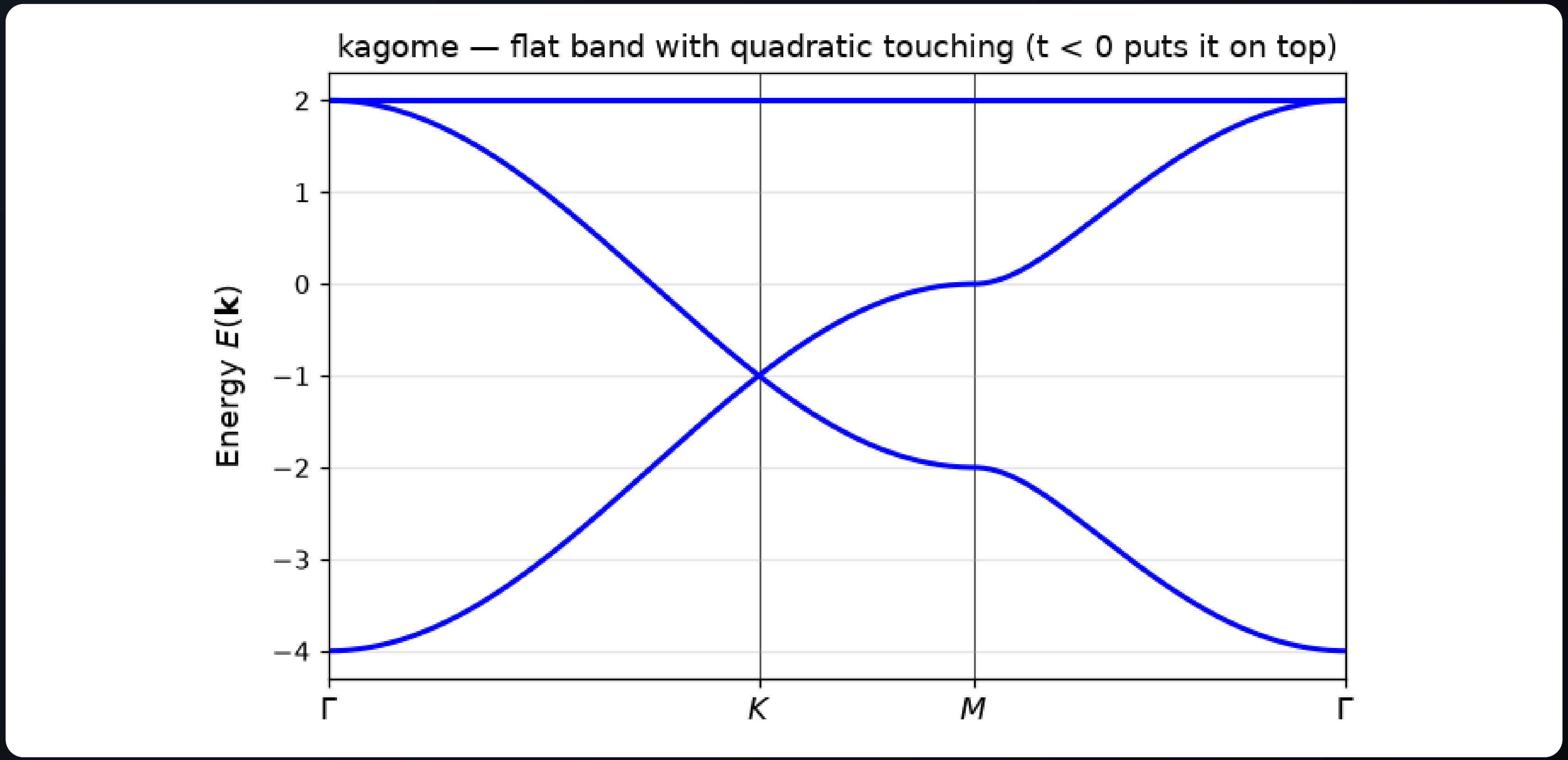
- Flat band ⇒ infinite effective mass ⇒ interactions win over kinetic energy: the playground for magnetism and correlated phases.
- The localized state lives on the four edge sites around a plaquette with alternating signs: every hop out of it cancels.
- Bipartite geometry (corner sites outnumbered by edge sites) guarantees at least one zero-energy state per cell.

The kagome lattice: flat band meets Dirac cone

Corner-sharing triangles, three sites per cell. The spectrum has both a Dirac crossing at K and a flat band touching a dispersive one quadratically at Γ.



The kagome lattice: corner-sharing triangles, three sites per cell. Geometric frustration — every closed loop of bonds has odd length 3 — is what breaks particle-hole symmetry and pins the flat band at the TOP of the spectrum for t < 0 rather than in the middle. — `Figure 13 · notebook §5 · chapter 1, cell 20`



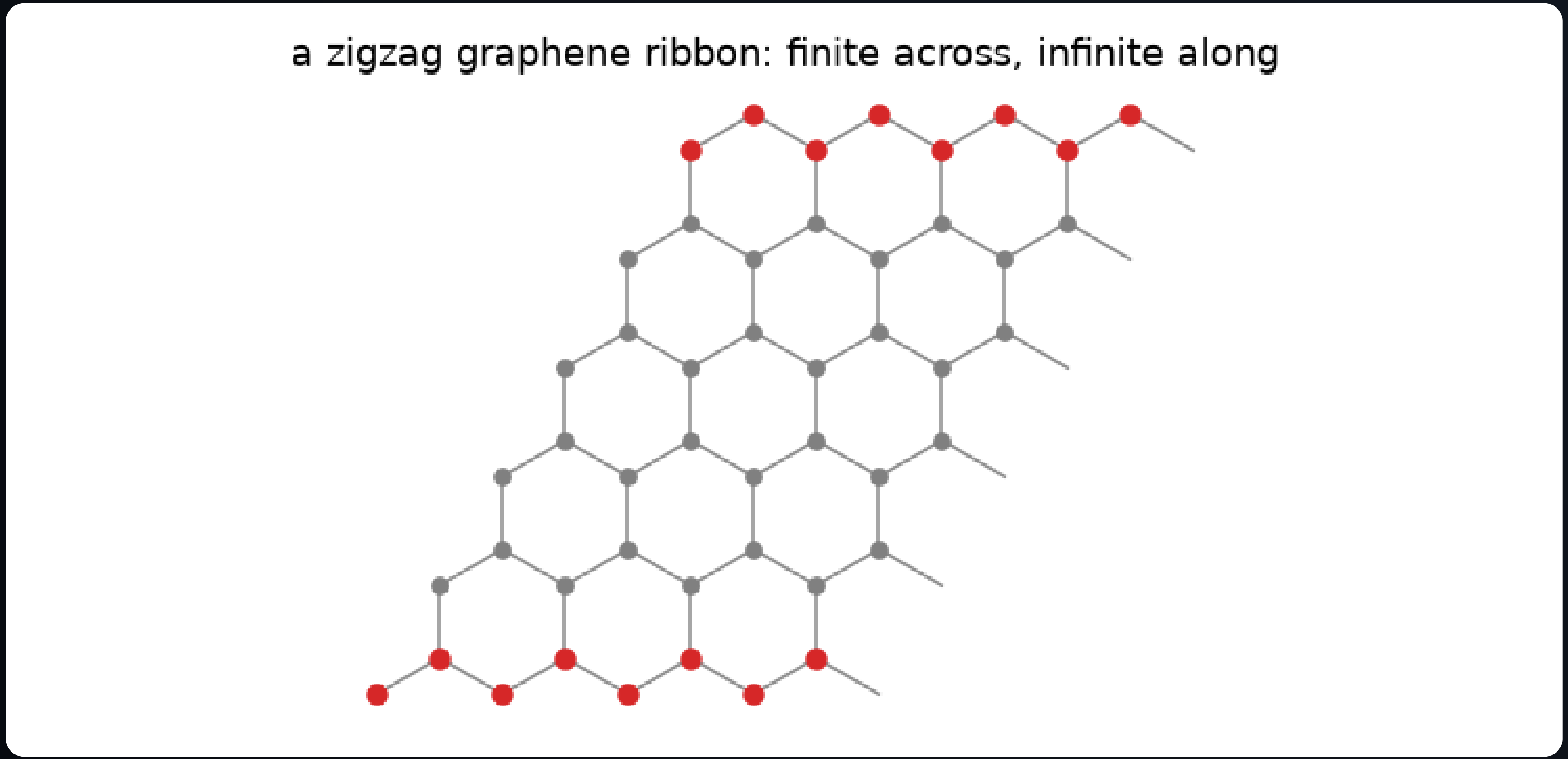
Kagome bands: a Dirac crossing at K (graphene-like, from the underlying triangular Bravais lattice) plus an exactly flat band touching the middle band quadratically at Γ. The kagome CLS lives on a hexagon with alternating signs; unlike the Lieb case the touching makes the flat band's topology singular — it cannot be spanned by exponentially localized Wannier functions. — `Figure 14 · notebook §5 · chapter 1, cell 20`

- Here the localized states are ring states around a hexagon, alternating in sign.
- The sign of t decides whether the flat band sits on top or at the bottom — a one-line experiment in the notebook.
- Real kagome metals (e.g. the AV_3Sb_5 family) show exactly these features in photoemission.

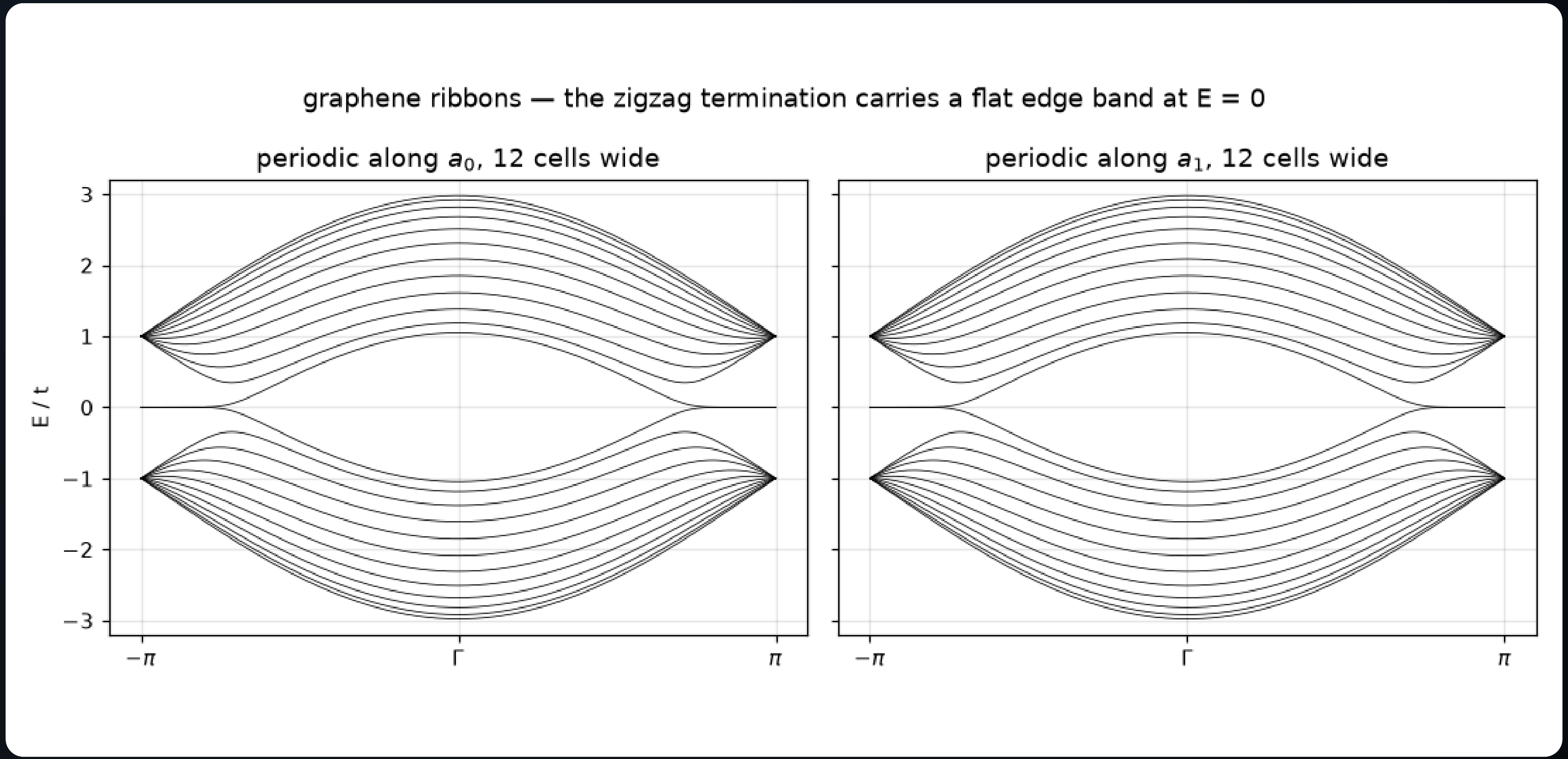
CORE

Ribbons: finite across, infinite along

`cut_piece` along one direction makes a strip: `dim_k = 1` inside `dim_r = 2`. Its band structure depends on the edge: the zigzag termination carries a **flat edge band** at $E = 0$.



The ribbon geometry produced by cut_piece: six cells wide, periodic along the horizontal direction (a few periods are drawn). The outermost atoms (red) form zigzag edges, and — crucially — the top and bottom edge atoms belong to OPPOSITE sublattices. The flat edge band in the next figure lives on exactly these red sites.
— Figure 17 · notebook §6 · chapter 2, cell 6

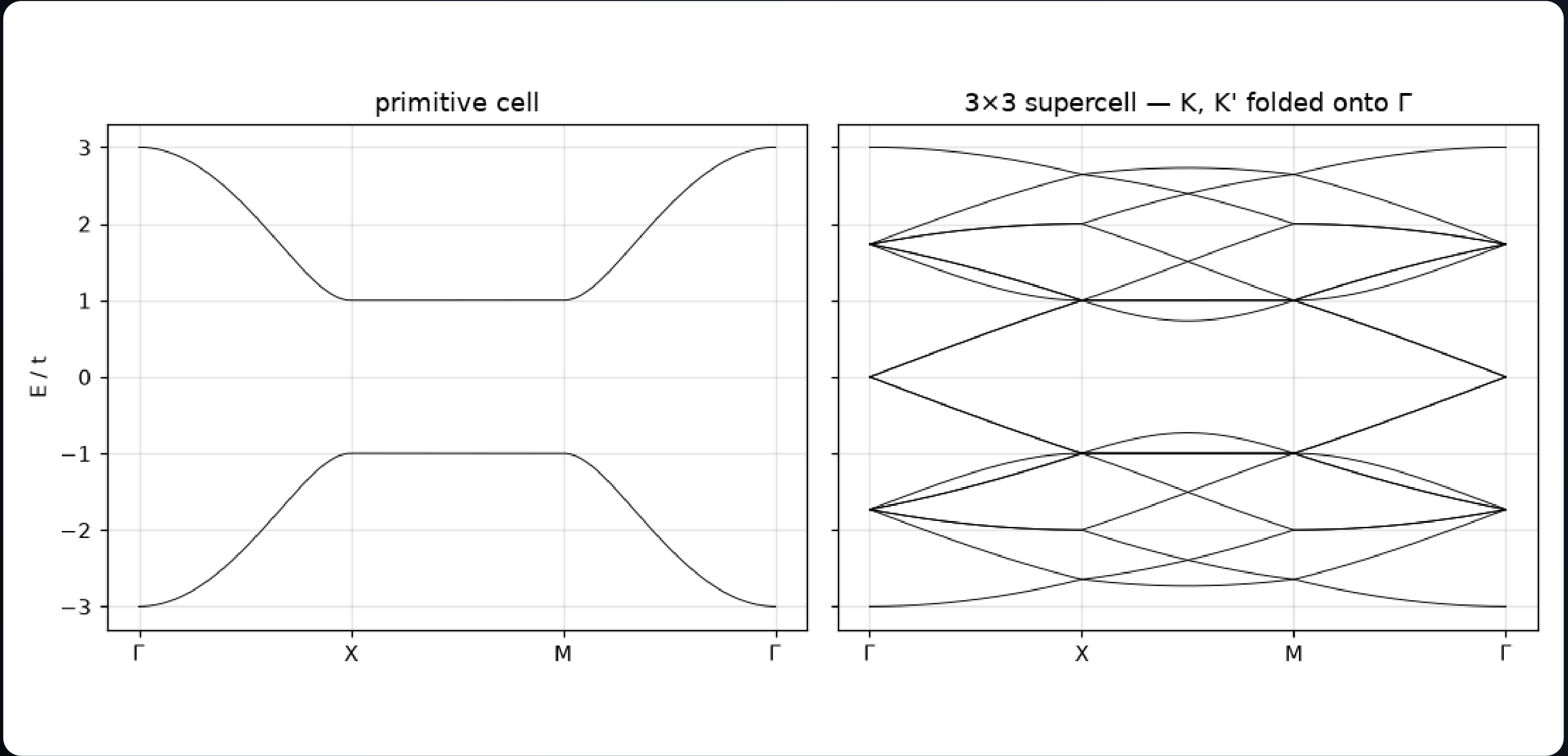


Ribbon band structures for the two cut directions. Between the projections of K and K' (around k = π) a nearly flat band sits at E = 0: these are the zigzag edge states, one per edge, living on a single sublattice. Their flatness (zero velocity) concentrates density of states at the Fermi level of neutral graphene — the seed of the edge magnetism computed in §29. — Figure 18 · notebook §6 · chapter 2, cell 7

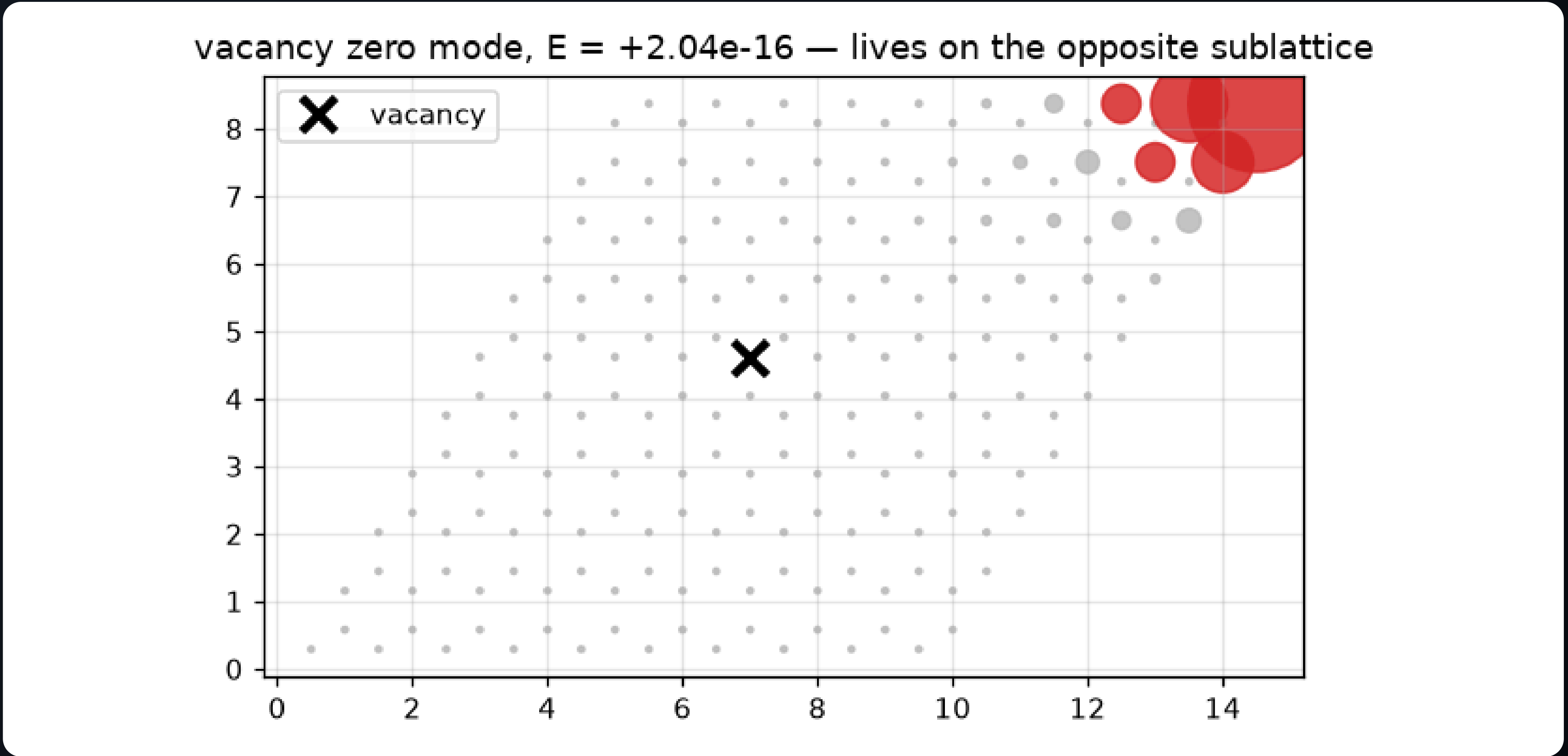
- Projecting the 2D bands onto the strip's k gives the shaded continuum; states outside it live on the edges.
- Zigzag edges end on one sublattice: the edge state is the SSH end mode in disguise, with k as a parameter.
- Armchair edges mix both sublattices and have no such band.

Supercells and defects

`make_supercell` repeats the cell; the bands fold into a smaller zone (left). `remove_orb` deletes a site: a vacancy in graphene binds a zero-energy state on the other sublattice (right).



Band folding: the same graphene crystal described in its primitive cell (left) and in a 3×3 supercell (right). Nothing physical changed — the supercell BZ is 9× smaller, so the bands fold back 9× denser, and both Dirac points land on Γ. Folding is the price of any supercell treatment (defects, disorder, magnetic flux) and must not be mistaken for new physics. — Figure 19 · notebook §7 · chapter 2, cell 9



A single vacancy in a graphene flake (black cross) and the zero mode it creates (red, symbol size ∝ $|\psi|^2$). Removing one A-site leaves the B sublattice with one extra site, and on any bipartite lattice that imbalance guarantees an exact E = 0 state confined to the majority sublattice — decaying as a power law around the missing atom. The proof is counting, not perturbation theory, so the mode survives at any hopping strength.
— Figure 20 · notebook §7 · chapter 2, cell 10

- Band folding creates no new physics — it re-labels the same states; the crossings at the new zone boundary are artefacts.
- Defects, disorder and edges all need supercells: this is how a periodic code describes a non-periodic object.
- The vacancy mode is a bipartite-lattice theorem (Lieb): sublattice imbalance ⇒ zero modes.

MATH

The honeycomb Bloch Hamiltonian

TWO-BAND MODEL

$$H(k) = (\Delta\ f(k)\ ;\ f^*(k)\ -\Delta\ ),\ f(k) = t\ \Sigma_{j=1}{}^{3}\ e^{\,i\,k\cdot\delta_j}$$

BANDS

$$E_\pm(k) = \pm\ \sqrt{(\ \Delta^2 + |f(k)|^2\ )}$$

NEAR K

$$f(K + q) \approx \hbar v_F\,(q_x - i\ q_y),\ \hbar v_F = 3ta_{cc}/2$$

- $f(K) = 0$ because the three phases $e^{iK\cdot\delta_j}$ are the cube roots of unity: the Dirac point is a cancellation forced by symmetry.
- $\Delta = 0$: massless cone, $E = \pm\hbar v_F|q|$. $\Delta \neq 0$: mass gap 2Δ (boron nitride).
- Lieb flat band: the state $|\psi\rangle = |e_1\rangle - |e_2\rangle + |e_3\rangle - |e_4\rangle$ around a plaquette satisfies $H|\psi\rangle = 0$ exactly.

L4 · NOTEBOOK §8, §11–12

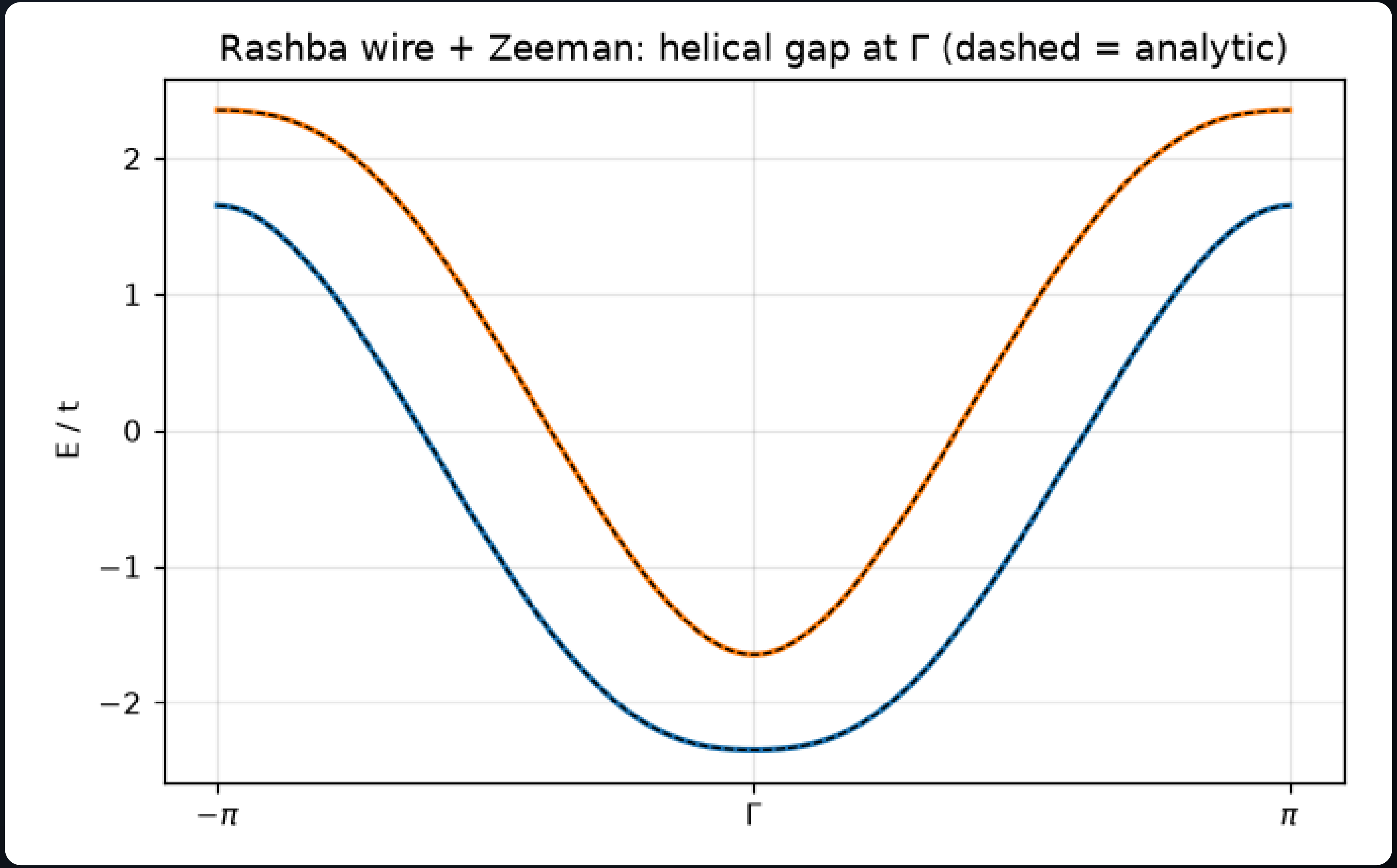
Spin, 3D, materials

Spinful models with Rashba and Zeeman terms, the simple cubic crystal and its density of states, and real silicon from Wannier90.

Spin, natively

`spinful=True` doubles every orbital and lets hoppings be 2×2 matrices in spin space. A one-dimensional wire with Rashba spin–orbit coupling and a Zeeman field shows what that buys.

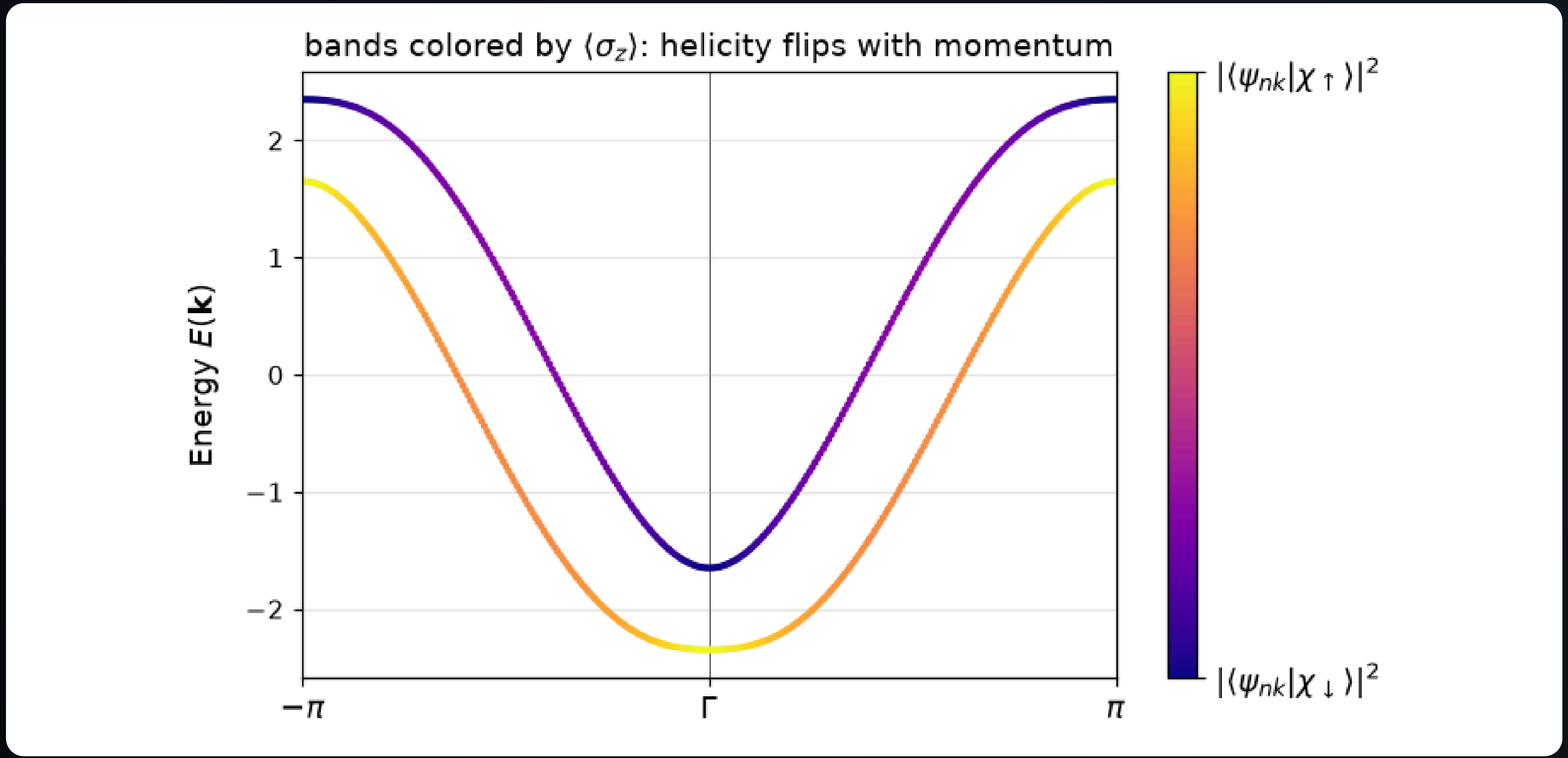
- Rashba: the spin rotates as the electron moves — two shifted parabolas with opposite spin.
- Zeeman: a field opens a gap where the two cross, at $k = 0$.
- Together: a helical gap — the recipe behind Majorana nanowires.



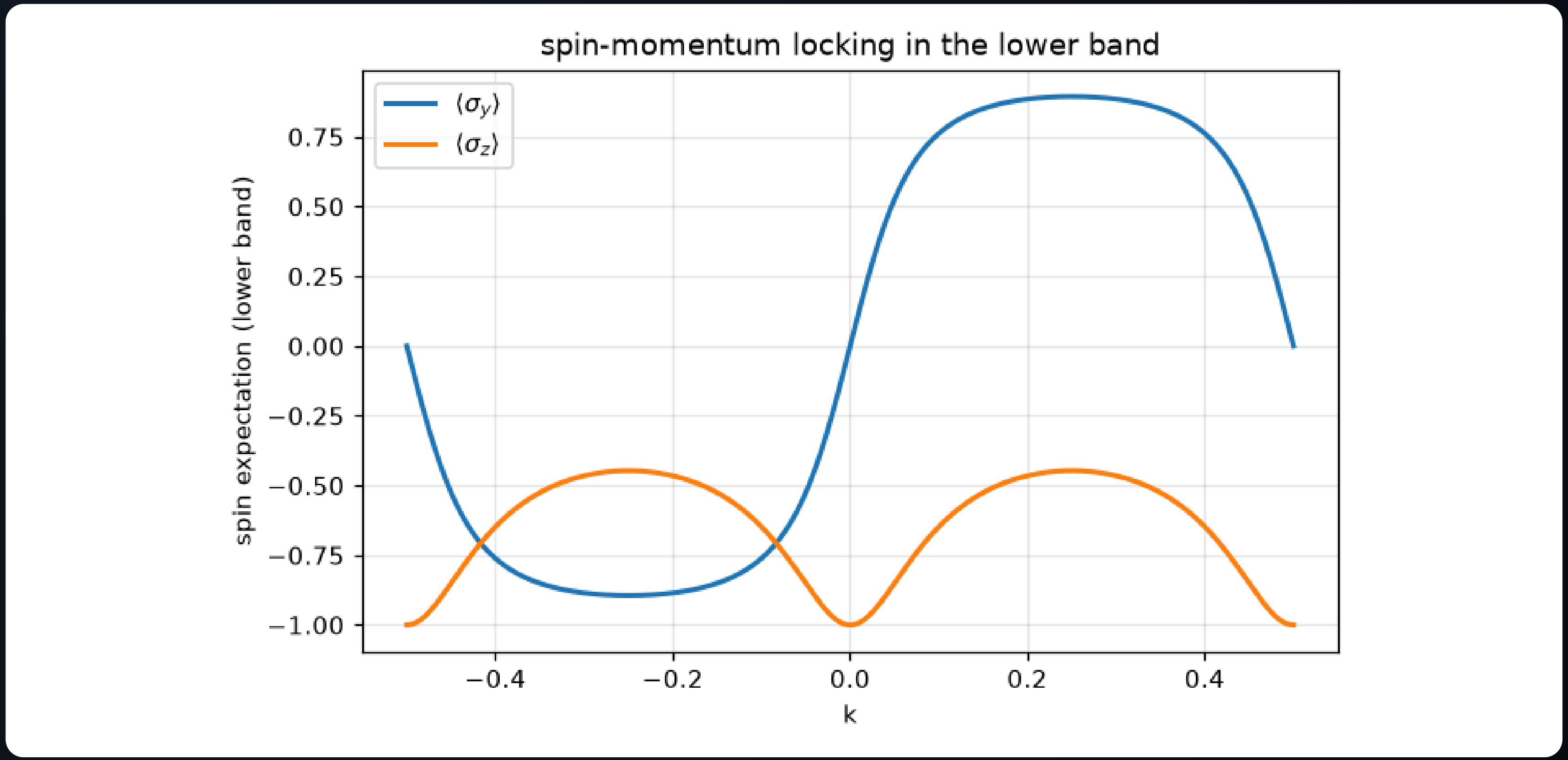
Bands of the spinful wire. Rashba coupling alone would give two shifted bands crossing at k = 0; the Zeeman field opens the 'helical gap' at that crossing. Tuning the chemical potential into this gap leaves a single pair of counter-propagating, spin-momentum-locked modes — the starting point of every Majorana nanowire proposal. – Figure 21 · notebook §8 · chapter 2, cell 12

Spin–momentum locking

Colour the bands by $\langle\sigma_z\rangle$ (left) or plot the full texture $\langle\sigma_y\rangle$, $\langle\sigma_z\rangle$ along the band (right): the spin direction is fixed by the momentum.



The same bands colored by the spin projection ⟨σz⟩ (plot_bands with proj_spin): near k = 0 the Zeeman term dominates and polarizes the spins along z; at larger |k| the Rashba field wins and rotates them into the y direction, the color fading toward the middle of the scale. – Figure 22 · notebook §8 · chapter 2, cell 13

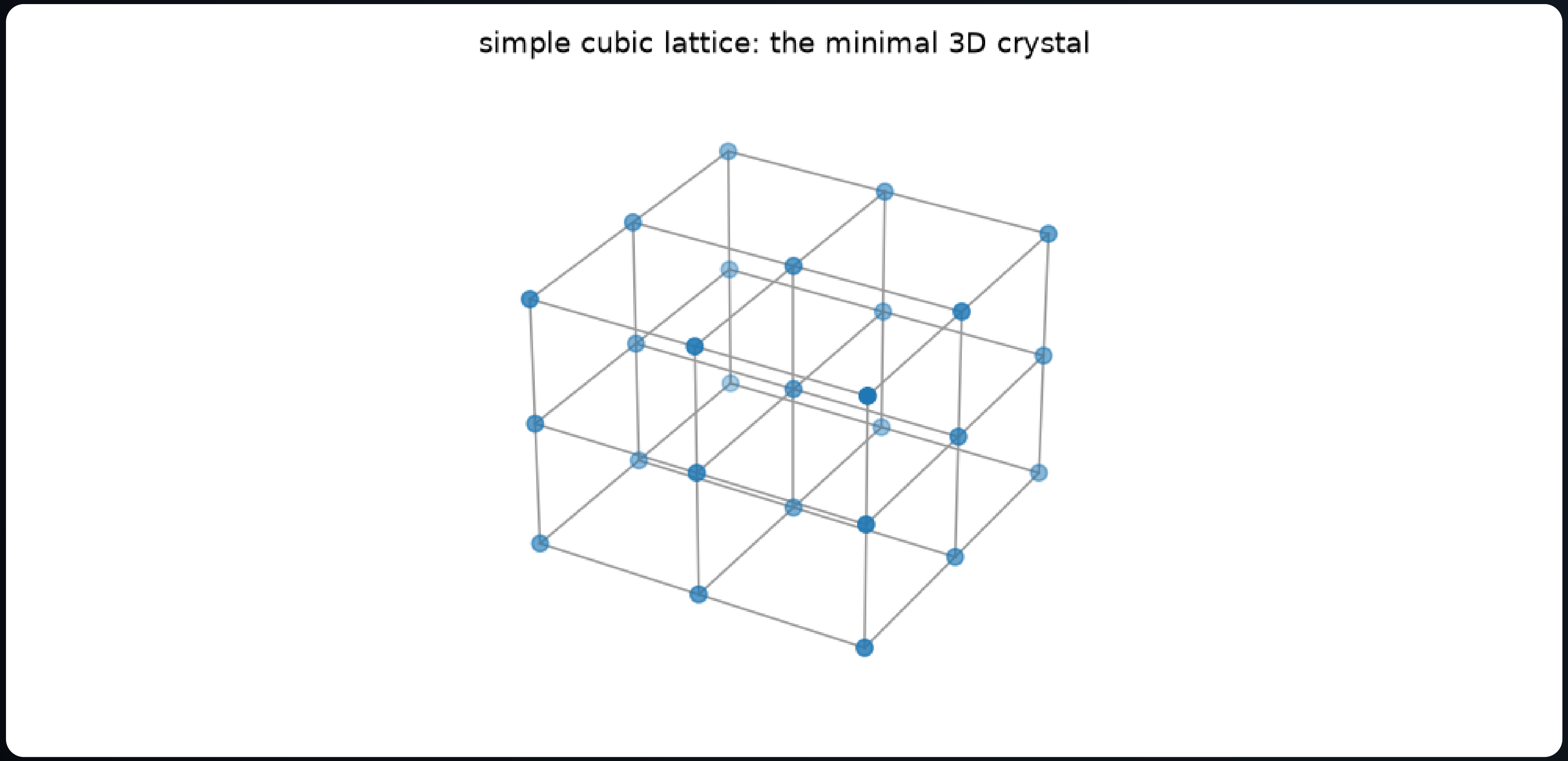


Spin texture of the lower band: ⟨σy⟩ is odd in k (the Rashba field is ∝ sin k · σy) while ⟨σz⟩ is even and peaks at k = 0 where the Zeeman term rules. A right-mover and a left-mover at the same energy inside the helical gap therefore carry nearly opposite spins — backscattering requires a spin flip, which is what protects helical transport.
– Figure 23 · notebook §8 · chapter 2, cell 14

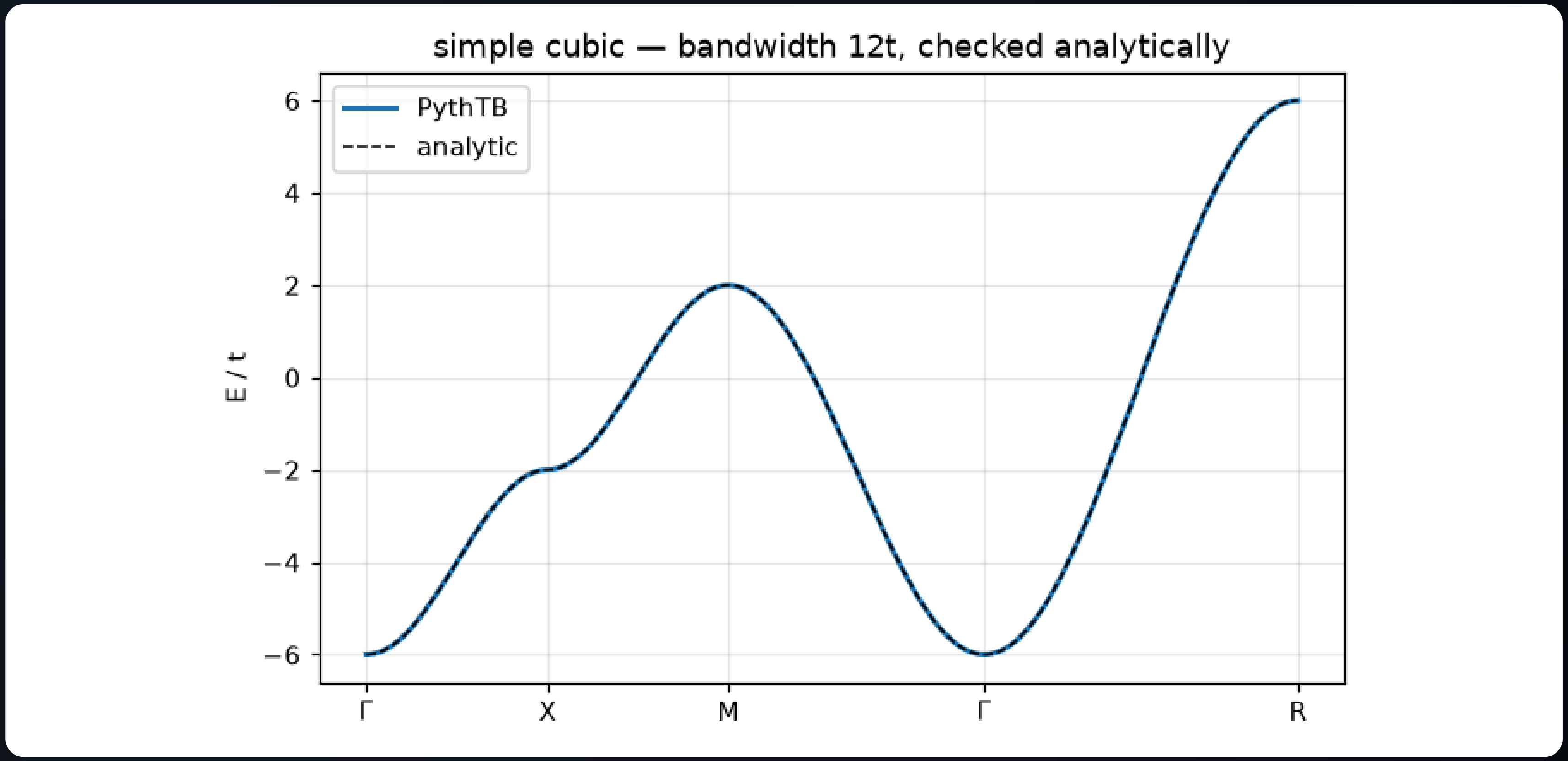
- `plot_bands` can colour by any expectation value: the eigenvectors are one call away.
- $\langle\sigma_y\rangle$ is odd in k (Rashba), $\langle\sigma_z\rangle$ is even and peaks at $k = 0$ (Zeeman).
- Inside the helical gap only one spin direction moves right and the other left: backscattering needs a spin flip.

Three dimensions: the simple cubic crystal

One orbital, six equal nearest-neighbour hops. The band is a sum of three cosines, bandwidth $12|t|$, checked analytically along a standard path.



The simple cubic lattice: one orbital per site, hopping −t along each of the three axes. Its band is the sum of three independent 1D cosine bands — the rare 3D problem with a closed-form answer, which is why it opens the 3D part of this notebook as the analytic benchmark. – Figure 28 · notebook §11 · chapter 3, cell 4



The simple-cubic band along a standard path. Reading it dimension by dimension: bandwidth 12t = 3 × 4t (one 4t per axis), minimum at Γ where all three cosines add, maximum at the zone corner R where all three subtract. Numerics and the analytic sum of cosines are indistinguishable. – Figure 29 · notebook §11 · chapter 3, cell 5

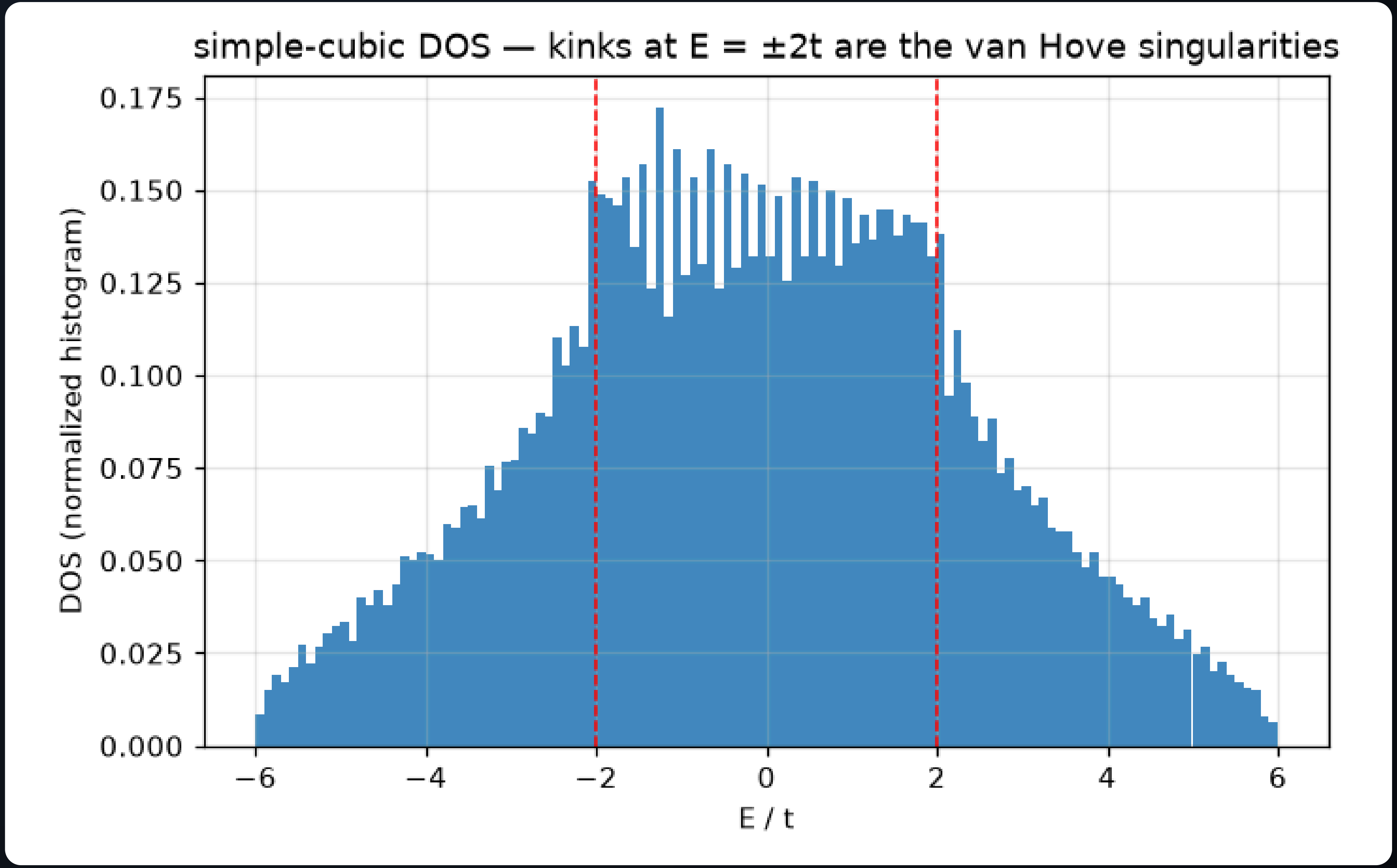
- `dim_k = dim_r = 3`; the k-path visits Γ, X, M, R: each segment turns on one more cosine.
- Nothing new conceptually — but the mesh needed for a 3D integral grows as n^3.
- Reading dimension by dimension: Γ–X is the 1D chain of lecture 1.

CORE

The density of states and its van Hove kinks

Histogram 216 000 eigenvalues from a full 3D mesh: the density of states has kinks at $E = \pm 2t$ where the band has saddle points.

- $g(E)\,dE$ counts states per energy interval; it is what photoemission, heat capacity and tunnelling actually measure.
- In 1D a saddle gives a divergence, in 2D a logarithm, in 3D only a kink: dimensionality is visible in the DOS.
- Square-root edges at $\pm 6t$: the free-electron behaviour returns near the band bottom.



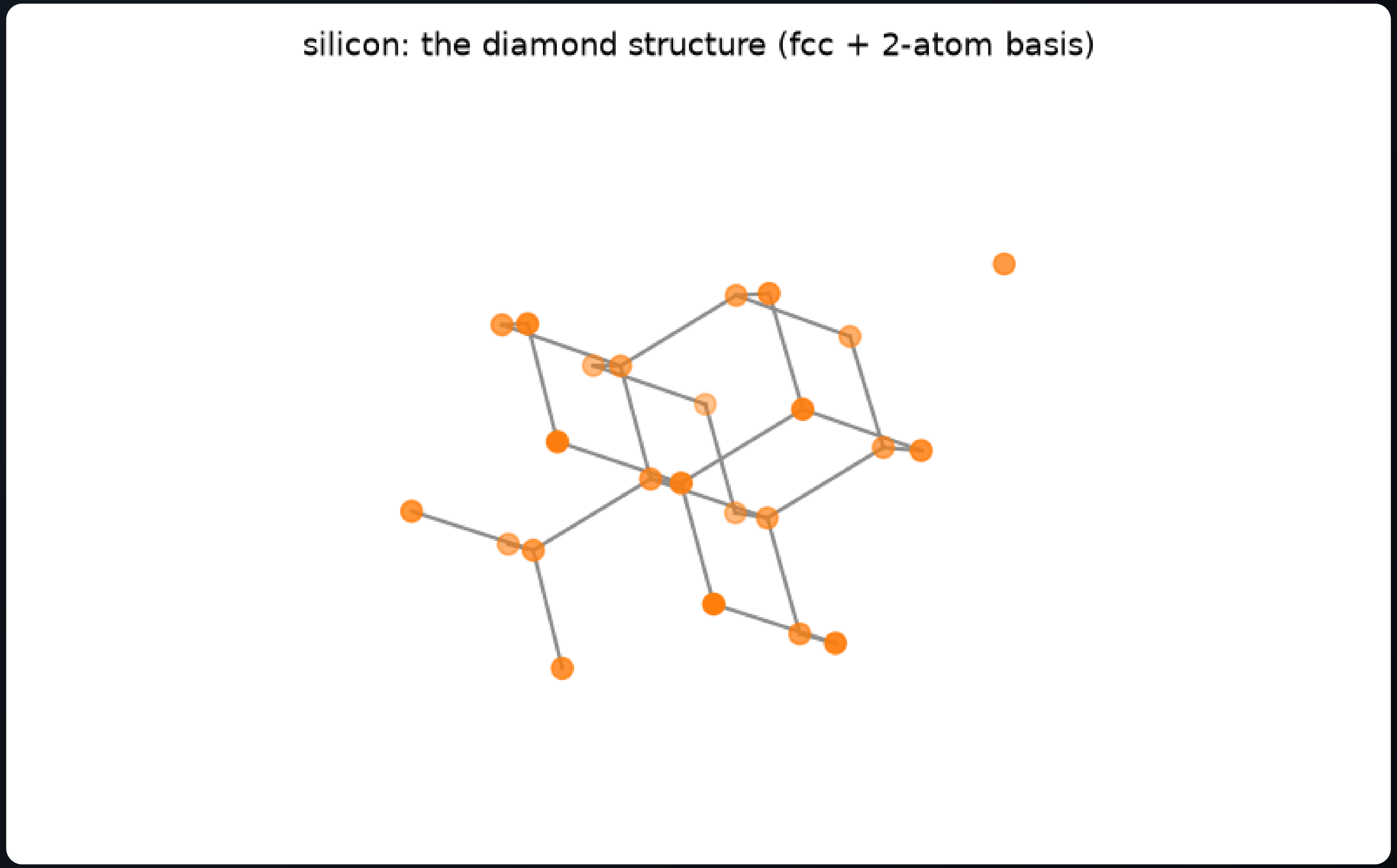
Density of states from a histogram of 216,000 eigenvalues. In 3D the van Hove singularities are kinks (discontinuous slope) rather than the divergences of 2D — the dimensionality of a material is legible in the SHAPE of its DOS anomalies. The band edges at ±6t rise as √|E − E_edge|, the free-electron-like signature of a parabolic band minimum. — `Figure 30 · notebook §11 · chapter 3, cell 6`

CORE

A real material: silicon in the diamond structure

Two interpenetrating fcc lattices, each atom tetrahedrally bonded to four neighbours. Tight-binding parameters for it can be computed, not guessed: Wannier90 turns a first-principles calculation into hoppings.

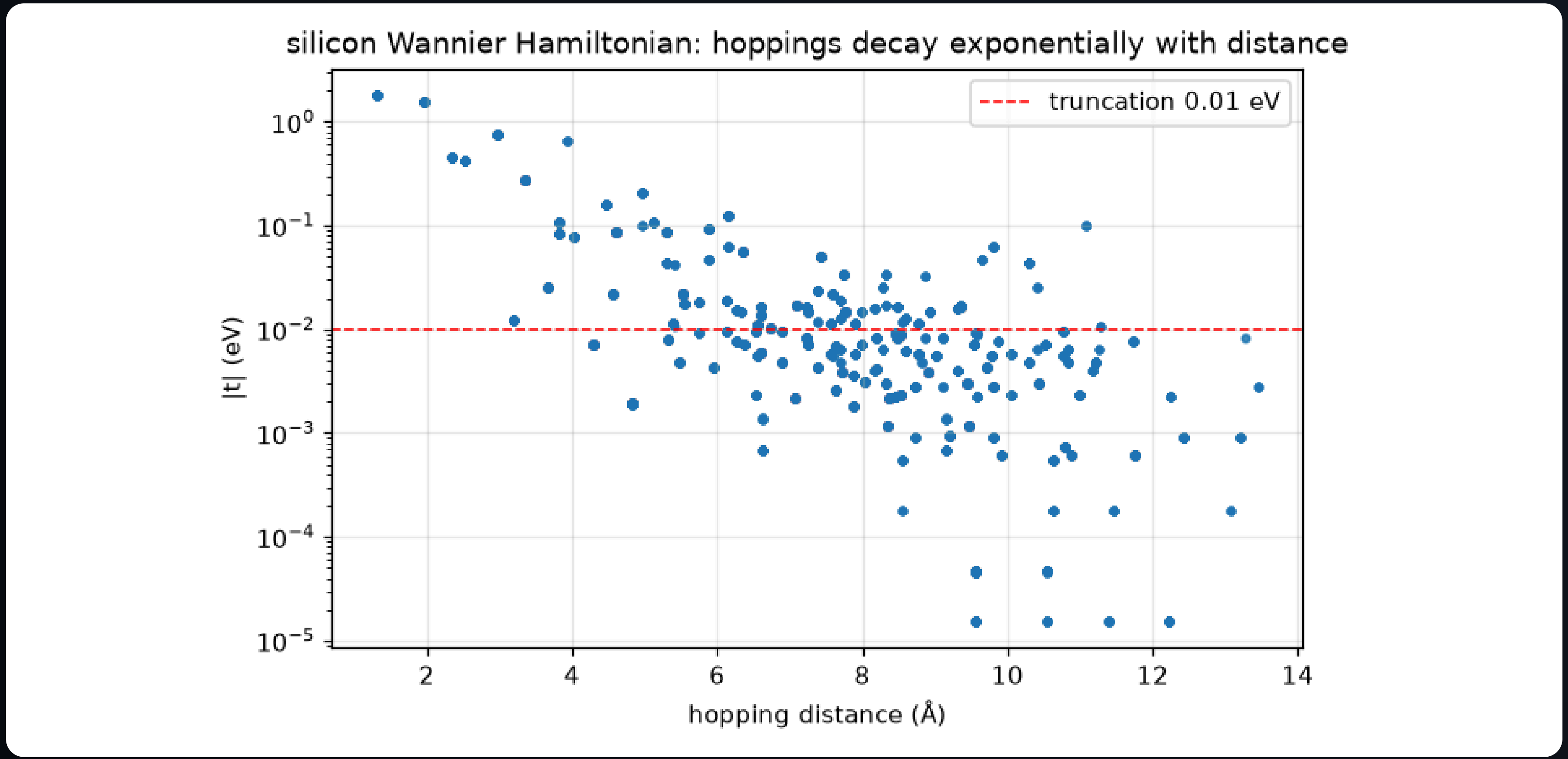
- The Wannier functions are sp^3-like bond orbitals; the hoppings between them are the matrix elements of the DFT Hamiltonian.
- PythTB's `W90` class reads the standard `_hr.dat`, `_centres.xyz` and `.win` files.
- The data ships with PythTB (GPL, attributed); no DFT code is needed to follow along.



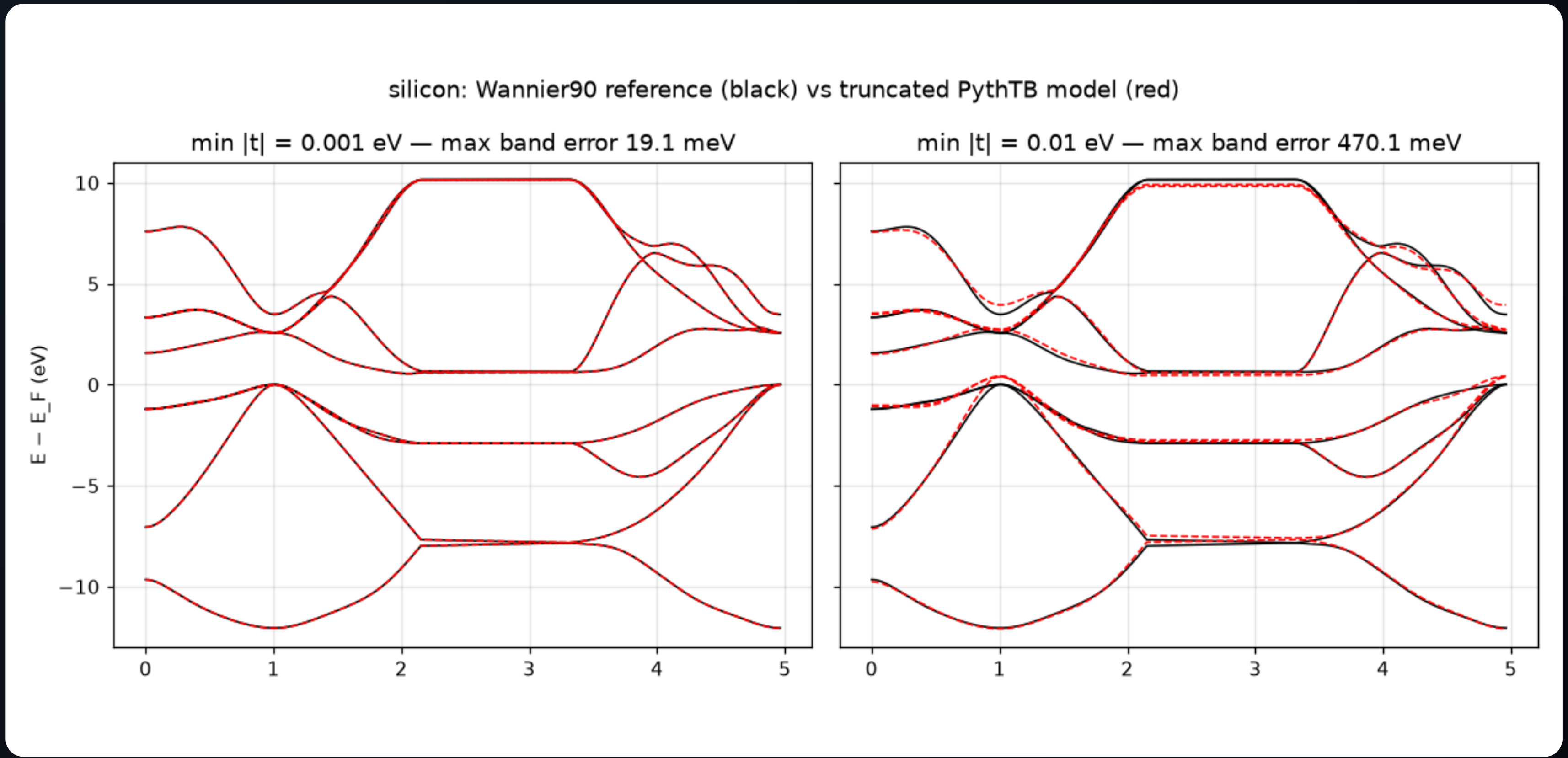
The diamond structure of silicon: two interpenetrating fcc lattices, each atom bonded tetrahedrally to four neighbours. Four sp^3 bonding orbitals per atom pair fill to make the valence bands; the Wannier90 dataset below compresses exactly those bands (plus four antibonding partners) into 8 localized functions whose hoppings PythTB reads back in. – `Figure 31 · notebook §12 · chapter 3, cell 8`

Hoppings decay; truncate and compare

Every hopping matrix element versus distance (left): exponential decay. Keep the hoppings above a threshold and compare with the Wannier90 reference bands (right): the truncated model tracks the first-principles bands.



Every hopping matrix element of the silicon Wannier Hamiltonian versus bond length, on a log scale: five orders of magnitude of clean exponential decay. This is the numerical face of Kohn's 'nearsightedness' — gapped bands have exponentially localized Wannier functions — and it is what makes the truncation studied next a controlled approximation rather than a leap of faith. — Figure 32 · notebook §12 · chapter 3, cell 9



First-principles silicon bands (black) against the truncated tight-binding model (red dashed) at two truncation levels. Keeping hoppings down to 1 meV reproduces DFT to ~20 meV across 8 bands and 17 eV of bandwidth — from a Hamiltonian with a few hundred numbers. The right panel shows the price of an aggressive cut: the conduction bands, whose Wannier functions spread further, degrade first (quantified in exercise I.6). — Figure 33 · notebook §12 · chapter 3, cell 10

- Exponential decay is the definition of a good tight-binding basis — and of a localized Wannier function.
- Truncation is a physical approximation with a controllable error: raise the cutoff, watch the bands converge.
- Below the gap the match is essentially perfect; the conduction bands are more sensitive.

MATH

Spin–orbit on a lattice, and the DOS

RASHBA–ZEEMAN WIRE

$$H(k) = -2t \cos k + 2\alpha \sin k\, \sigma_y + B\, \sigma_z$$

BANDS

$$E_\pm(k) = -2t \cos k \pm \sqrt{(2\alpha \sin k)^2 + B^2}$$

DENSITY OF STATES

$$g(E) = \Sigma_n \int_{BZ} d^3k/(2\pi)^3\, \delta(E - E_n(k))$$

- The lattice Rashba term is $i\alpha\, \sigma_y$ on the bond (Hermitian partner $-i\alpha\, \sigma_y$): a 2×2 hopping matrix in `set_hop`.
- Helical gap $2B$ at $k = 0$; for $B = 0$ the bands cross there — Kramers pairs, no gap allowed.
- Van Hove: $g(E)$ is singular where $\nabla_k E = 0$; the type of singularity depends on the dimension.

Berry phase & pumping

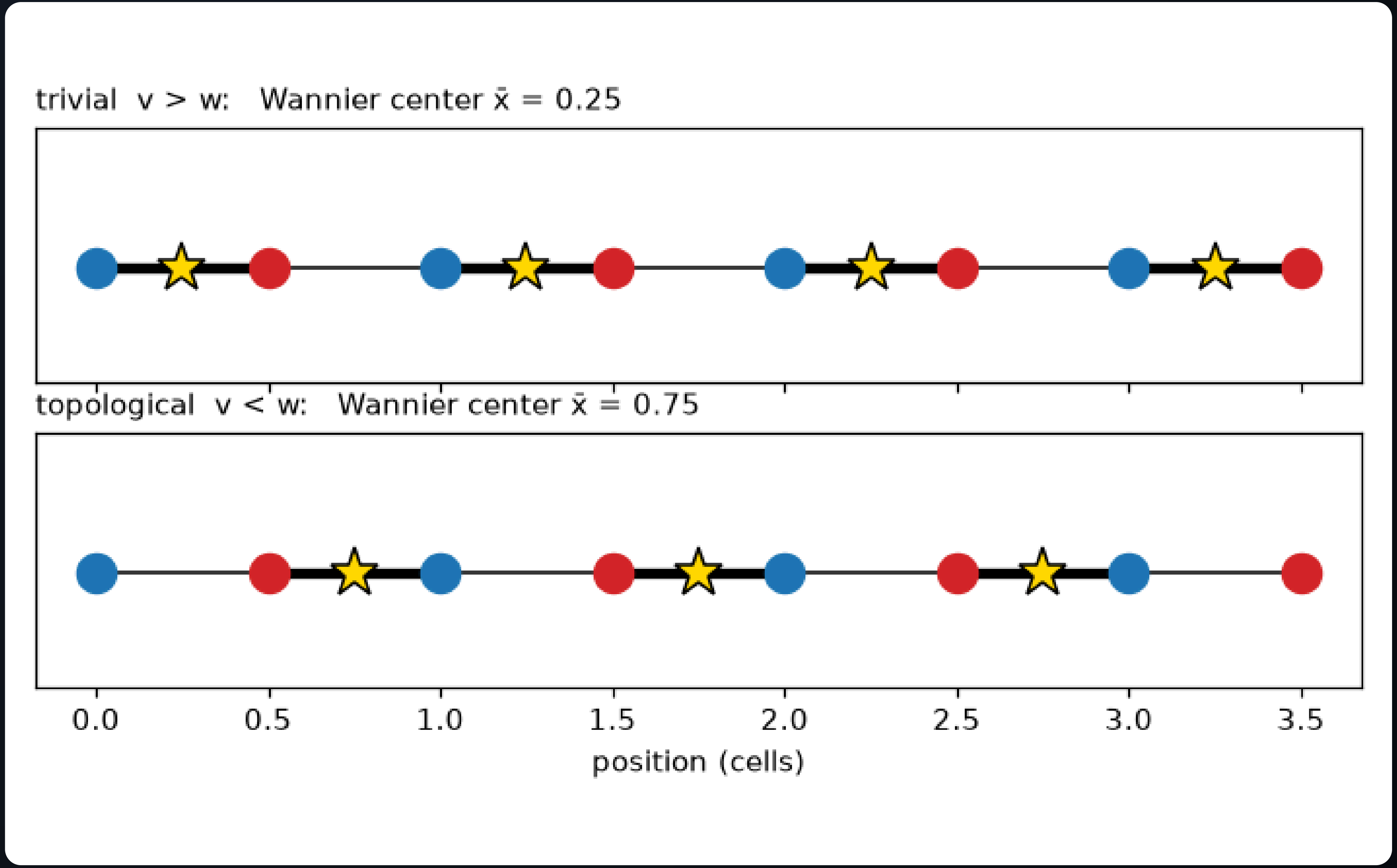
The Berry phase as the position of charge, polarization as a Wannier centre, and the Thouless pump as a quantized flow.

1. Where is the electron's charge?
2. The Rice–Mele cycle
3. The Wannier centre winds once per cycle
4. The pump seen from the ends
5. Berry phase, polarization, pumped charge

Where is the electron's charge?

In an insulator the occupied band is a cloud of charge in each cell. Its centre — the **Wannier centre** — is the electric polarization. Move the atoms, and the centre moves by a definite amount.

- The figure draws the polarization difference between two dimerizations: the charge centre shifts along the bond.
- PythTB's `Mesh` + `WFArray` compute the shift from the Bloch eigenvectors alone.
- Only differences of polarization are physical — the absolute value depends on the choice of unit cell.

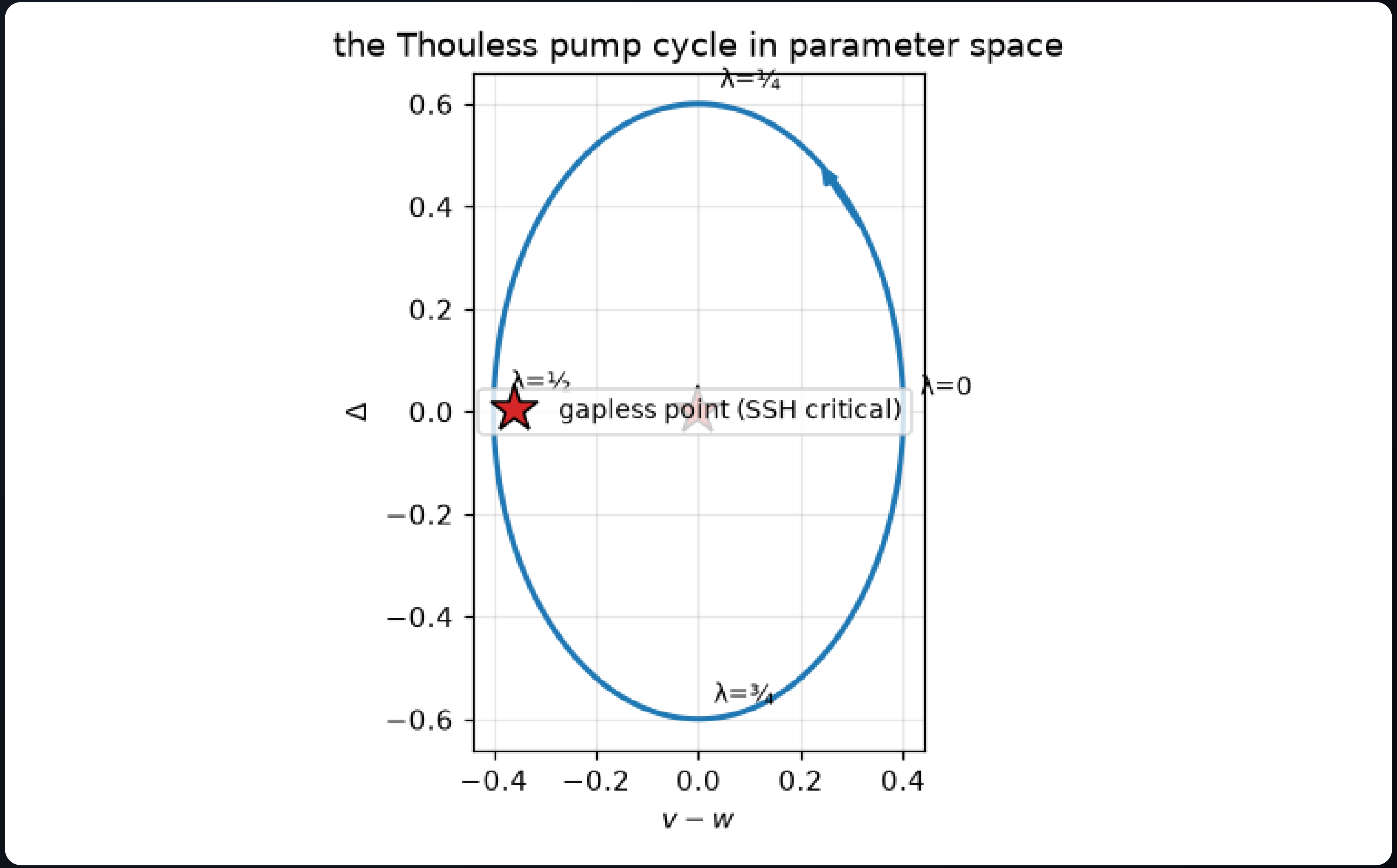


The polarization difference, drawn: the occupied band's Wannier center (star) sits at the middle of whichever bond is STRONG — $\bar{x}$ = ¼ inside the cell in the trivial phase, $\bar{x}$ = ¾ (on the intercell bond) in the topological one. Cutting the chain through a strong bond costs an end state: comparing the two rows shows immediately why the topological chain, terminated at a cell boundary, must leave half an electron's worth of charge behind at each end. — `Figure 24 · notebook §9 · chapter 2, cell 17`

The Rice–Mele cycle

Add a staggered onsite energy $\pm\Delta$ to the SSH chain. In the plane of (dimerization, Δ) the gap closes only at the origin. A cycle that **encircles** it is a Thouless pump.

- Along the loop the chain is always gapped: an adiabatic process.
- Yet after one full cycle charge has moved by exactly one electron per cell.
- The loop is a path in parameter space, and the pumped charge is a winding number around the gapless point.



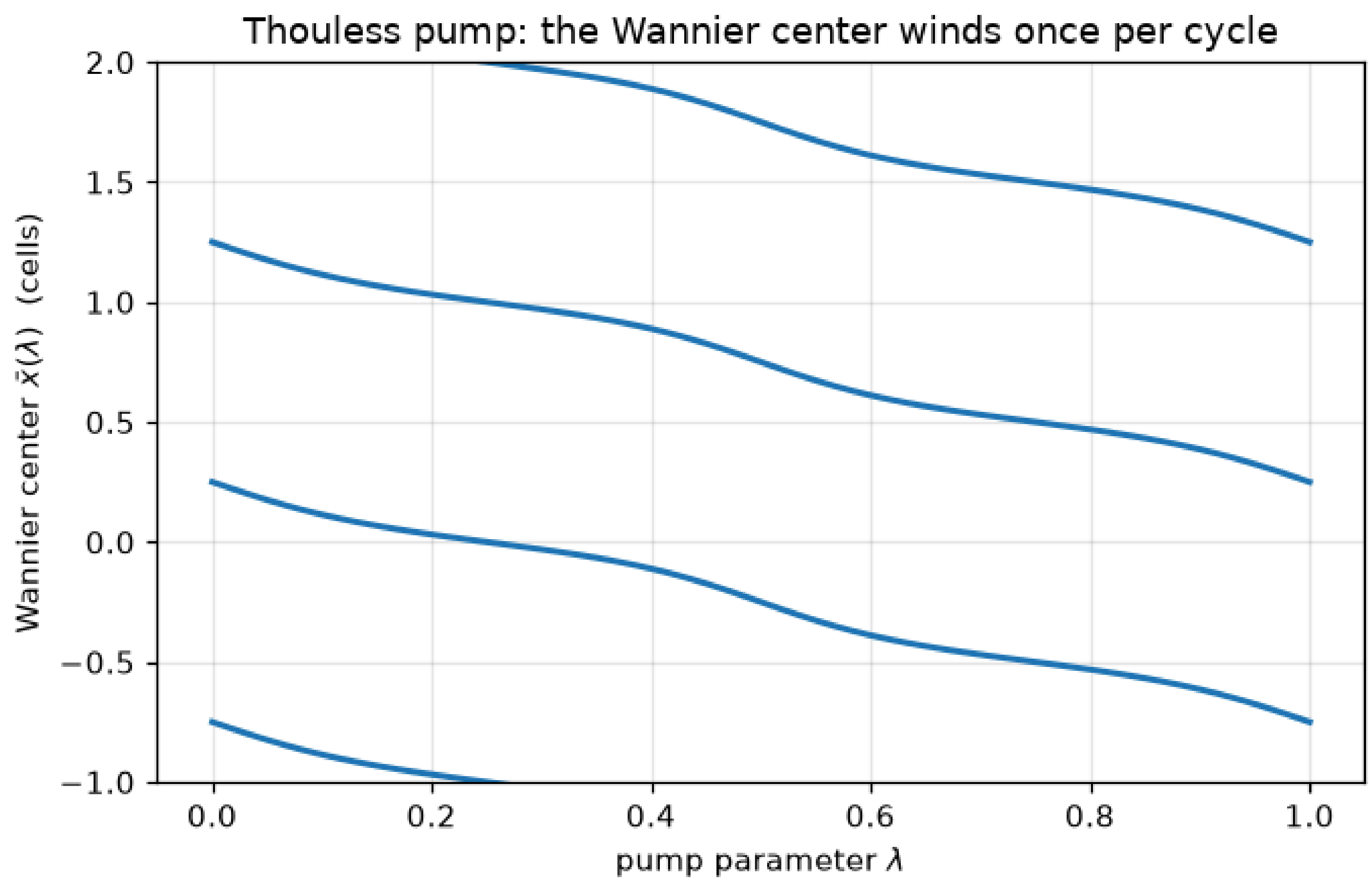
The pump protocol: over one period the dimerization (v − w) and the staggered onsite Δ trace a loop that encircles the SSH critical point (star), where the bulk gap would close. The pumped charge equals the winding number of this loop around the star — which is why the off-center cycle of exercise I.5 pumps nothing. Enclosing a gapless point is to a pump what enclosing a solenoid is to an Aharonov–Bohm phase.

— Figure 25 · notebook §10 · chapter 2, cell 20

CORE

The Wannier centre winds once per cycle

Track the Wannier centre as the pump parameter λ advances: it flows continuously across one full unit cell and returns to its starting value mod a. That integer is the pumped charge.



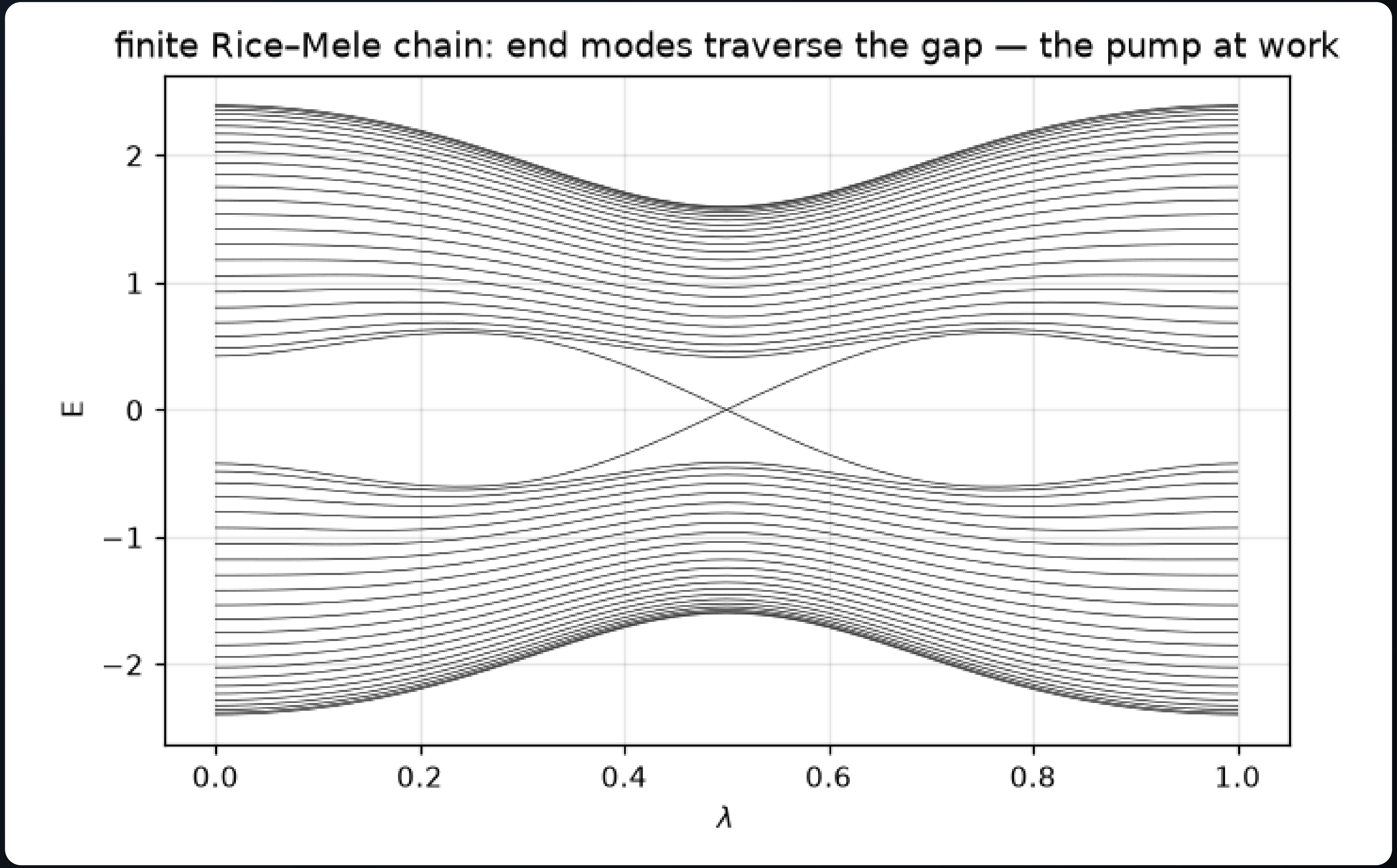
Wannier-center flow over one pump cycle (several unit cells shown). The charge center advances smoothly and ends one full lattice constant from where it started: exactly one electron per cell has been transported, with no bias voltage — quantized by the Chern number of the (k, λ) torus, not by any fine-tuning of the drive. — Figure 26 · notebook §10 · chapter 2, cell 21

CORE

The pump seen from the ends

In a finite chain the pumped charge has to go somewhere: as λ advances, a bound state peels off the valence band at one end, crosses the gap and joins the conduction band — one state per cycle.

- The spectral flow of the end states is the bulk Chern number made visible in a finite system.
- Bulk–boundary correspondence again: an integer computed from the bulk eigenvectors counts a boundary process.
- The same picture, with λ replaced by a transverse momentum, is the chiral edge state of lecture 6.



The pump seen from a finite chain: as λ advances, bound states peel off one band edge, cross the gap on the chain's ends, and merge into the other band. That gap traversal is how the transported charge physically enters and leaves the sample boundary — the spectral flow that MUST accompany a nonzero Chern number, here resolved state by state. — Figure 27 · notebook §10 · chapter 2, cell 22

MATH

Berry phase, polarization, pumped charge

BERRY PHASE OF A BAND

$$\varphi = i \oint dk\, \langle u_k |\, \partial_k\, u_k \rangle = -\mathrm{Im}\,\ln \Pi_j\, \langle u_{k_j} |\, u_{k_{j+1}} \rangle$$

POLARIZATION

$$P = e\, a\, \varphi \,/\, 2\pi\ (\mathrm{mod}\ e\ a),\ \bar{x} = a\, \varphi \,/\, 2\pi$$

THOULESS PUMP

$$\Delta Q = e \oint d\lambda\, \partial_\lambda(\varphi/2\pi) = e \cdot C,\ C = (1/2\pi) \iint dk\, d\lambda\, \Omega_{k\lambda} \in \mathbb{Z}$$

- The discrete formula (King-Smith–Vanderbilt) is gauge invariant: an arbitrary phase on each u_{k_j} cancels in the product of overlaps.
- `WFArray.berry_phase` implements exactly that product; `berry_flux` the plaquette version of Ω.
- The pump integer is the Chern number of the occupied band on the (k, λ) torus — the same object as in lecture 6 with $\lambda \to k_y$.

Chern insulators

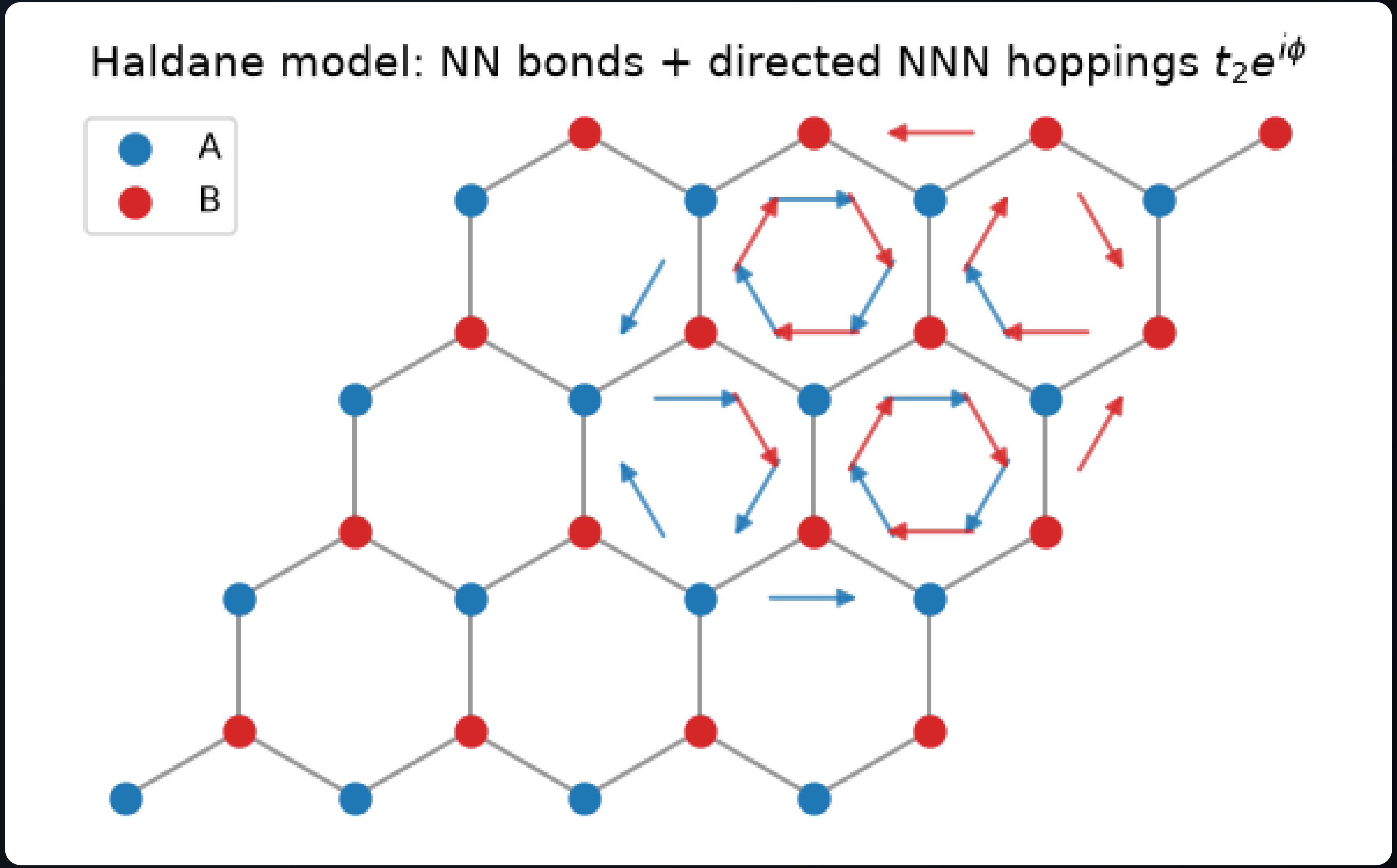
The Haldane model: Berry curvature, the Chern number, the phase diagram, chiral edge states and a real-space marker.

INTRO

Haldane's trick: complex hoppings, no magnetic field

Graphene plus directed second-neighbour hoppings $t_2 e^{i\varphi}$, arrows clockwise around each hexagon. The net flux through the cell is zero — yet time reversal is broken.

- A complex hopping is a phase an electron picks up on a bond; going round a triangle, it accumulates 3φ — a local flux.
- Opposite triangles carry opposite flux: no net field, no Landau levels, still a quantum Hall effect.
- Plus a sublattice potential M to compete with it: two masses, one phase diagram.

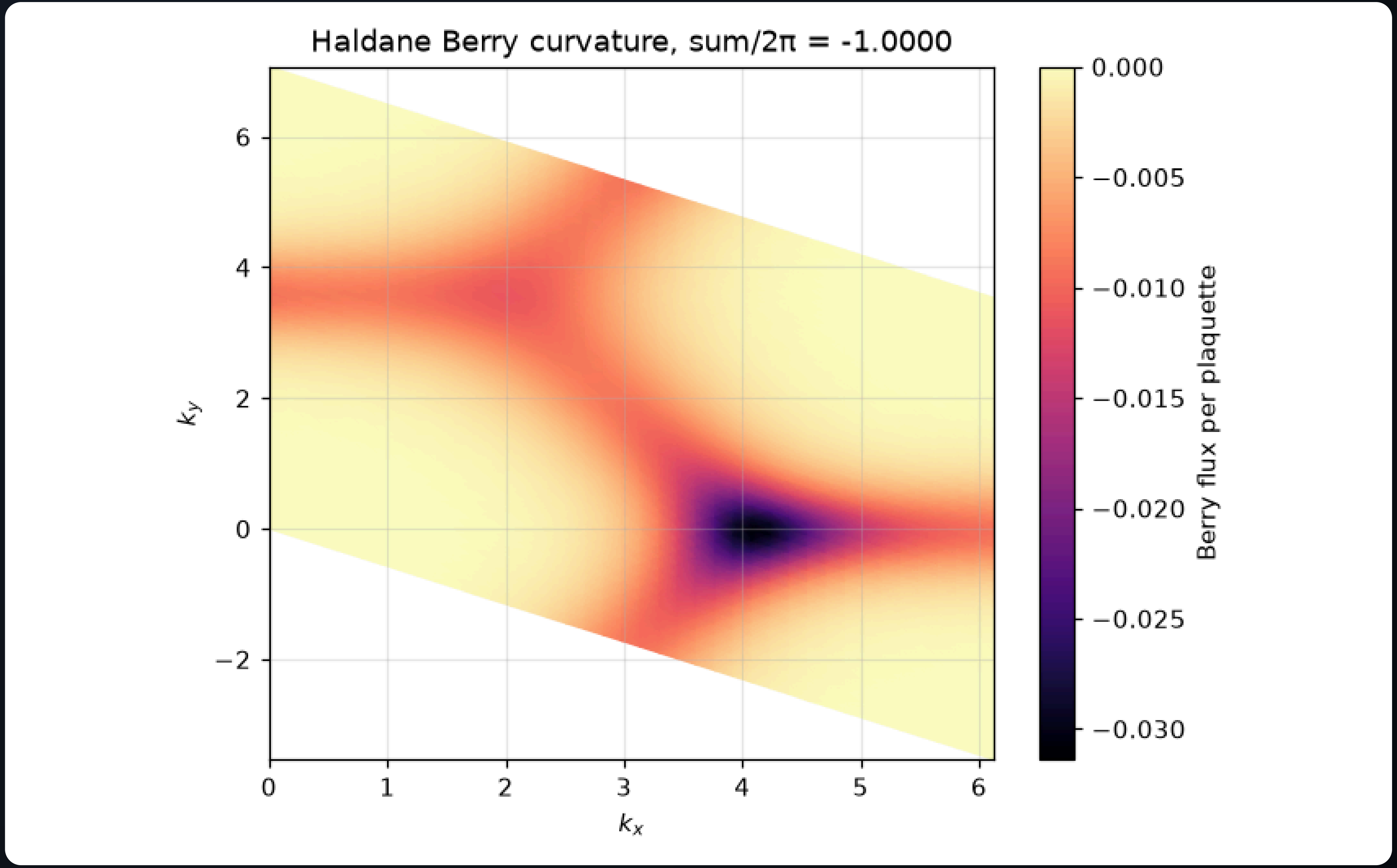


The Haldane model's key ingredient: complex second-neighbour hoppings whose positive-phase direction circulates counter-clockwise around A sublattice triangles (blue arrows) and clockwise around B (red). An electron hopping around one small triangle picks up phase φ, around a full hexagon zero — a staggered magnetic flux pattern that breaks time-reversal with NO net field, which is why this is a 'quantum Hall effect without Landau levels'. — Figure 35 · notebook §14 · chapter 4, cell 3

Berry curvature: where the geometry lives

The Berry curvature $\Omega(k)$ of the occupied band over the Brillouin zone. It is concentrated near K and K', where the gap is smallest — and it integrates to 2π times an integer.

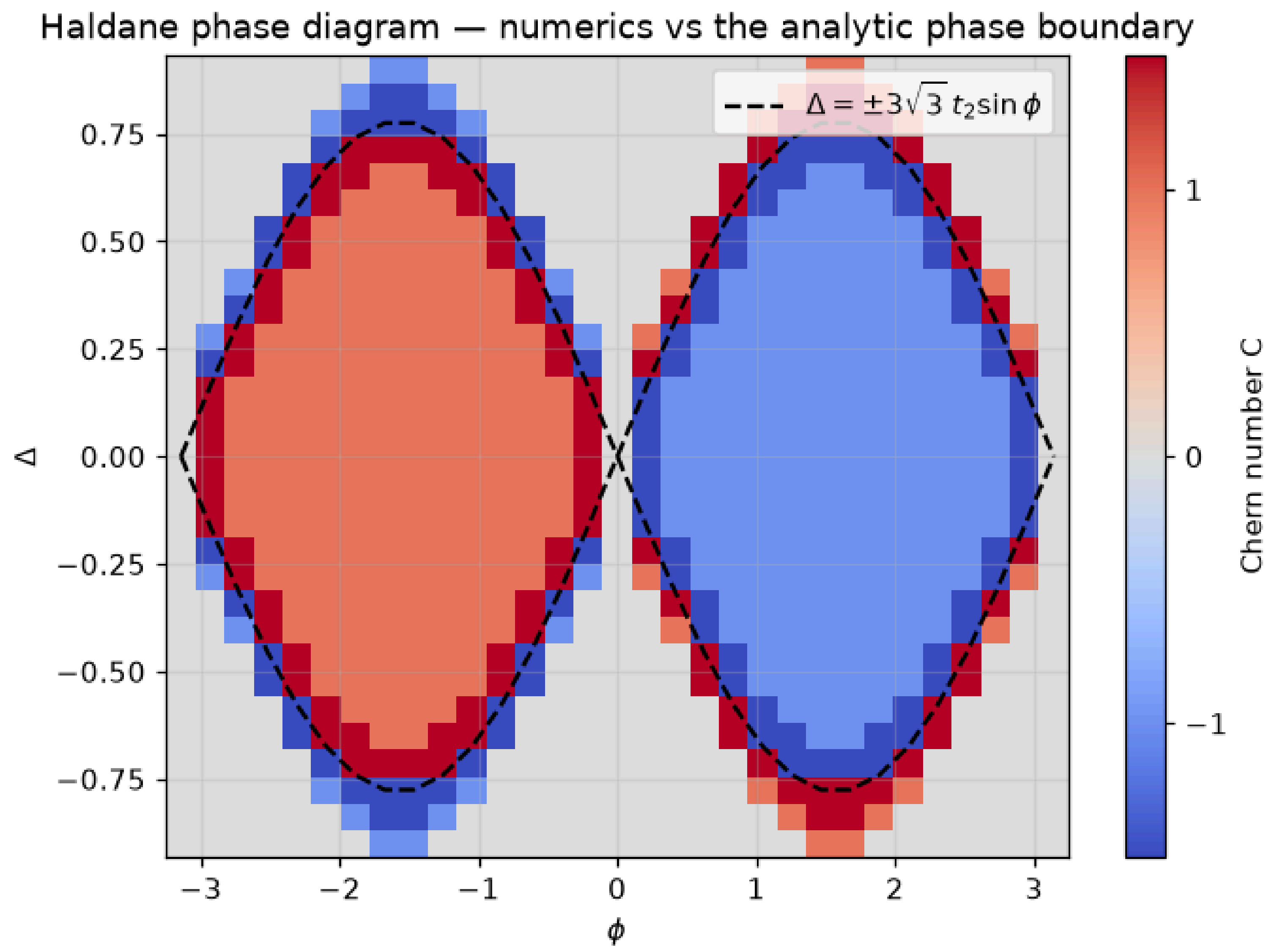
- Ω is the 2D analogue of the Berry phase per unit area; PythTB's `berry_flux` computes it plaquette by plaquette.
- Both valleys contribute with the same sign in the topological phase, opposite signs in the trivial one.
- The integral is the **Chern number** C; the Hall conductance is $C\,e^2/h$.



Berry curvature of the occupied band over the Brillouin zone. It is not spread uniformly: it concentrates at the (gapped) Dirac points, where the band wavefunction twists fastest — each former Dirac cone contributes close to half the total. The integral over the BZ is quantized to -2π regardless of how the curvature is distributed: that rigidity IS the Chern number. — `Figure 36 · notebook §14 · chapter 4, cell 5`

The Haldane phase diagram

Chern number over the $(\varphi, M/t_2)$ plane, computed point by point, against the analytic boundary $M = \pm 3\sqrt{3}\, t_2 \sin \varphi$. Inside the lobes $C = \pm 1$; outside, $C = 0$.



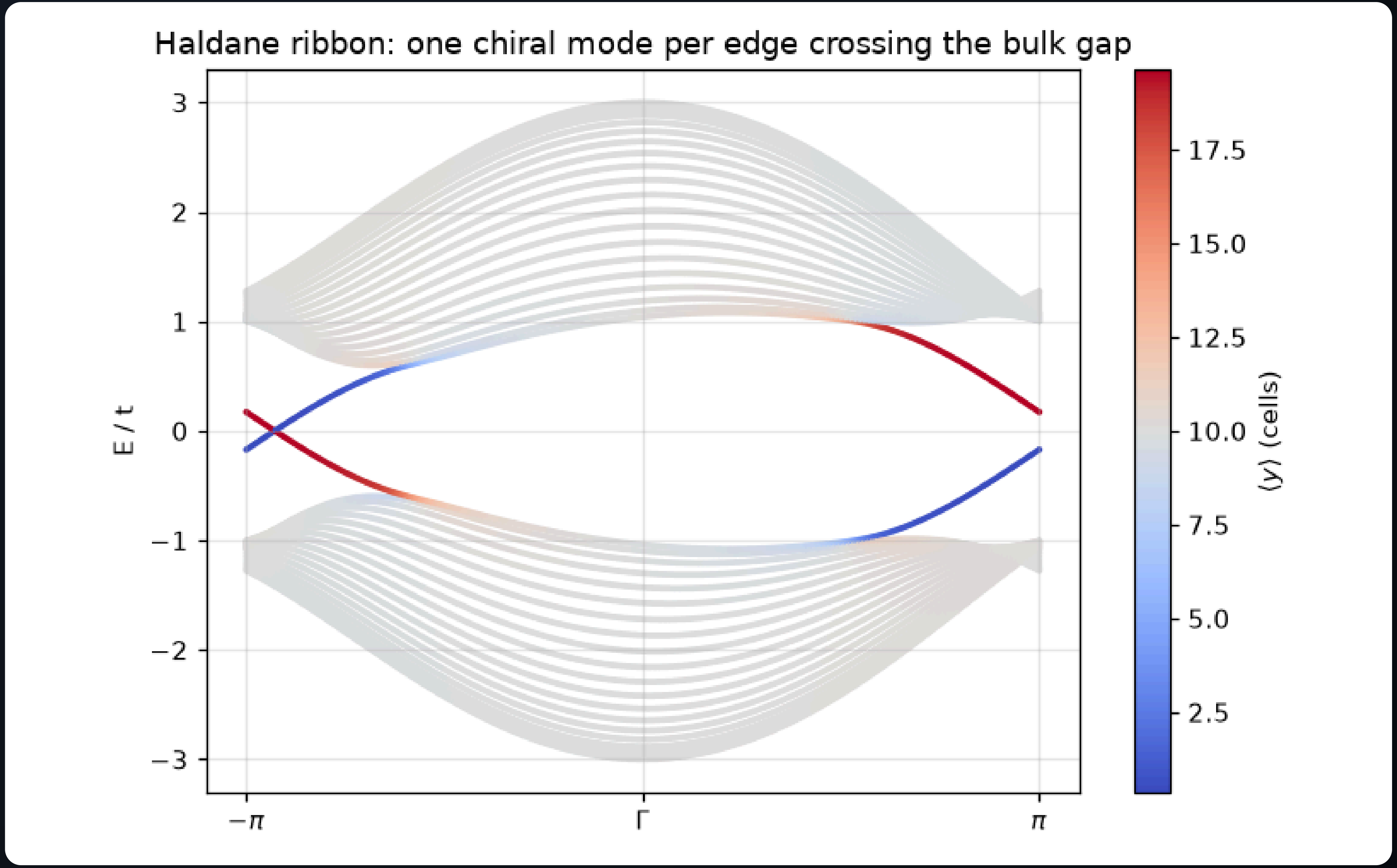
The Haldane phase diagram computed point by point: Chern number as a function of the TRS-breaking phase φ and the inversion-breaking mass Δ. The two competing masses fight at the two valleys independently, giving the celebrated lobes; the numerical boundary lands exactly on the analytic curve $\Delta = \pm 3\sqrt{3}\, t_2 \sin \varphi$ (dashed). Points ON the boundary have no Chern number at all — the gap is closed there, and PythTB rightly refuses. — `Figure 37 · notebook §14 · chapter 4, cell 6`

CORE

One chiral edge mode per edge

Cut a ribbon in the $C = 1$ phase and colour the bands by edge weight: a single state crosses the gap on each edge, moving in opposite directions on opposite edges.

- Chiral: one direction only. There is no state to backscatter into — the edge conducts without dissipation.
- The number of crossings (net, with sign) equals C: bulk–boundary correspondence in 2D.
- This is the quantum anomalous Hall effect: quantized σ_{xy} with no external field.

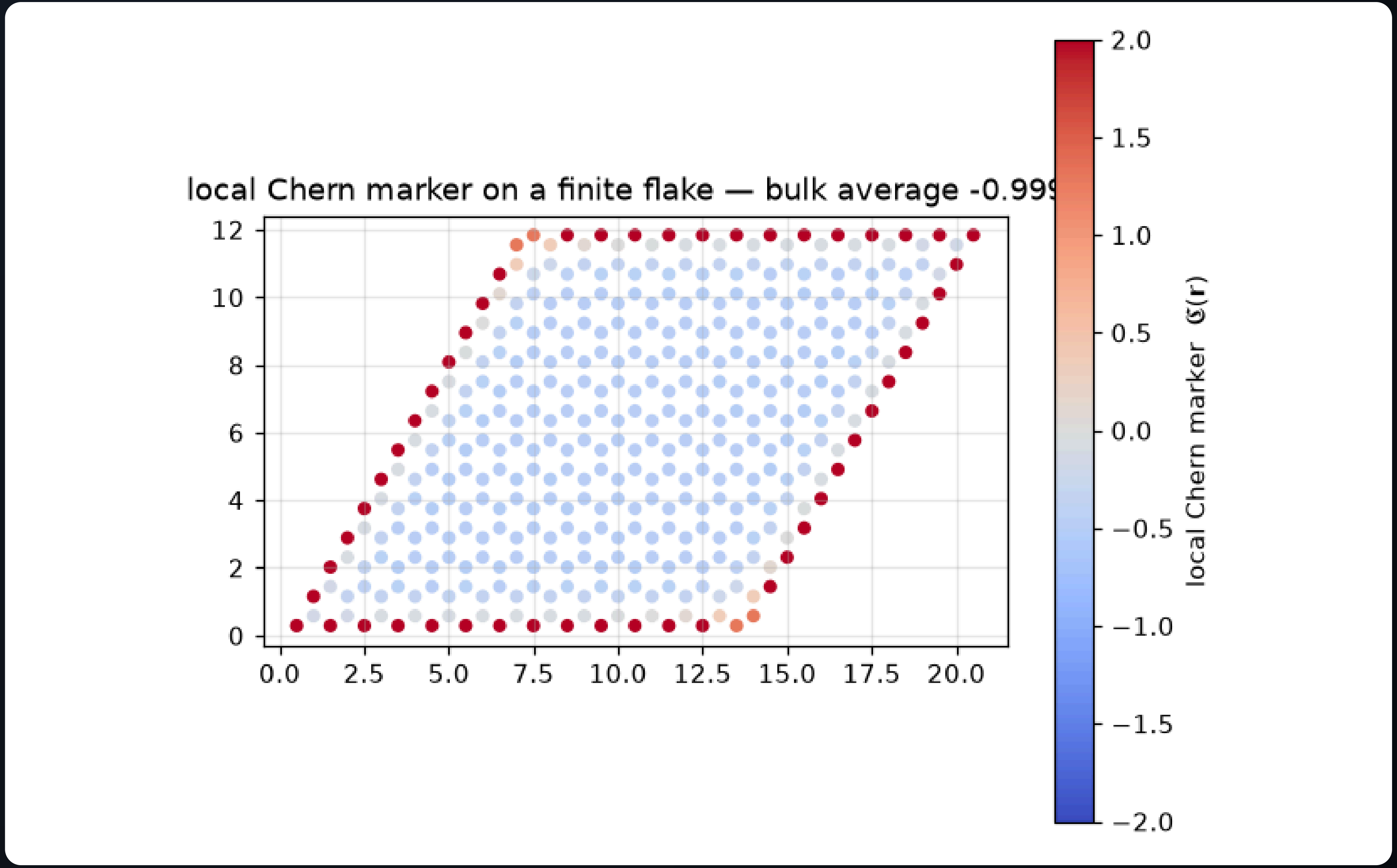


Bulk-boundary correspondence, resolved in space: ribbon bands colored by the transverse position ⟨y⟩ of each state. The bulk bands (mixed colors) are gapped; two branches cross the gap, one confined to each edge (deep red / deep blue), with opposite velocities — chiral edge channels. Their number equals |C|, and no edge disorder can backscatter them: the counter-propagating partner lives on the other side of the sample. — Figure 38 · notebook §14 · chapter 4, cell 7

Topology without k-space: the local Chern marker

The Bianco–Resta marker evaluates C at each site of a finite flake from the projector onto occupied states and the position operators. In the bulk it reads the Chern number; at the edges it reverses sign so the total is zero.

- Needs no periodicity: works for disordered, amorphous or finite samples.
- Costs a full diagonalization of the flake — dense, $O(N^3)$.
- PythTB gives the eigenvectors; the marker is twenty lines of numpy on top (in the notebook).



The Chern number without k-space: the Bianco–Resta local marker evaluated on every site of a finite flake. Deep in the interior it locks to C = −1; at the boundary it swings hard positive so that the sample-wide average is exactly zero (the marker is a trace of a commutator). The edge is not noise — it is the same spectral flow that carries the chiral edge states. — Figure 39 · notebook §14 · chapter 4, cell 8

MATH

Chern number and the Haldane masses

CURVATURE AND CHERN NUMBER

$$\Omega = \partial_{k_x} A_y - \partial_{k_y} A_x,\ A = i\langle u|\nabla_k u\rangle,\ C = (1/2\pi)\int_{BZ} \Omega\, d^2k \in \mathbb{Z}$$

LATTICE (FUKUI–HATSUGAI–SUZUKI)

$$C = (1/2\pi)\, \Sigma_{plaquettes}\, \mathrm{Im}\ln\,(\,U_{12}\, U_{23}\, U_{34}\, U_{41}\,),\ U_{ij} = \langle u_i|u_j\rangle/|\langle u_i|u_j\rangle|$$

HALDANE MASSES

$$m_K = M - 3\sqrt{3}\, t_2 \sin\varphi,\ m_{K'} = M + 3\sqrt{3}\, t_2 \sin\varphi,\ C = ½[\mathrm{sgn}(m_{K'}) - \mathrm{sgn}(m_K)]$$

- Each gapped Dirac cone contributes $\pm½$ to C, the sign being that of its mass; the total is an integer because the cones come in pairs.
- The plaquette formula is gauge invariant and exactly integer on any mesh — PythTB's `berry_flux` is this product.
- $\sigma_{xy} = C\, e^2/h$ (TKNN): a transport coefficient equal to a topological integer.

Z_2 & spin Hall

Time reversal changes the rules: Kramers pairs, band inversion, the Z_2 invariant from Wannier flow, helical edges.

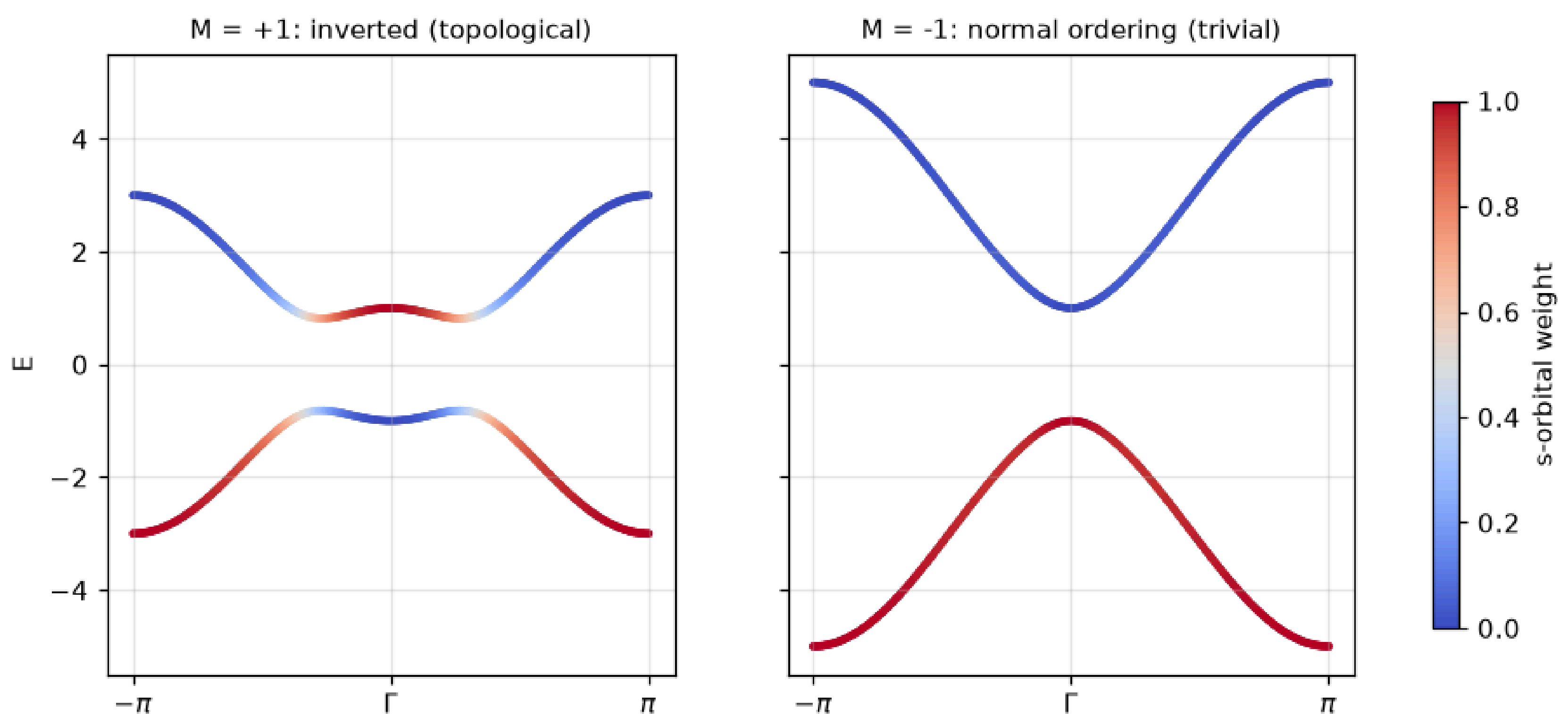
1. Band inversion: the switch
2. The Z_2 invariant from Wannier flow
3. Helical edge states: a Kramers pair per edge
4. Kane–Mele, Kramers, and the two ways to Z_2

INTRO

Band inversion: the switch

In a HgTe quantum well the ordering of the s-like and p-like bands at Γ flips with the well thickness. Colour the BHZ bands by orbital character: for $M > 0$ the characters are swapped near Γ — the **inverted** regime.

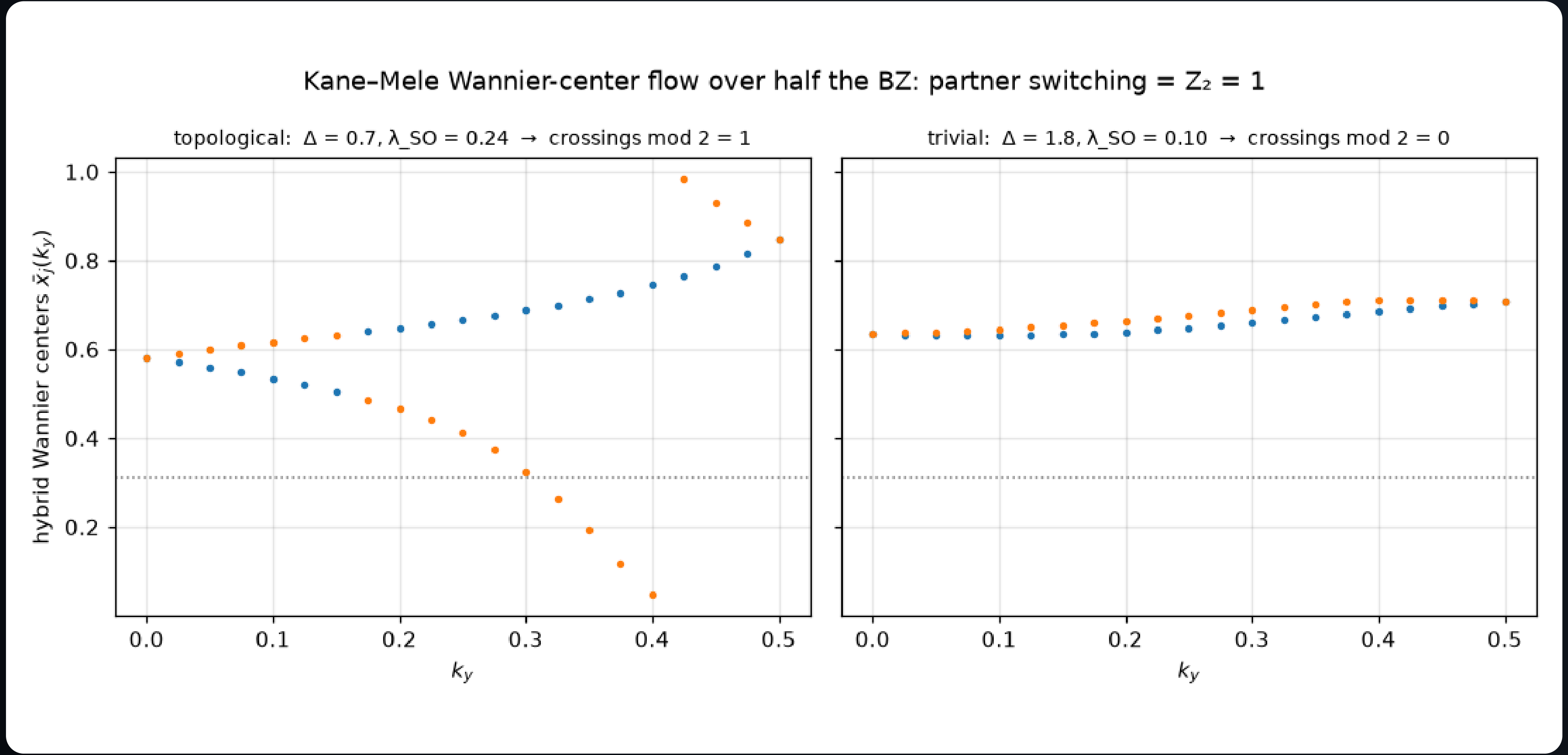
- Inversion is a property of eigenvectors, invisible in the energies alone.
- It is the two-dimensional, time-reversal-symmetric cousin of the Haldane mass changing sign.
- Predicted 2006 (Bernevig, Hughes, Zhang), measured 2007 (König et al.): the quantum spin Hall effect.



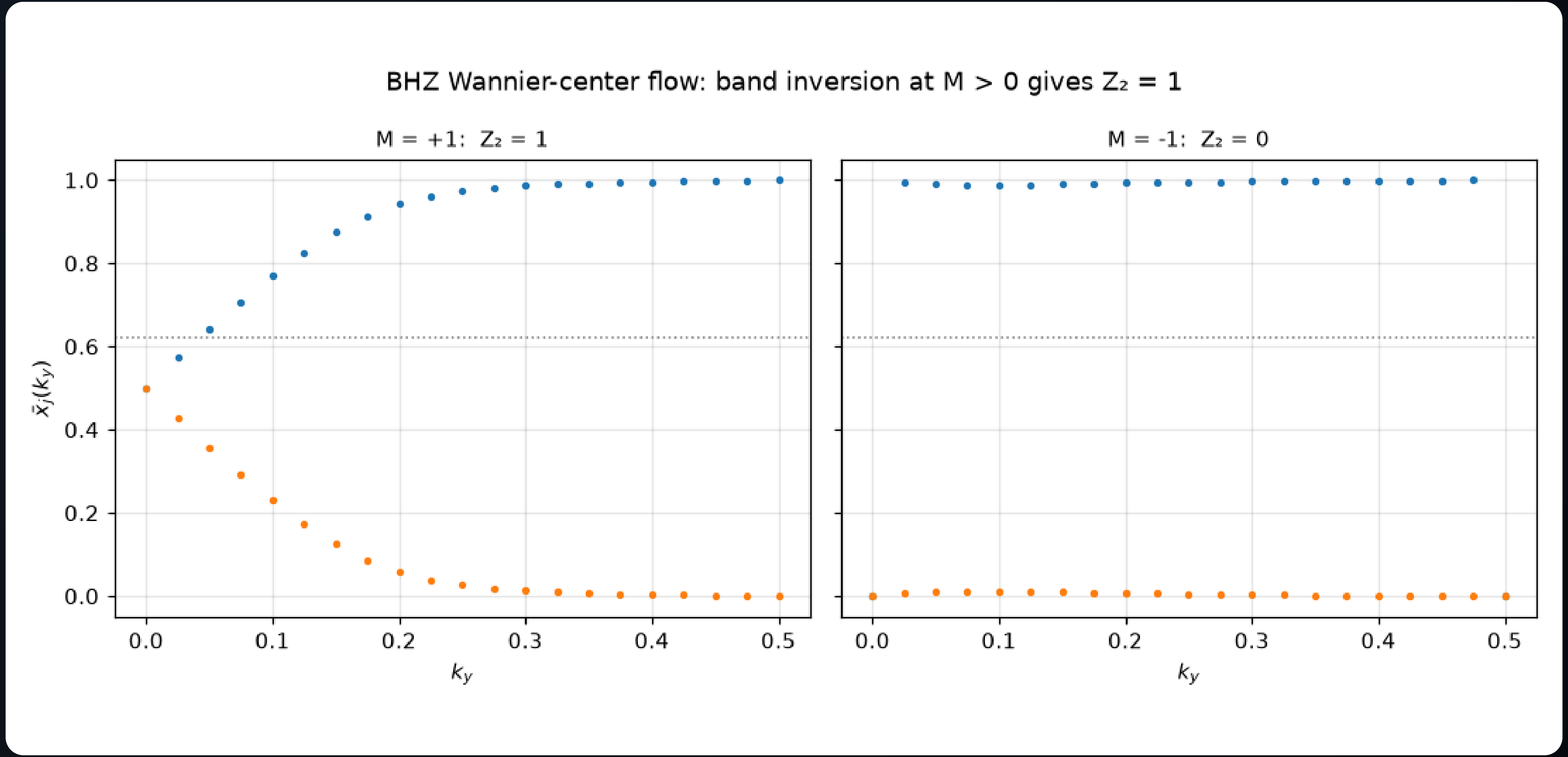
Band inversion, the mechanism behind the BHZ prediction: bands along kx colored by s-orbital character. For M 0 (left) the characters are exchanged in a neighbourhood of Γ: the bands have inverted, and no local perturbation can undo the exchange without closing the gap. In HgTe quantum wells the knob M is the well thickness, and the inversion happens at d ≈ 6.3 nm. — Figure 42 · notebook §16 · chapter 4, cell 14

The Z_2 invariant from Wannier flow

Hybrid Wannier centres $\bar{x}(k_y)$ over half the zone, for Kane–Mele (left) and BHZ (right). In the topological phase the two centres **switch partners**: any horizontal line crosses them an odd number of times.



The Z_2 diagnostic: hybrid Wannier centers x̄(ky) of the two occupied bands, followed over half the Brillouin zone. At the time-reversal-invariant momenta ky = 0 and ½ the centers come in Kramers-degenerate pairs; in between they may exchange partners. Left: the pair switches — any reference line (dotted) is crossed an odd number of times, Z_2 = 1. Right: the centers return to their partners, even crossings, trivial. This counting is gauge-invariant and needs no symmetry beyond time reversal. — Figure 40 · notebook §15 · chapter 4, cell 10

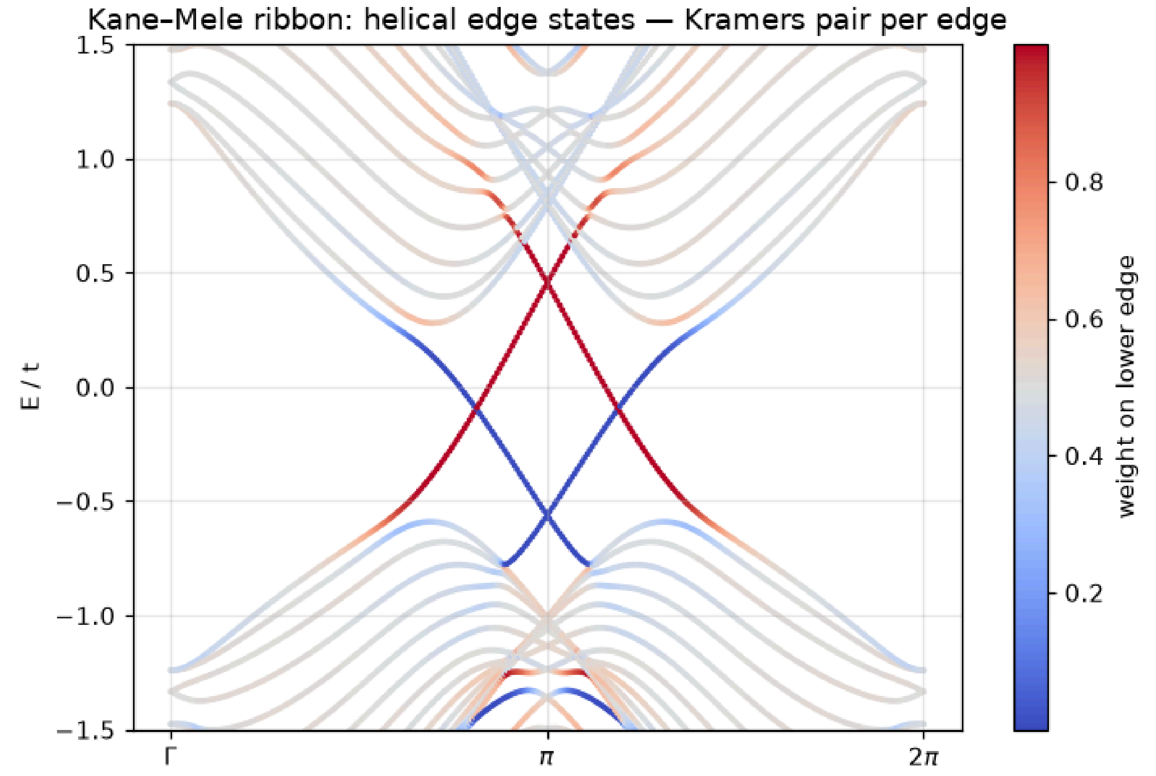


The same Wannier-flow diagnostic applied to the BHZ model: partner switching (left, odd crossings of the reference line) for the inverted regime, none for the normal one. The Z_2 index sees the band inversion of the previous figure without ever looking at orbital character — invariants do not care how the topology was engineered. — Figure 43 · notebook §16 · chapter 4, cell 15

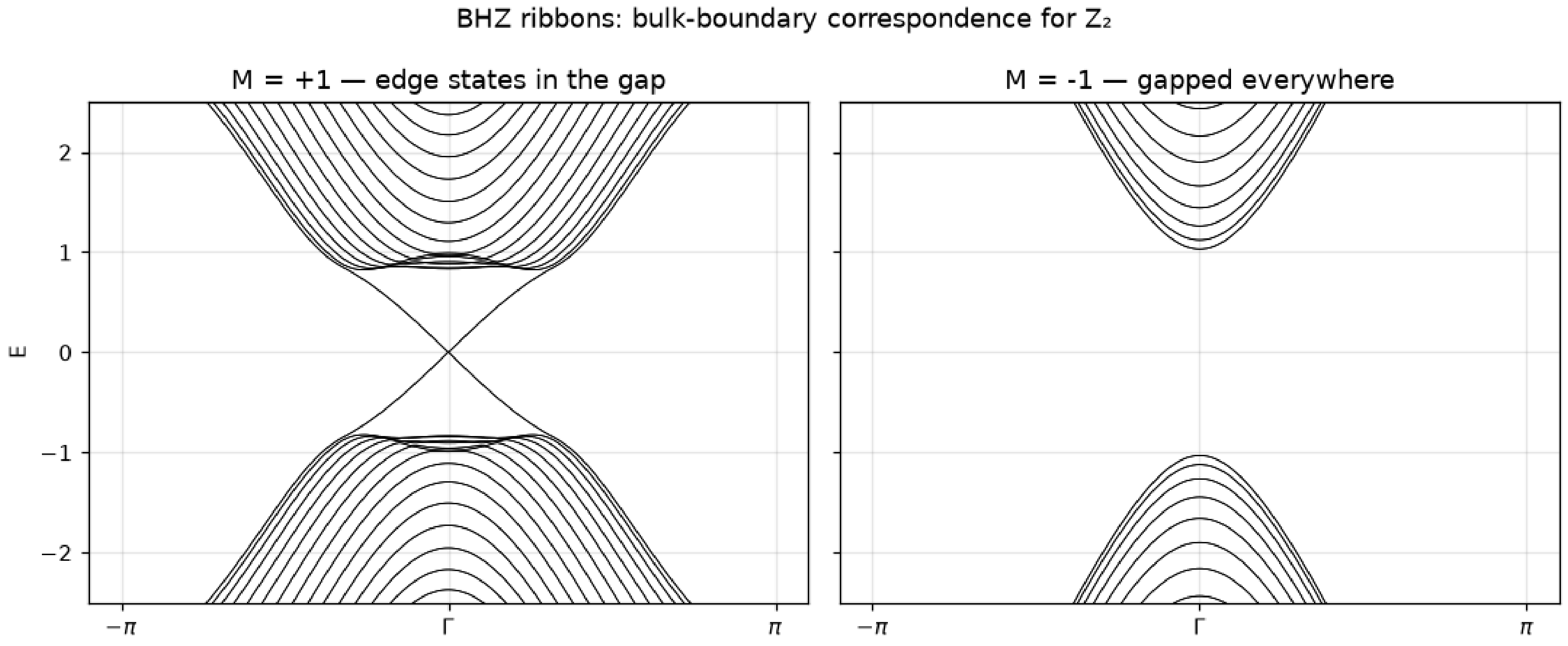
- Time reversal pairs the centres at $k_y = 0$ and π; in between they may swap. Z_2 counts whether they do (mod 2).
- PythTB: `WFArray.berry_phase(..., berry_evals=True)` along k_x for each k_y.
- The Wannier flow is the pump of lecture 5 with k_y as the pump parameter, restricted by time reversal to half a period.

Helical edge states: a Kramers pair per edge

Ribbons in the topological phase carry, on each edge, two counter-propagating states with opposite spin (left, Kane–Mele). BHZ ribbons (right): edge states only in the inverted regime.



Kane–Mele ribbon bands colored by weight on the lower edge. Unlike the Haldane chiral edge (one direction per edge), each edge here carries TWO counter-propagating modes forming a Kramers pair with opposite spins — helical states. Time reversal forbids elastic backscattering between Kramers partners, so the edges conduct perfectly as long as TRS holds; a magnetic impurity would break the protection. — Figure 41 · notebook §15 · chapter 4, cell 11



BHZ ribbon spectra in the two phases. Only the inverted regime (left) carries states crossing the bulk gap — the helical edge channels whose quantized conductance $2e^2/h$ was measured by the Würzburg group in 2007, the first experimental topological insulator. The trivial ribbon (right) is gapped through and through. — Figure 44 · notebook §16 · chapter 4, cell 16

- Helical, not chiral: both directions exist on the same edge, but tied to opposite spins.
- Backscattering needs a spin flip, which time-reversal-invariant disorder cannot provide: the crossing is protected.
- Two edge pairs ($Z_2 = 0$) can gap each other; one pair ($Z_2 = 1$) cannot — that is why the invariant is mod 2.

Kane–Mele, Kramers, and the two ways to Z_2

KANE–MELE

$$H = t\,\Sigma_{\langle ij\rangle}\, c^{\dagger}{}_{i}c_j + i\,\lambda_{SO}\,\Sigma_{\langle\langle ij\rangle\rangle}\, \nu_{ij}\, c^{\dagger}{}_{i}\, s_z\, c_j + \lambda_v\,\Sigma_i\, \xi_i\, c^{\dagger}{}_{i}c_i$$

TIME REVERSAL

$$\Theta = i\, s_y\, K,\ \Theta^2 = -1 \Rightarrow \text{Kramers: } E_n(k) = E_{n'}(-k), \text{ degenerate at TRIM}$$

Z_2

$$\nu = (\ \#\text{ crossings of } \bar{x}_n(k_y) \text{ with any line, } k_y \in [0, \pi]\ ) \bmod 2 = \Pi_{i=1}{}^{4}\, \delta_i,\ \delta_i = \Pi_{n\ \mathrm{occ}}\, \xi_{2n}(\Gamma_i) \text{ (with inversion)}$$

- Kane–Mele = two copies of Haldane with opposite φ for spin up and down (s_z conserved): $C_\uparrow = -C_\downarrow = 1$, $C = 0$, $\nu = 1$.
- Rashba coupling breaks s_z conservation; the spin Chern numbers lose meaning but ν survives — that is its point.
- With inversion symmetry, Fu–Kane's parity product at the four TRIM gives ν from four numbers; without it, use the Wannier flow.

Corners, Majoranas, Weyl

Higher-order topology, superconductivity as extra orbitals, Weyl monopoles with Fermi arcs, and the axion angle.

Four more kinds of topological matter

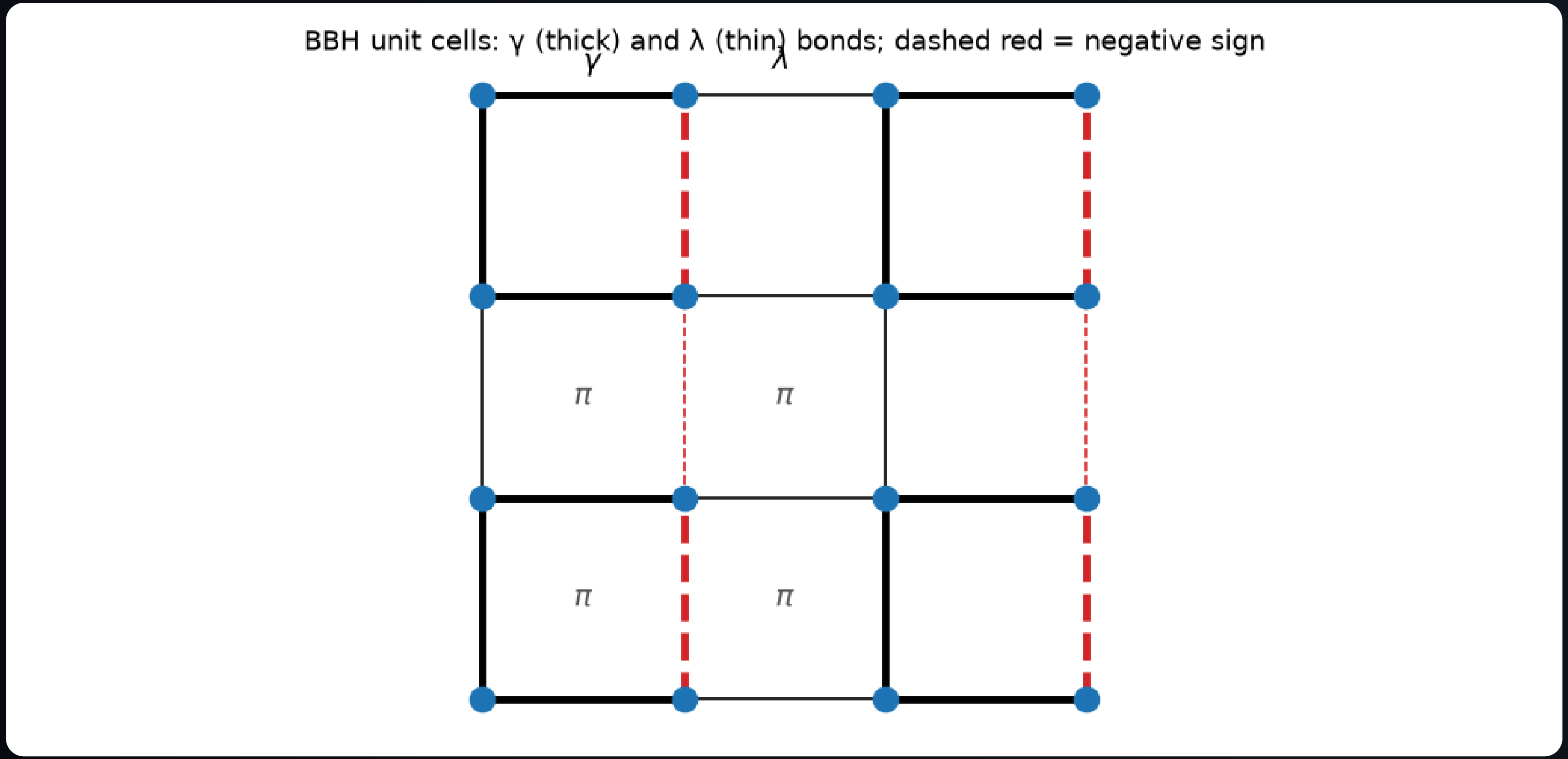
The same machinery — a lattice model, its eigenvectors, a Berry-phase quantity — classifies far more than Chern and Z_2 insulators. This lecture is a tour; each stop is a full notebook section.

Model	What is new	Invariant	Boundary signature
BBH quadrupole (§17)	topology of the Wannier bands themselves	nested Wilson loop, $q_{xy} = ½$	four corner states at zero energy
Kitaev chain (§18)	superconductivity as extra (hole) orbitals	Pfaffian / winding, $\lvert\mu\rvert < 2t$	one Majorana per end
Weyl semimetal (§19)	gapless but topological: monopoles in 3D	sliced Chern number $C(k_z)$	Fermi arcs on the surface
Fu–Kane–Mele + axion (§20)	3D strong TI; magnetoelectric angle	$\theta \in \{0, \pi\}$, second Chern number	half-quantized surface Hall effect

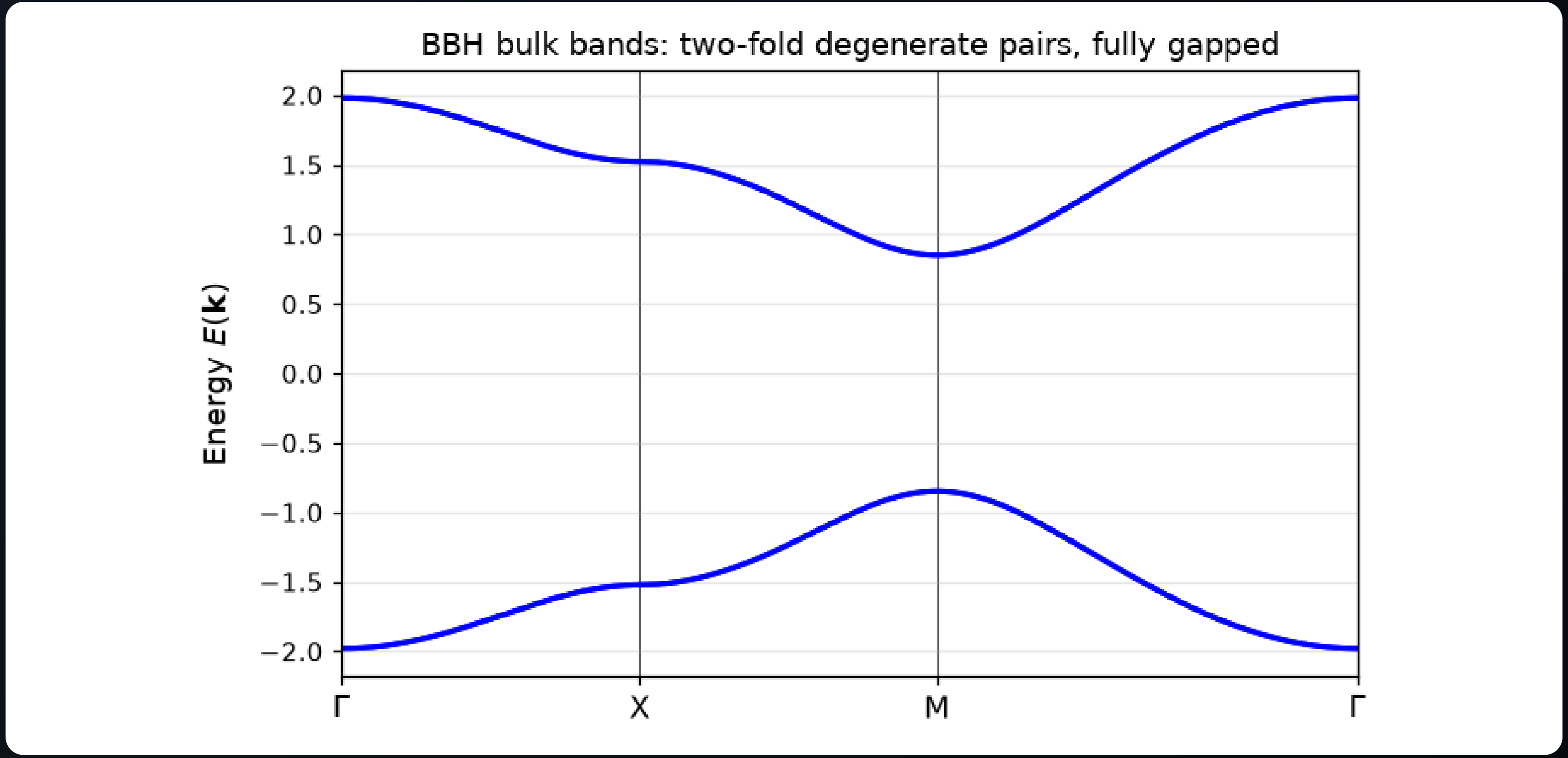
CORE

The BBH model: two SSH chains crossed

Four orbitals per square cell, intracell bonds γ and intercell bonds λ, with one negative bond per plaquette — a π flux (left). The bulk bands come in two doubly-degenerate pairs with a full gap (right).



The BBH quadrupole model: four orbitals per cell coupled by intracell γ (thick) and intercell λ (thin) bonds, with the vertical bonds of one column negated (dashed red). Multiply the bond signs around any elementary square: always −1, i.e. π flux per plaquette. The flux gaps the edge spectrum too — that is what demotes the topological states from edges to corners, one dimension down. – Figure 45 · notebook §17 · chapter 5, cell 3

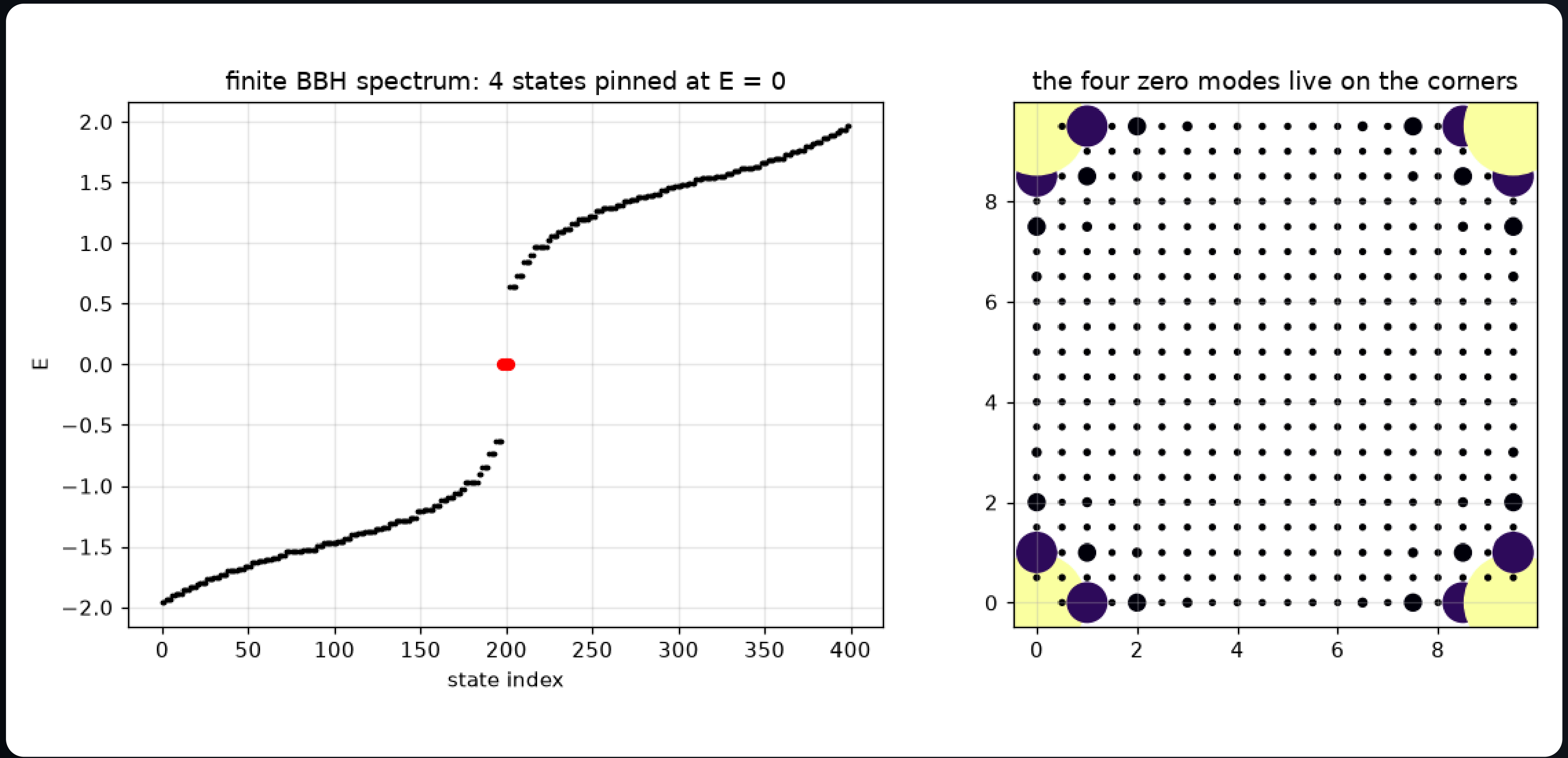


BBH bulk bands: two doubly-degenerate pairs separated by a full gap. Nothing here betrays the topology — bulk dipole moments vanish by symmetry, and only the quadrupole moment, extracted by the nested Wilson loop below, distinguishes this insulator from a trivial one. – Figure 46 · notebook §17 · chapter 5, cell 3

- SSH logic in both directions at once: $\lambda > \gamma$ is the candidate topological phase.
- The π flux per plaquette is essential: it keeps the bands degenerate and the Wannier bands gapped.
- Nothing in these bulk bands betrays the topology — the dipole moment vanishes by symmetry in every direction.

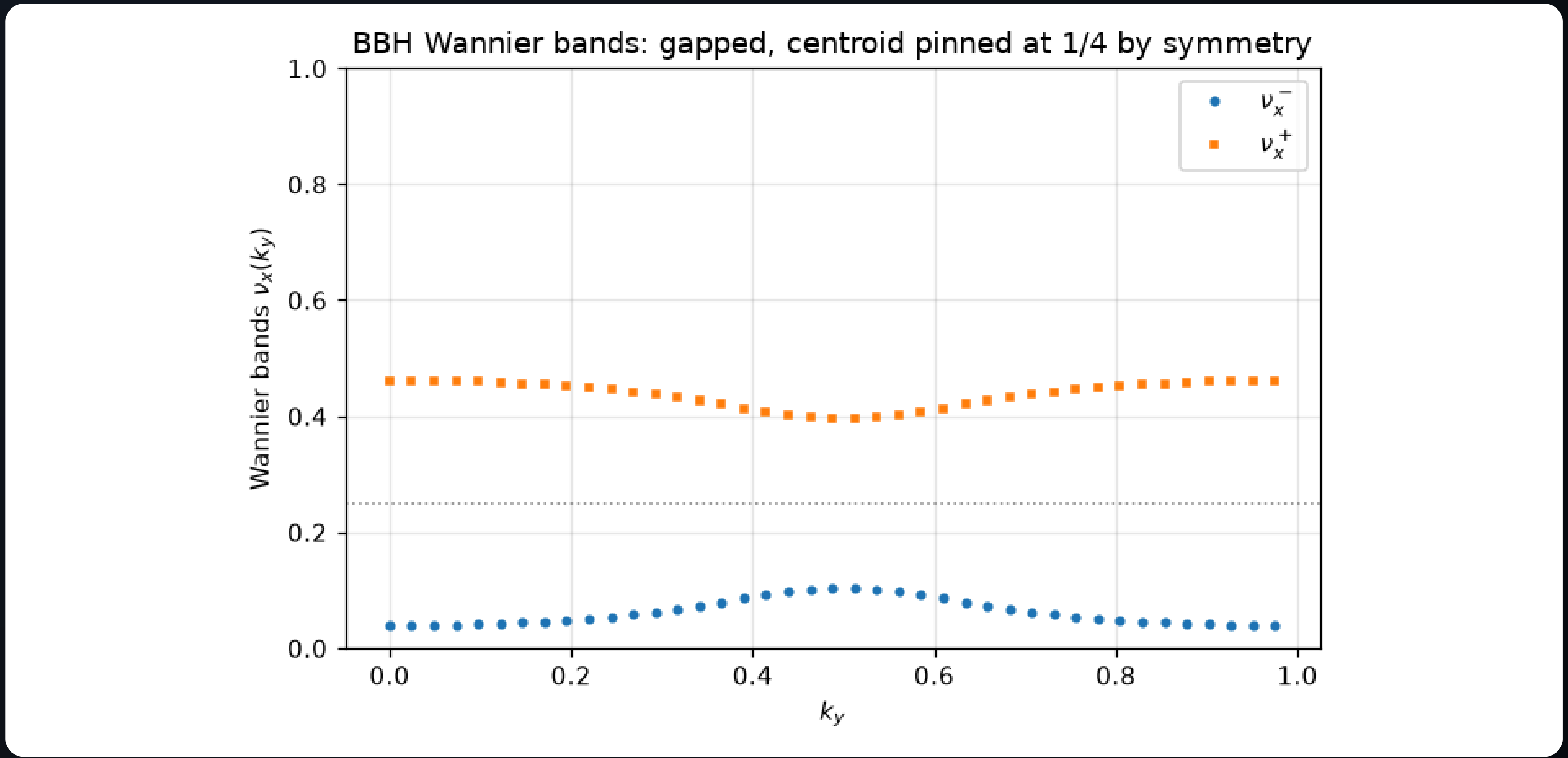
Higher-order topology: the quadrupole insulator

The Benalcazar–Bernevig–Hughes model has a gapped bulk, gapped edges, and **four zero modes on the corners** (left). The invariant lives one level down: the Wilson-loop eigenphases form Wannier bands, gapped and pinned at ±¼ by symmetry (right).



Higher-order bulk-boundary correspondence: the finite BBH sample has a gapped bulk AND gapped edges, yet four states sit pinned at E = 0 (red in the left panel), one exponentially localized on each corner (right). Each corner binds charge ±e/2 — the 2D analogue of the SSH end states, one dimension of boundary further down.

— Figure 47 · notebook §17 · chapter 5, cell 4



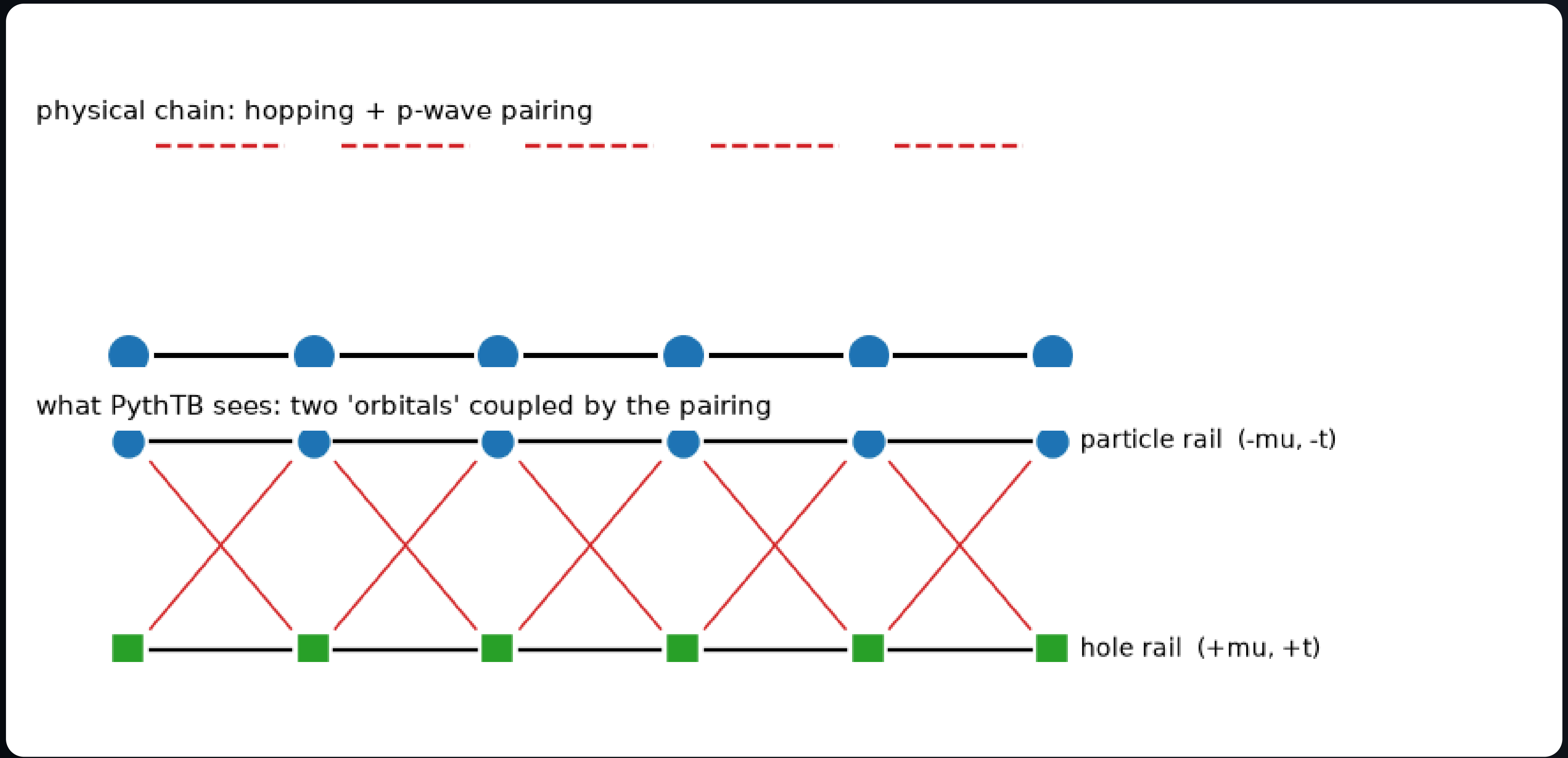
First layer of the nested construction: the Wilson-loop eigenphases (Wannier bands) ν±(ky) of the occupied pair. They disperse but never touch, and mirror symmetry pins their centroid to exactly ¼. A gapped Wannier spectrum is itself a band structure one level up — with its own Berry phase, computed next.

— Figure 48 · notebook §17 · chapter 5, cell 5

- Four orbitals per cell, intracell γ and intercell λ bonds, one negative sign per plaquette (a π flux): SSH in two directions at once.
- Corner charge $\pm e/2$, quadrupole moment q_{xy} = ½: the boundary of the boundary carries the charge.
- Built by hand in the notebook from `berry_phase(berry_evals=True)` — a nested Wilson loop is a Wilson loop of a Wilson loop.

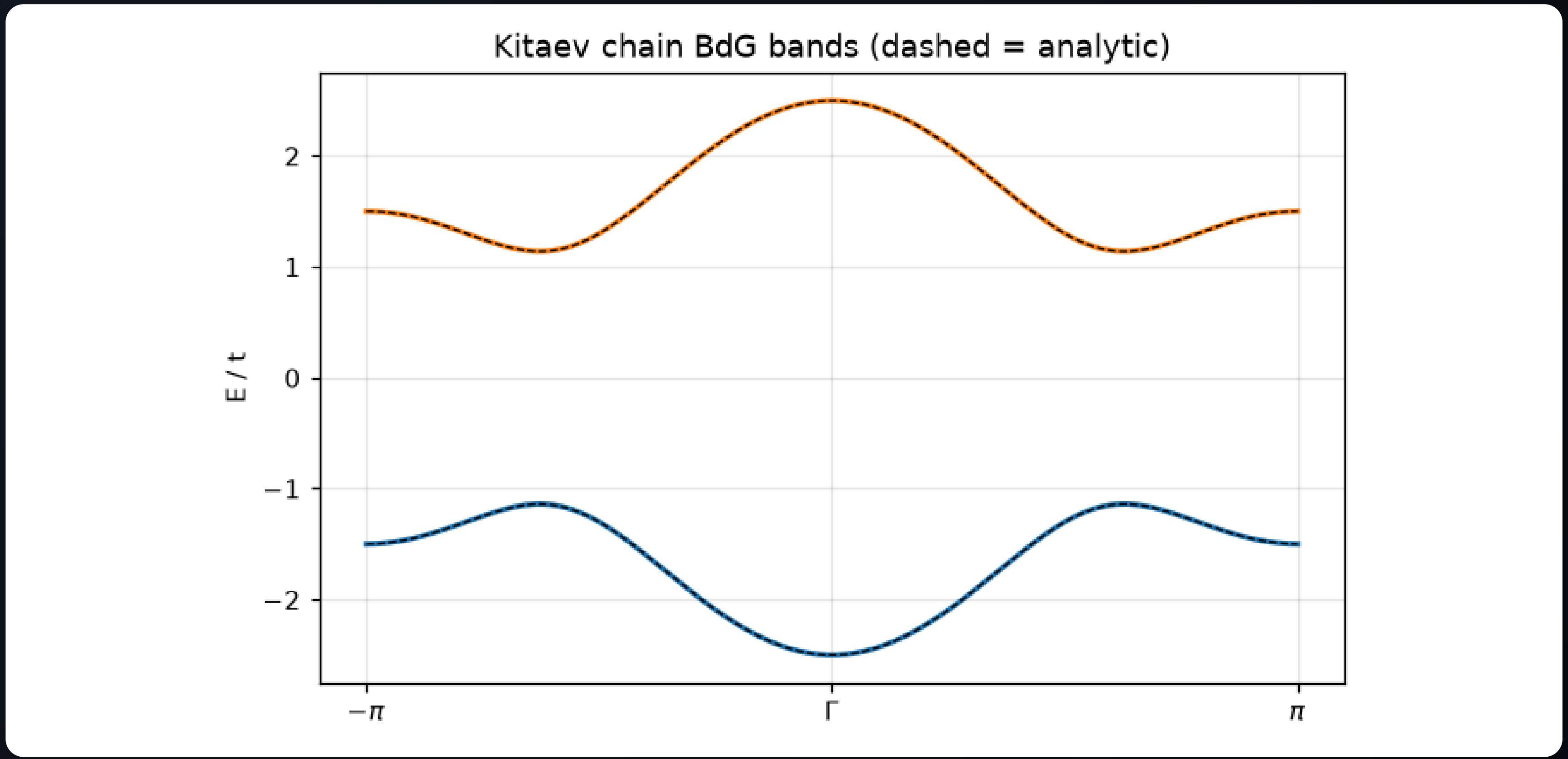
Writing pairing as hopping: the BdG trick

The physical chain has hopping t and p-wave pairing Δ that creates and destroys electron pairs (left, top) — not a hopping at all. Double the orbitals into a particle and a hole copy and it becomes one (left, bottom); the resulting BdG bands match the analytic dispersion exactly (right).



Top: the physical Kitaev chain — spinless fermions with hopping t and p-wave pairing Δ that creates/destroys PAIRS on neighbouring sites. Bottom: the Bogoliubov–de Gennes doubling PythTB actually diagonalizes — a particle rail and a hole rail (with negated parameters), the anomalous pairing recast as an ordinary inter-rail 'hopping' (red). The doubling is exact but redundant: every physical excitation appears twice, at ±E.

– Figure 49 · notebook §18 · chapter 5, cell 8



BdG bands of the Kitaev chain (solid) on top of the analytic dispersion $\pm\sqrt{(\xi^2 + \Delta^2_k)}$ (dashed). The pairing gaps the would-be Fermi points of the normal chain; the gap at $k = \pm\pi/2$ is purely superconducting ($\propto \Delta$), while the gap at $k = 0$ and π is set by $|\mu \mp 2t|$ — and it is the latter that closes and reopens at the topological transition $\mu = 2t$.

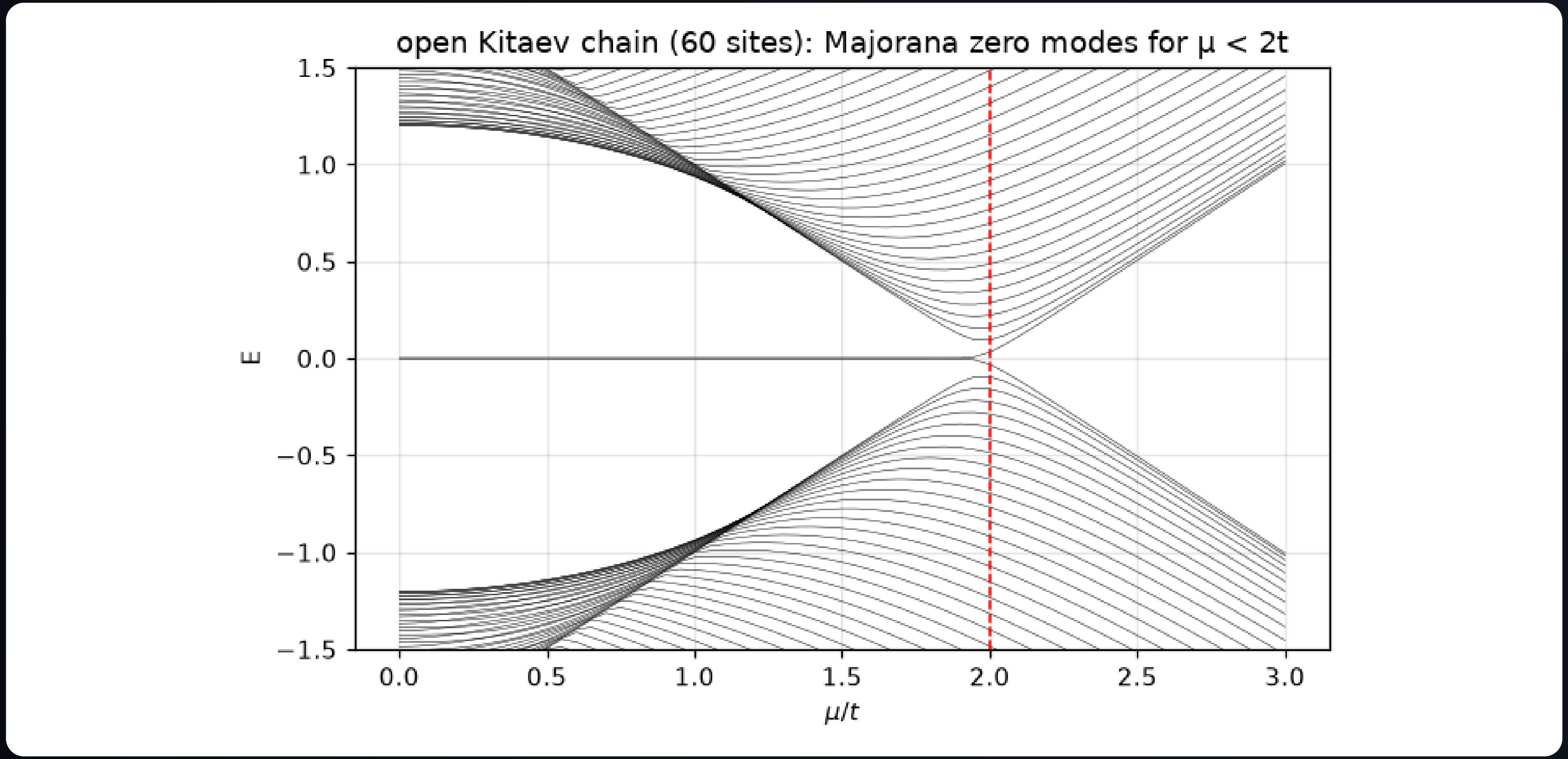
– Figure 50 · notebook §18 · chapter 5, cell 9

- PythTB never learns about superconductivity: it sees an ordinary two-orbital tight-binding model.
- The price of the trick: every state appears twice, at E and $-E$; only half the spectrum is physical.
- The gap closes at $k = 0$ when $\mu = -2t$ and at $k = \pi$ when $\mu = +2t$ — the boundaries of the topological phase.

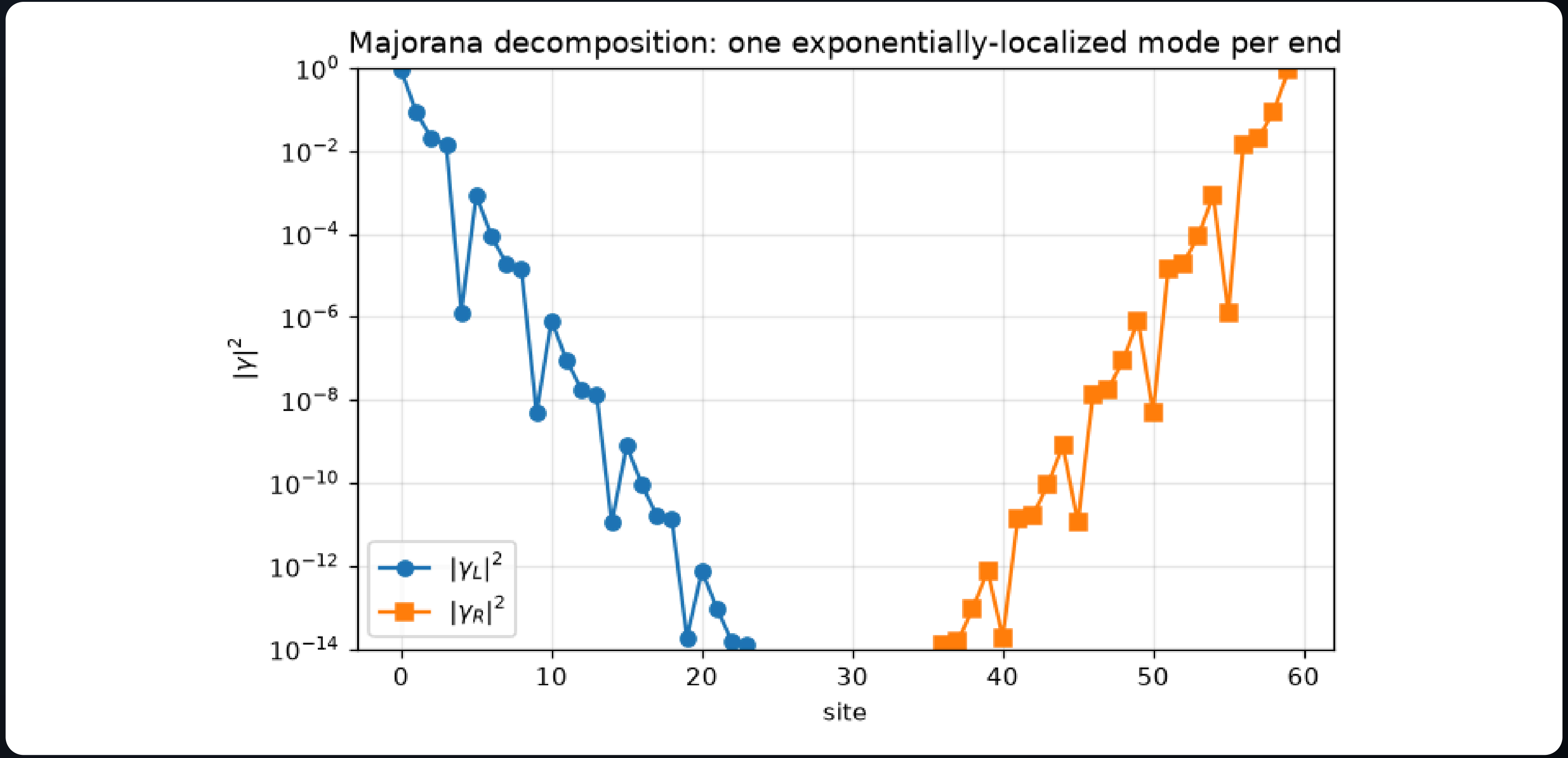
Superconductivity smuggled in: the Kitaev chain

CORE

PythTB knows nothing about pairing. Double the orbitals — particles and holes — and write the Bogoliubov–de Gennes matrix as hoppings. Majorana end modes appear exactly for $|\mu| < 2t$ (left), one per end (right).



Open-chain BdG spectrum versus chemical potential: throughout the topological phase $|\mu| < 2t$ a pair of states clings to $E = 0$, splitting only exponentially in system size; past the bulk transition they merge into the continuum. This zero-bias pinning is exactly the feature hunted in tunneling experiments on proximitized nanowires — with the caveat that a tunneling CONDUCTANCE calculation is Kwant's territory, not PythTB's.
— Figure 51 · notebook §18 · chapter 5, cell 10



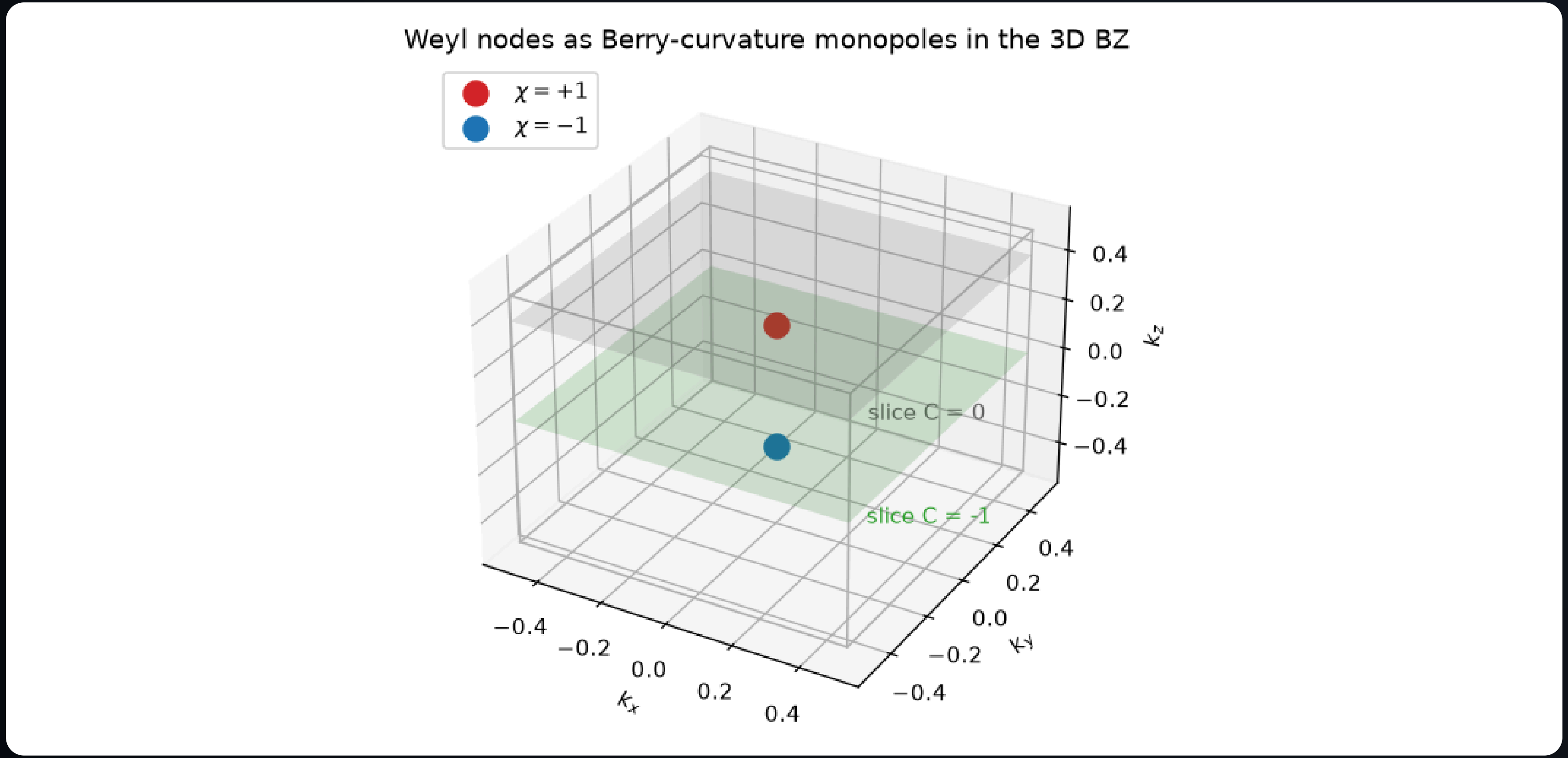
The two $E \approx 0$ BdG eigenstates recombined into maximally end-localized combinations: each is a MAJORANA mode — equal particle and hole content, exponentially confined to one end. A single fermionic level (occupied or empty) is physically split across the two chain ends; no local probe can measure its occupation, which is the storage principle of topological qubits. — Figure 52 · notebook §18 · chapter 5, cell 11

- The BdG spectrum is symmetric by construction (particle–hole); a state at $E = 0$ is its own antiparticle: a Majorana.
- Two Majoranas at opposite ends make one ordinary fermion that costs no energy to occupy: a two-fold degenerate ground state, the qubit.
- The trap: PythTB does not enforce particle–hole symmetry. Get the hole block wrong and the code runs happily — lecture 10.

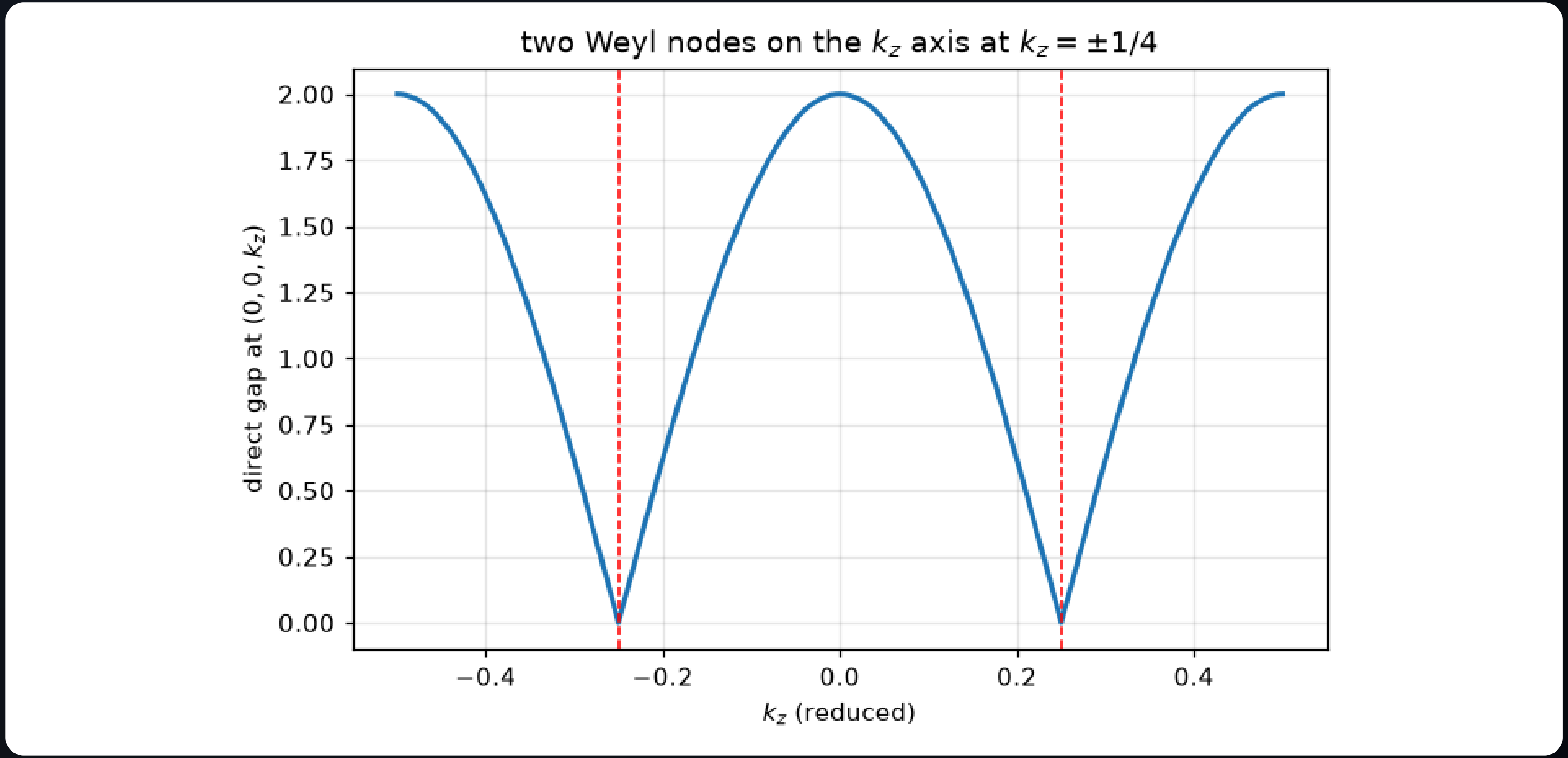
CORE

Weyl semimetals: monopoles in the Brillouin zone

Two bands touch at isolated points in a 3D zone (left): each node is a **monopole of Berry curvature**, with flux $\pm 2\pi$ through any surface around it, and nodes come in pairs of opposite charge. The direct gap along the k_z axis (right) collapses linearly at $k_z = \pm 1/4$ and nowhere else.



The playing field: the 3D Brillouin zone with the two Weyl nodes on the kz axis — a monopole (red) and an antimonopole (blue) of Berry curvature. Every fixed-kz plane is a 2D system with a well-defined Chern number, which changes by the enclosed chirality when the plane sweeps past a node: C = −1 between the nodes (green slice), 0 outside (gray slice). — Figure 53 · notebook §19 · chapter 6, cell 3

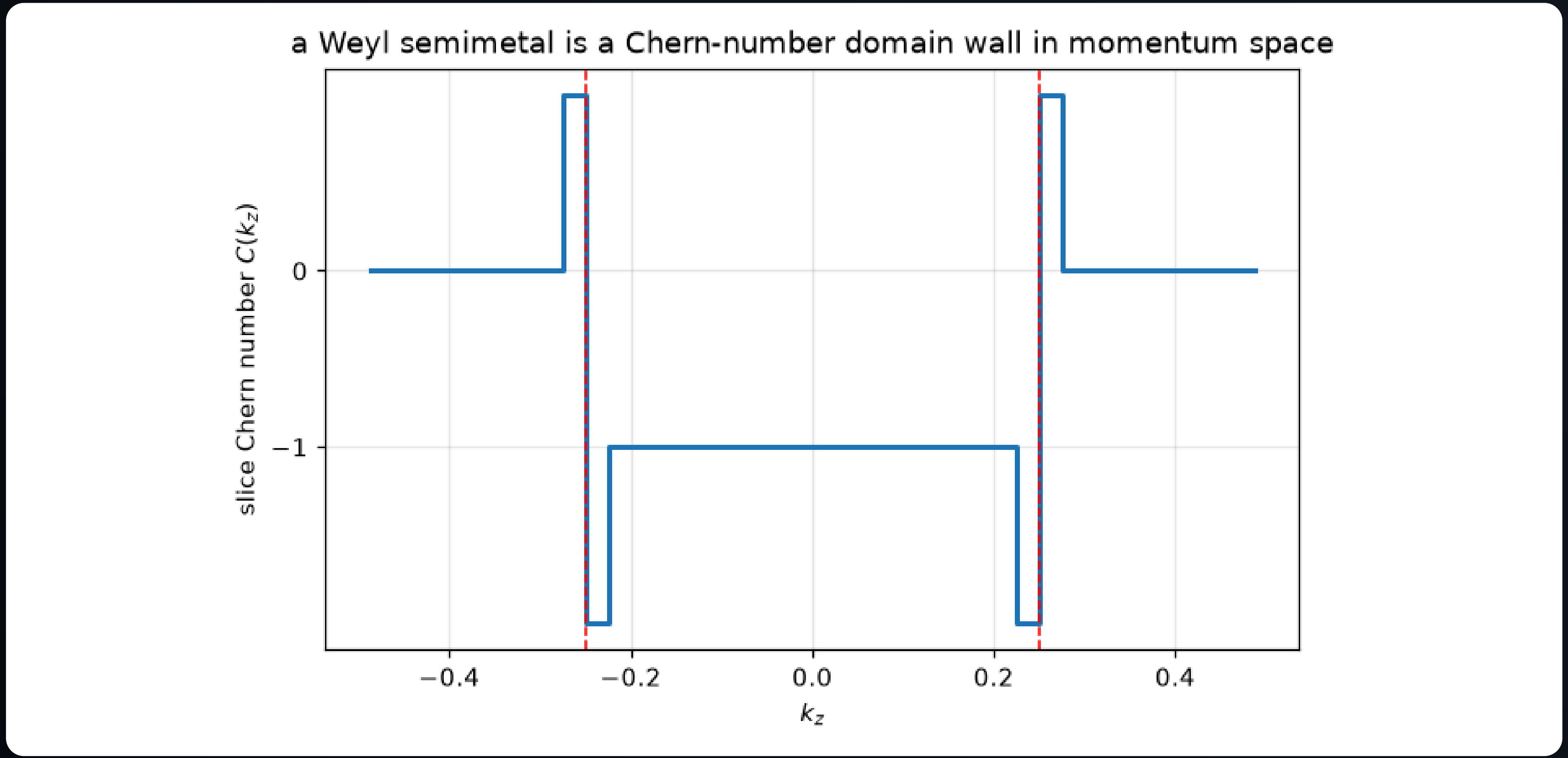


The direct gap along the kz axis: it collapses linearly to zero at kz = ±¼ and nowhere else — point touchings with conical dispersion in all three directions. Linear closure in 3D is generic (no fine-tuning): a Weyl node can only be removed by merging it with a partner of opposite chirality, which is why the phase is robust without any protecting symmetry beyond translation. — Figure 54 · notebook §19 · chapter 6, cell 4

- Gapless by topology, not by fine tuning: a point crossing in 3D cannot be removed by small perturbations (three parameters, three constraints).
- The minimal model breaks time reversal or inversion; here a two-orbital model with the nodes at $k_z = \pm 1/4$.
- A monopole pair is the 3D image of the Haldane phase transition, stretched along k_z.

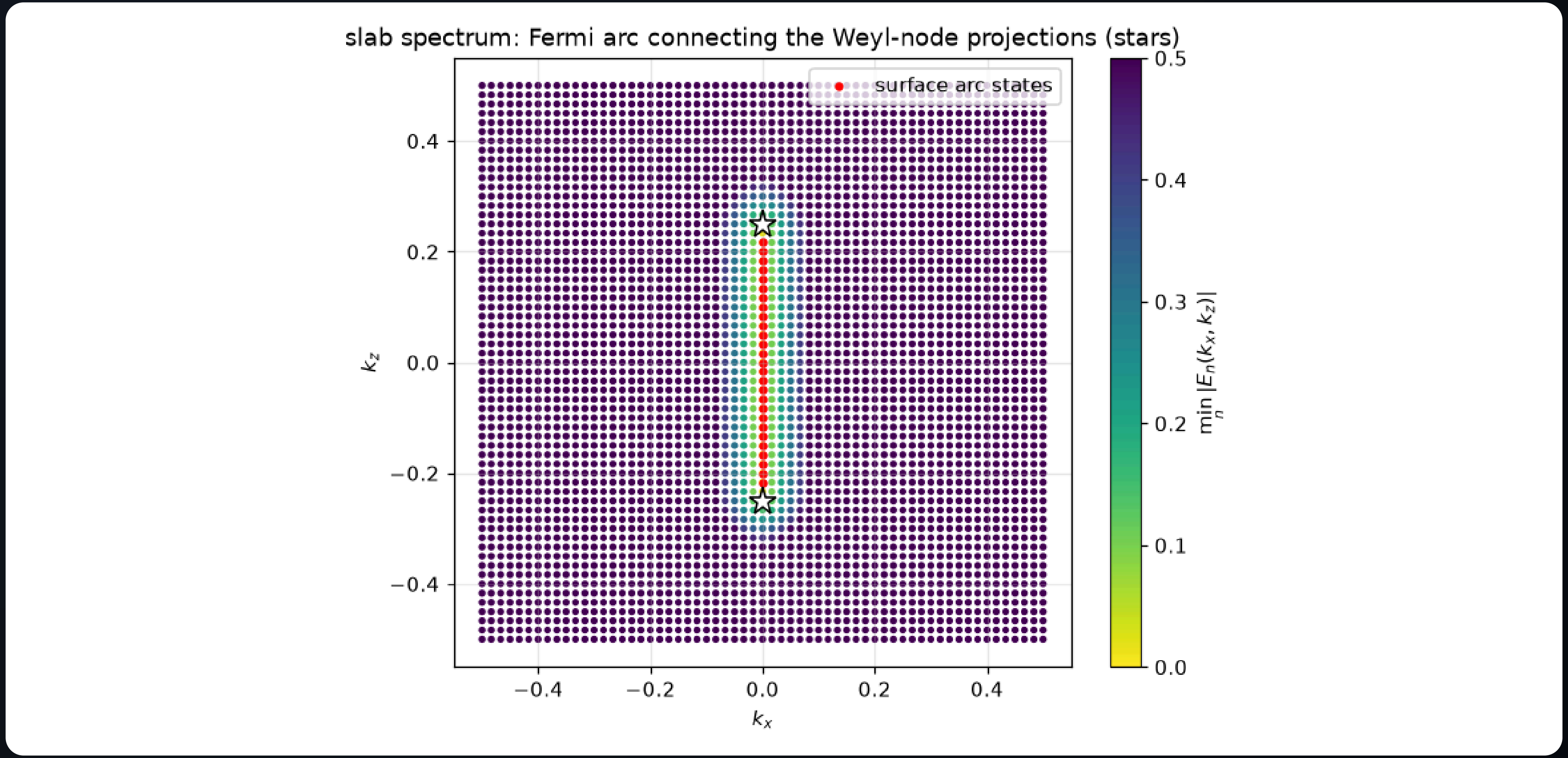
Sliced Chern numbers and Fermi arcs

Fix k_z and you have a 2D insulator with a Chern number; $C(k_z)$ jumps by one at each node (left). Every slice with $C = 1$ has a chiral edge state — stacked along k_z, they form a **Fermi arc** on the surface connecting the node projections (right).



The sliced Chern number C(kz), each point computed from a 2D model in which kz enters as a symbolic parameter: quantized plateaus with unit jumps exactly at the node positions. The jump measures the chirality of the node the slice crossed — Weyl nodes are literally the points where Chern numbers are created and destroyed.

— Figure 55 · notebook §19 · chapter 6, cell 5



Surface physics of the slab: the minimum |E(kx, kz)| over the slab bands (background) with surface-localized zero-energy states marked in red. They trace an open ARC connecting the projections of the two nodes — impossible for any isolated 2D metal, whose Fermi lines must close. The arc is the collection of chiral edge modes of all the C = −1 slices between the nodes, seen end-on; its partner lives on the opposite surface.

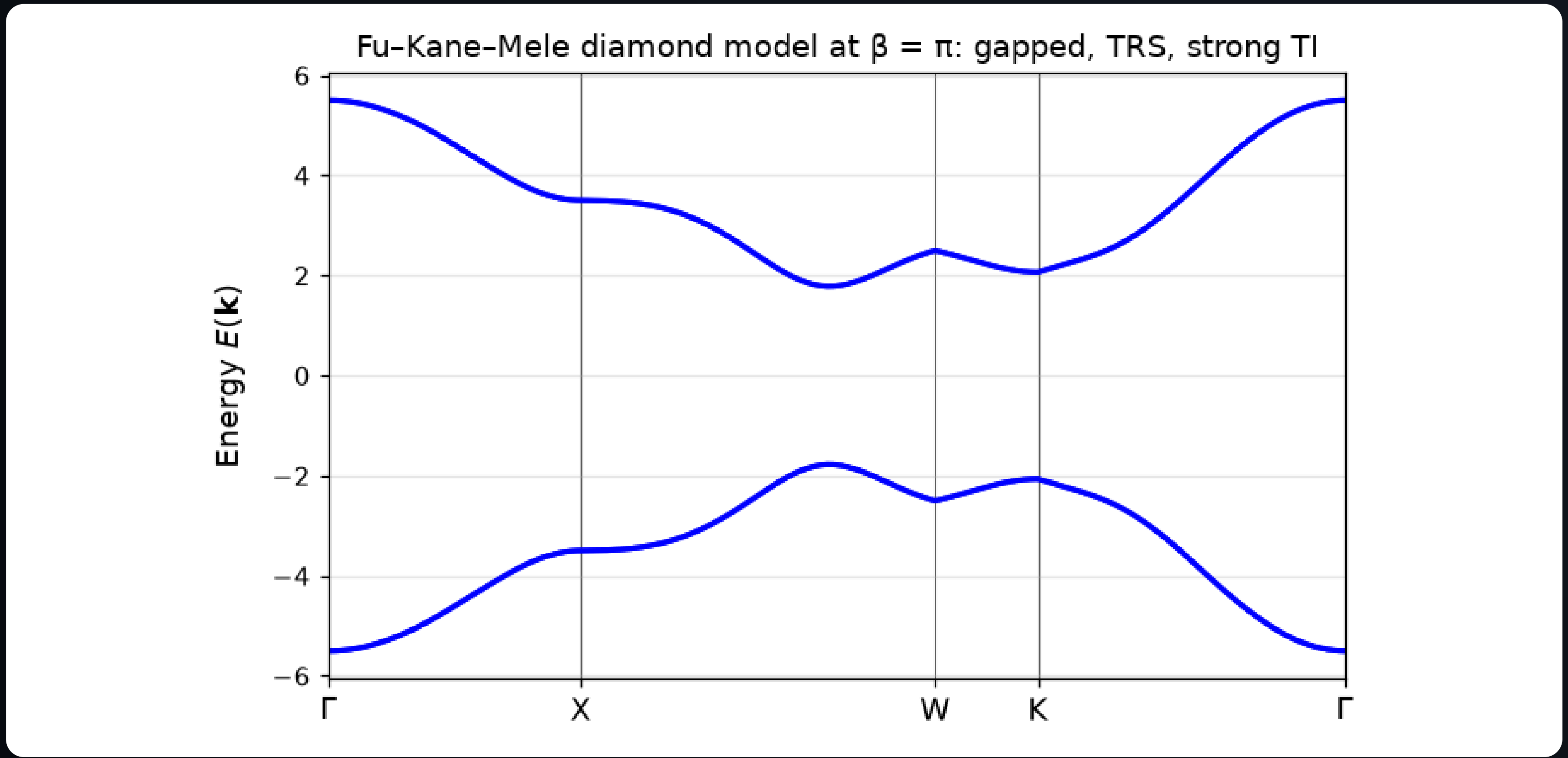
— Figure 56 · notebook §19 · chapter 6, cell 6

- A Weyl semimetal is a Chern-number domain wall in momentum space.
- The arc is an open Fermi surface — impossible for any purely 2D system, the fingerprint seen in ARPES on TaAs.
- PythTB: a 2D model with k_z as a symbolic parameter, `berry_flux` per slice, then `cut_piece` for the slab.

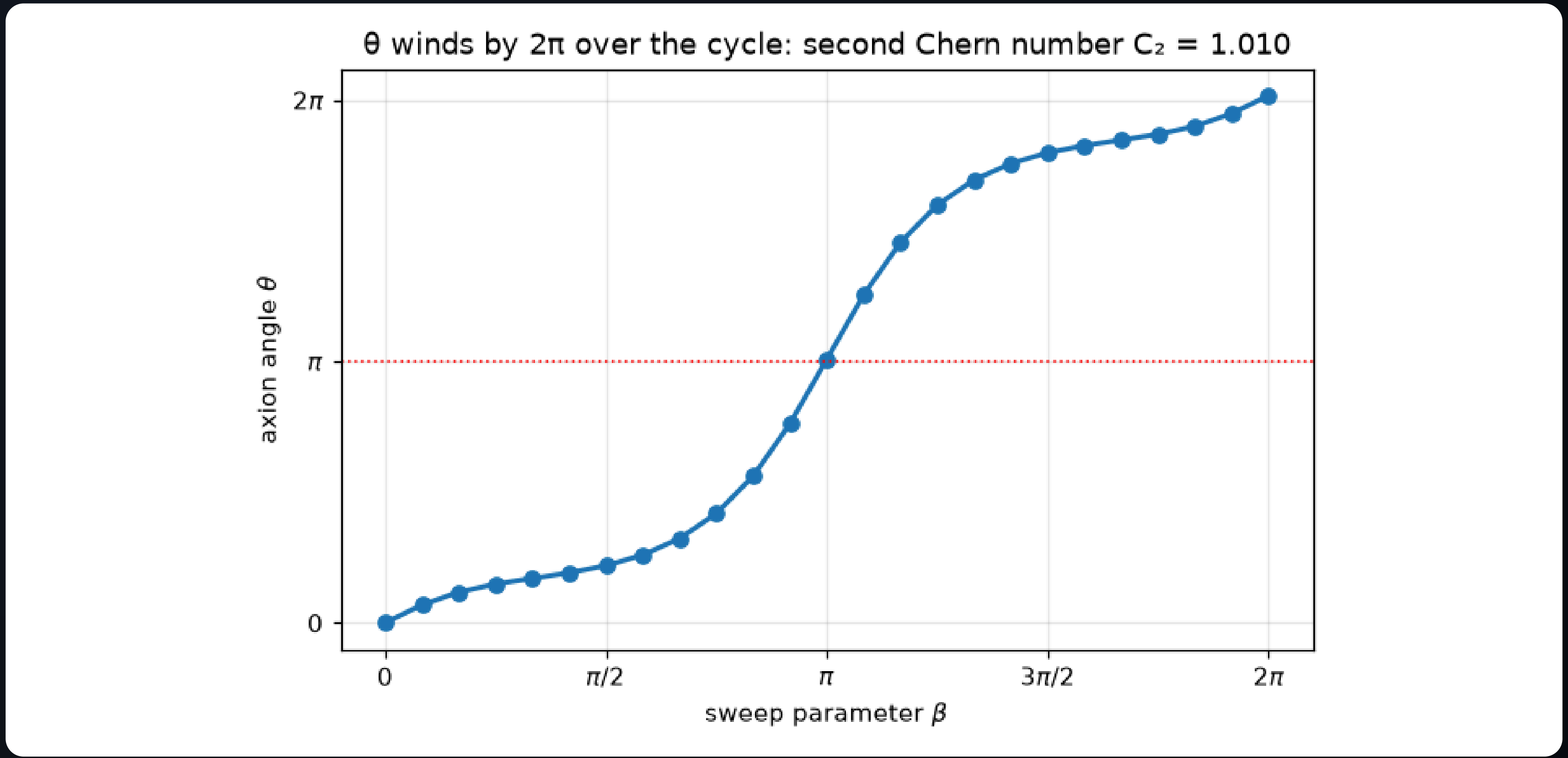
CORE

The axion angle θ

The Fu–Kane–Mele model on the diamond lattice at the time-reversal-symmetric point $\beta = \pi$: gapped bulk bands of a strong topological insulator (left). Driving β through a full adiabatic cycle winds the magnetoelectric angle θ by 2π (right): a **second Chern number** equal to one.



Bulk bands of the Fu–Kane–Mele model on the diamond lattice at the time-reversal-symmetric point of the sweep ($\beta = \pi$), where the (111) bond strengthened by the modulation makes the system a strong topological insulator. Every band is doubly degenerate (inversion × time reversal); the topology hides in the wavefunctions, not in this spectrum — it takes the axion angle to see it. — Figure 57 · notebook §20 · chapter 6, cell 8



The axion angle θ tracked along the adiabatic cycle β: it starts at 0 (trivial), passes exactly through π at the TRS point $\beta = \pi$ — the quantized magnetoelectric response of a strong TI — and returns having wound by 2π. The total winding is the second Chern number $C_2 = 1$ of the 4D (k, β) space: the same mathematics as the Thouless pump, two dimensions higher. θ-odd physics (surface half-integer Hall response, image magnetic monopoles) lives on the π plateau.
— Figure 58 · notebook §20 · chapter 6, cell 9

- θ is the 3D analogue of the polarization Berry phase: an angle, defined mod 2π, quantized to 0 or π by time reversal or inversion.
- $\theta = \pi$ is the strong topological insulator; the surface carries a half-quantized Hall conductance $e^2/2h$.
- Computed here from the non-Abelian Berry connection on a 3D mesh — PythTB's `WFArray` in its most demanding use.

MATH

BdG, Weyl, axion: the formulas

KITAEV BDG

$$H = ½\, \Sigma_k\, \Psi^\dagger{}_k\, H_{BdG}(k)\, \Psi_k,\ E(k) = \pm\sqrt{(\,(2t\cos k + \mu)^2 + 4\Delta^2 \sin^2 k\,)},$$
$$\textit{topological for } |\mu| < 2t$$

WEYL NODE

$$H(k_0 + q) = \chi\, \hbar\, v\, q\cdot\sigma,\ \oint_{S^2} \Omega\cdot dS = 2\pi\,\chi,\ \chi = \pm 1$$

AXION ANGLE

$$\theta = -(1/4\pi) \int_{BZ} d^3k\, \varepsilon^{abc}\, \mathrm{Tr}[\, A_a \partial_b A_c - (2i/3)\, A_a A_b A_c \,],\ \Delta\theta \textit{ over a cycle} = 2\pi\, C_2$$

- Particle–hole symmetry $\Xi\, H_{BdG}(k)\, \Xi^{-1} = -H_{BdG}(-k)$: the hole block must be $-H^*(-k)$. PythTB cannot check it for you.
- Chirality $\chi = \mathrm{sgn}\, \det(\partial H/\partial k)$ at the node; $C(k_z)$ changes by χ across it.
- A is the non-Abelian Berry connection of all occupied bands; only differences and the mod-2π value of θ are gauge invariant.

L9 · NOTEBOOK §21–24

Stretching PythTB

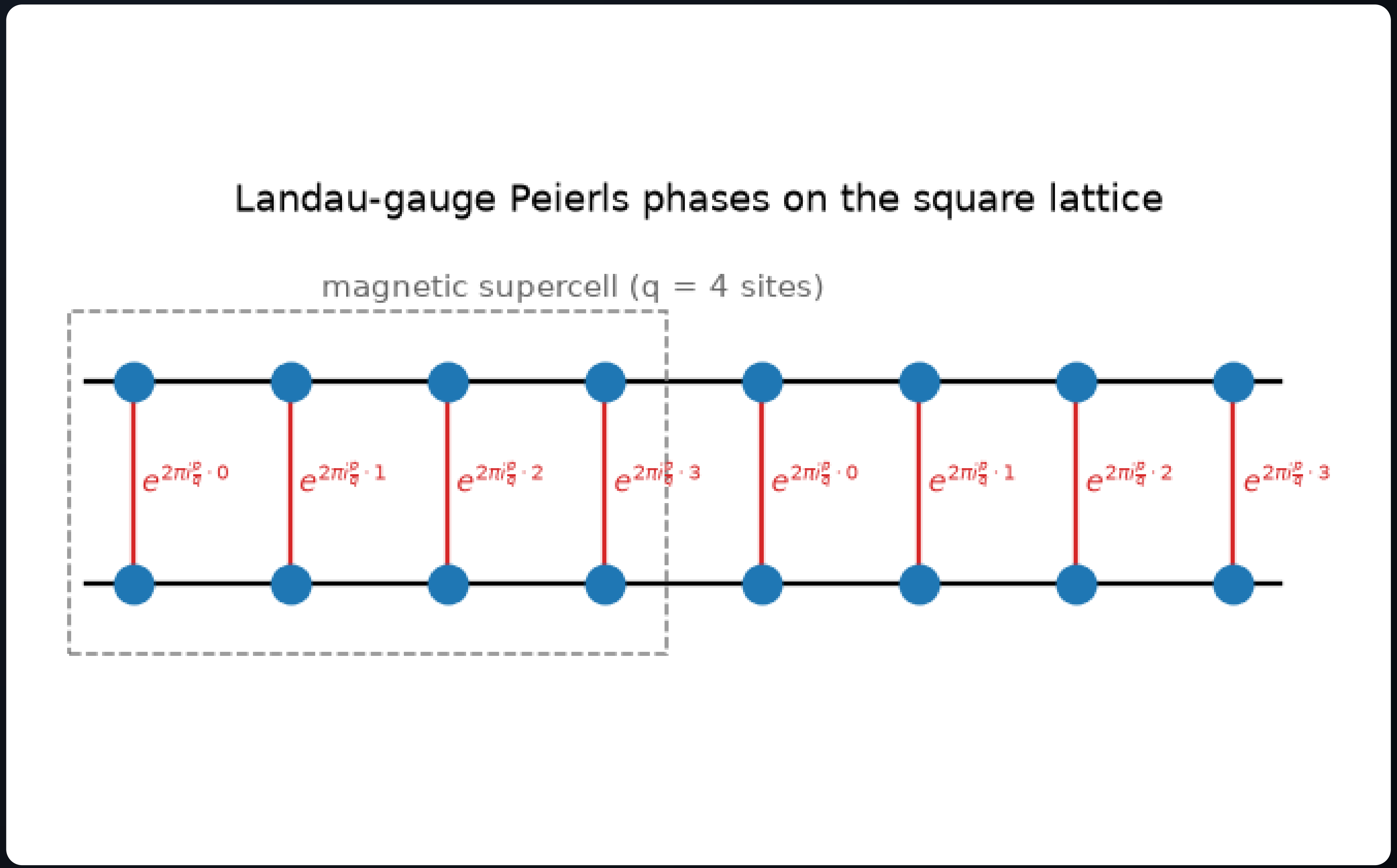
Magnetic fields by Peierls phases, Anderson localization, a Penrose quasicrystal, and the $O(N^3)$ wall.

1. A magnetic field as phases on bonds
2. The Hofstadter butterfly
3. Anderson localization, one eigenvector at a time
4. A Penrose quasicrystal: order without periodicity
5. Spiky spectrum and confined states
6. The performance wall
7. Peierls phases, IPR, and the cost of dense algebra

A magnetic field as phases on bonds

PythTB has no magnetic field. It does have complex hoppings. The **Peierls substitution** turns a uniform field into a phase on every bond — in Landau gauge, hops along x stay real and hops along y pick up $e^{2\pi i\,\varphi\,x}$.

- φ = flux per plaquette in units of the flux quantum h/e.
- Going round one plaquette accumulates $2\pi\varphi$: the Aharonov–Bohm phase, gauge independent.
- For rational $\varphi = p/q$ the phases repeat after q cells: a magnetic unit cell of q sites.

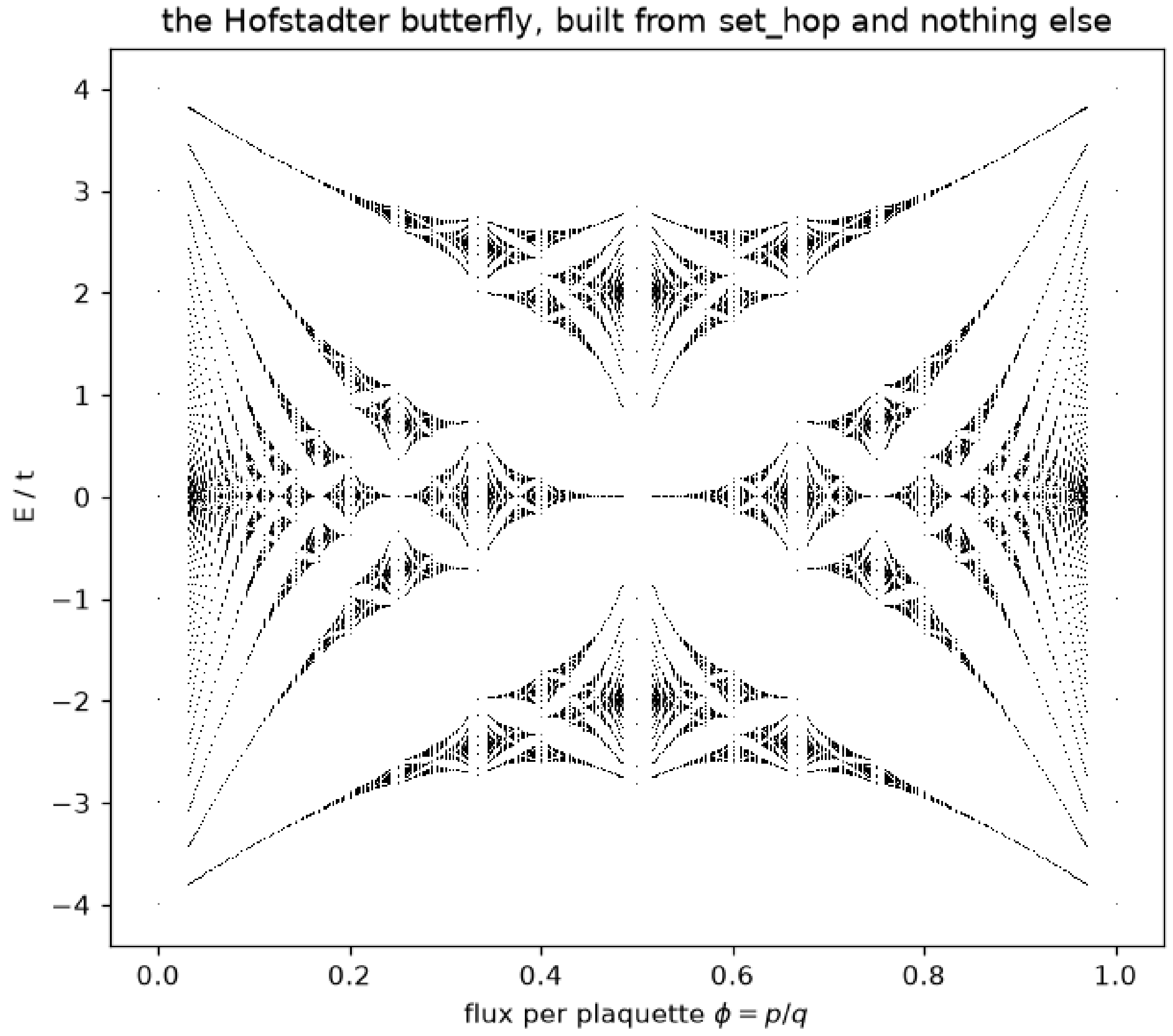


How a uniform field enters a lattice model: hoppings along x stay real (black), hoppings along y acquire the Peierls phase exp(2πi·(p/q)·x) that grows with the column index (red labels). The pattern repeats after q columns — the magnetic supercell (dashed box) — which is why rational flux p/q restores Bloch's theorem with a q-site cell and q subbands. — Figure 59 · notebook §21 · chapter 7, cell 3

CORE

The Hofstadter butterfly

Every eigenvalue of the square lattice at every rational flux p/q with $q \leq 32$, from `set_hop` and nothing else. Nested gaps at every scale; each gap carries a Chern number.



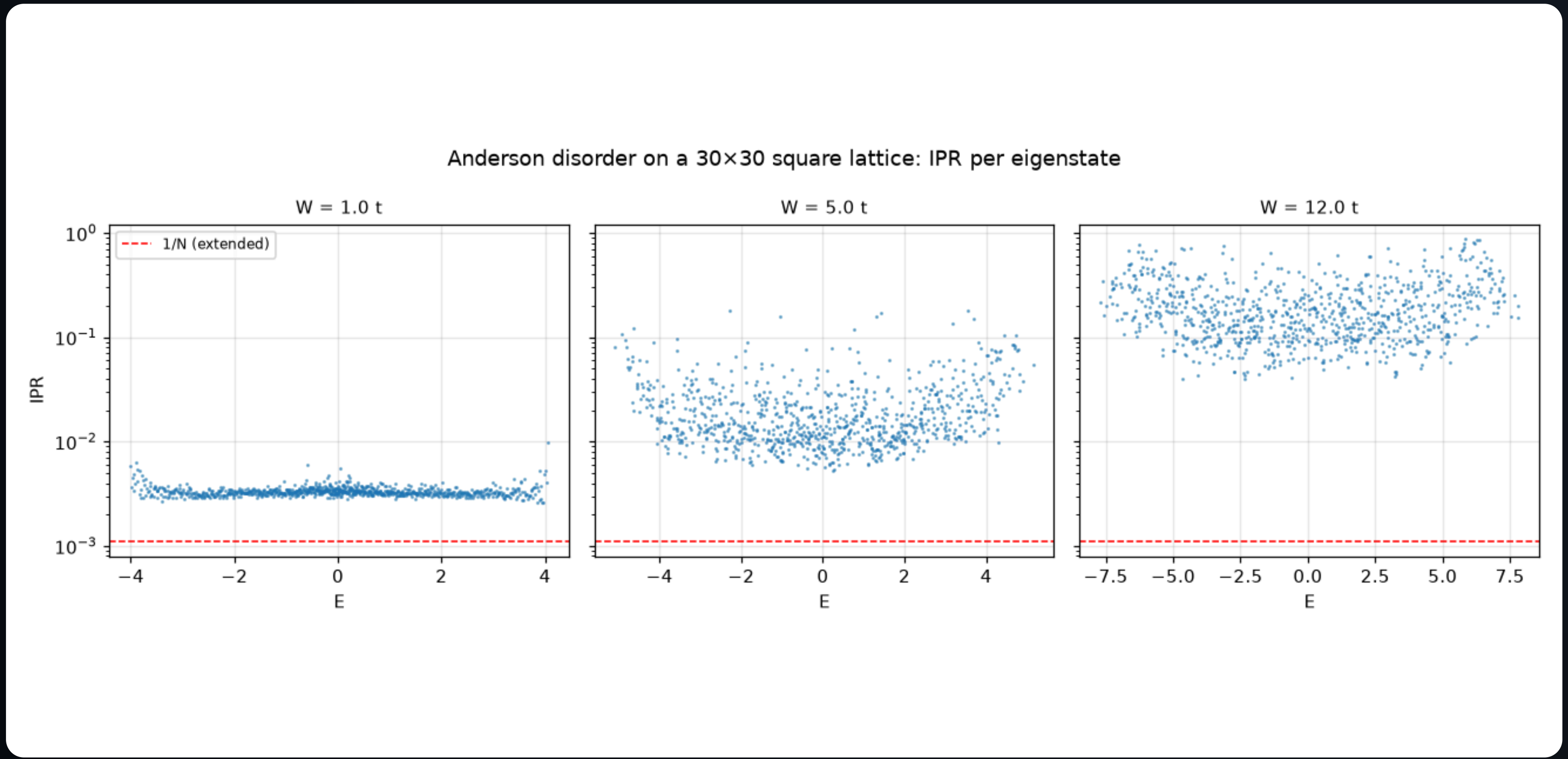
The Hofstadter butterfly: every eigenvalue of the square lattice at every rational flux p/q with q ≤ 32. The spectrum is a Cantor set — the pattern inside any gap-bounded wing repeats the whole. Each gap carries a quantized Hall conductance (the Diophantine labels of exercise III.1), and the giant central gaps at simple fractions are the ones seen in moiré superlattice experiments.

— Figure 60 · notebook §21 · chapter 7, cell 3

CORE

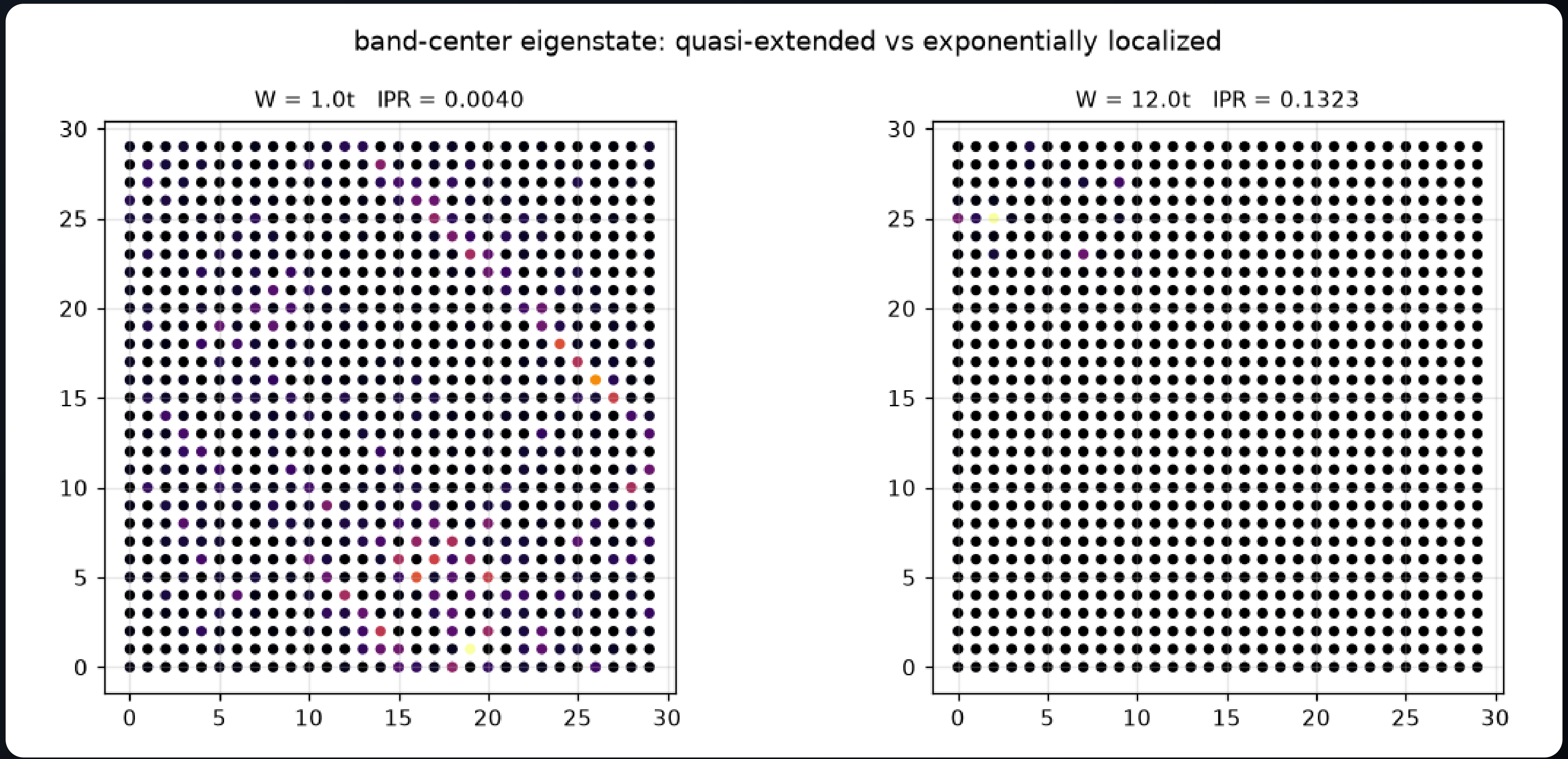
Anderson localization, one eigenvector at a time

Random onsite energies on a 30×30 supercell. The inverse participation ratio of every eigenstate (left) separates extended from localized states; one band-centre state at weak versus strong disorder (right) shows what localization looks like.



Inverse participation ratio of every eigenstate of a disordered square-lattice box, for three disorder strengths. At W = t most states hug the extended-state line 1/N (dashed red) — their localization length exceeds the sample. As W grows the whole spectrum lifts off by orders of magnitude, band-edge states first: localization sets in from the spectral edges inward, the standard mobility-edge phenomenology compressed into a 30×30 box.

— Figure 61 · notebook §22 · chapter 7, cell 6



One band-center eigenstate at weak (left) and strong (right) disorder, $|\psi|^2$ on the actual lattice. The weak-disorder state still spreads over the whole box in an irregular interference pattern; the strong-disorder state has collapsed onto a few sites around one favourable potential fluctuation. In 2D the left state IS localized too — its ξ merely exceeds the box, a distinction our finite sample cannot resolve, and (§24) dense diagonalization cannot push much further.

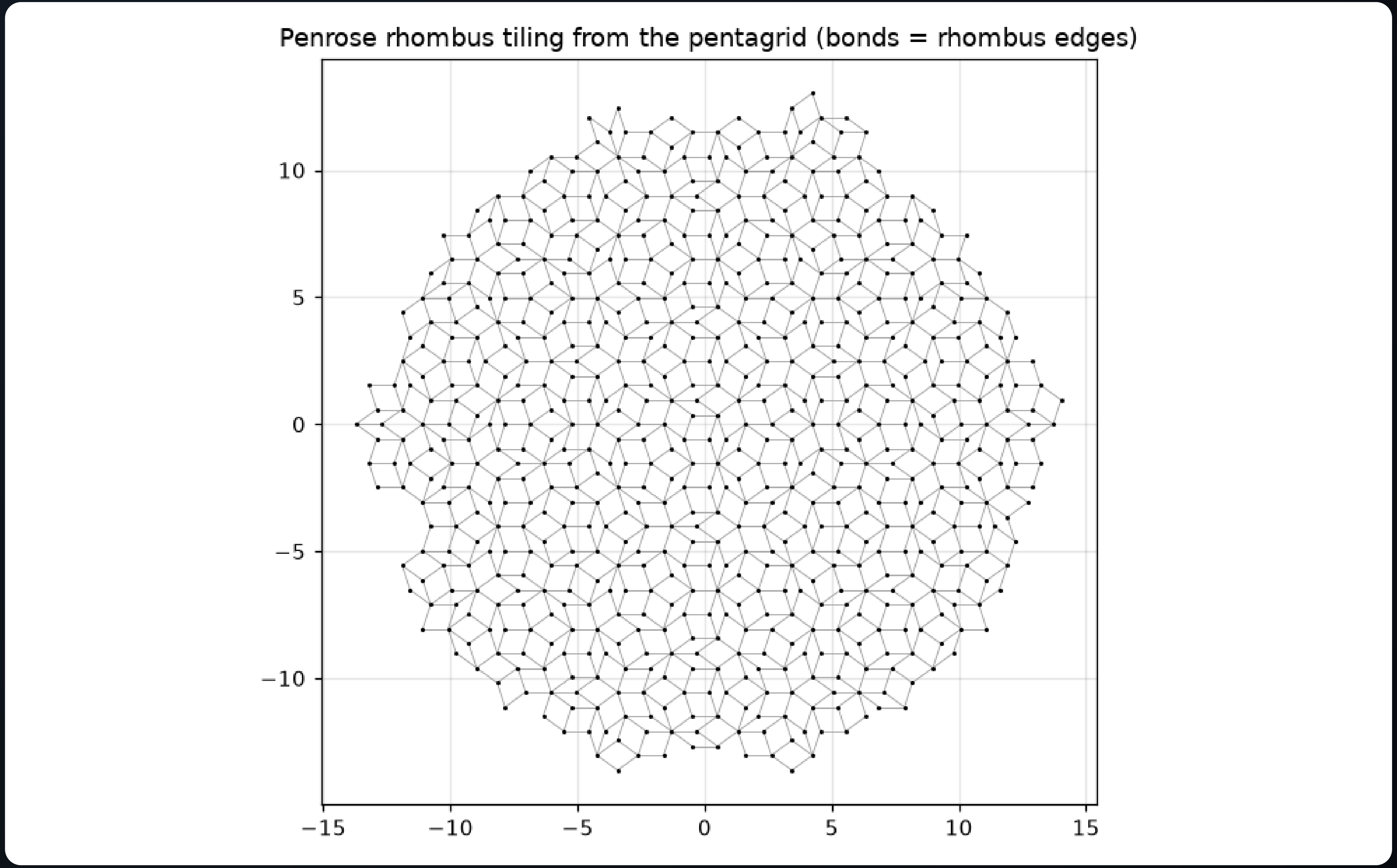
— Figure 62 · notebook §22 · chapter 7, cell 7

- $IPR = \Sigma_i |\psi_i|^4$: $\sim 1/N$ for an extended state, $\sim 1/\xi^2$ for one localized on ξ^2 sites.
- Band edges localize first; in 2D, in the thermodynamic limit, everything localizes — but the length can exceed the sample.
- No k-space: disorder is a supercell with one `set_onsite` per site, and a full dense diagonalization.

A Penrose quasicrystal: order without periodicity

De Bruijn's pentagrid generates a Penrose rhombus tiling; each vertex becomes a site, each rhombus edge a bond. Five-fold symmetry, no unit cell — a `dim_k = 0` model with a few thousand orbitals.

- Quasicrystals are ordered (sharp diffraction peaks) but not periodic: Bloch's theorem does not apply.
- PythTB treats it as one big finite molecule: geometry in, Hamiltonian out.
- The tiling is bipartite: the spectrum is symmetric about $E = 0$.

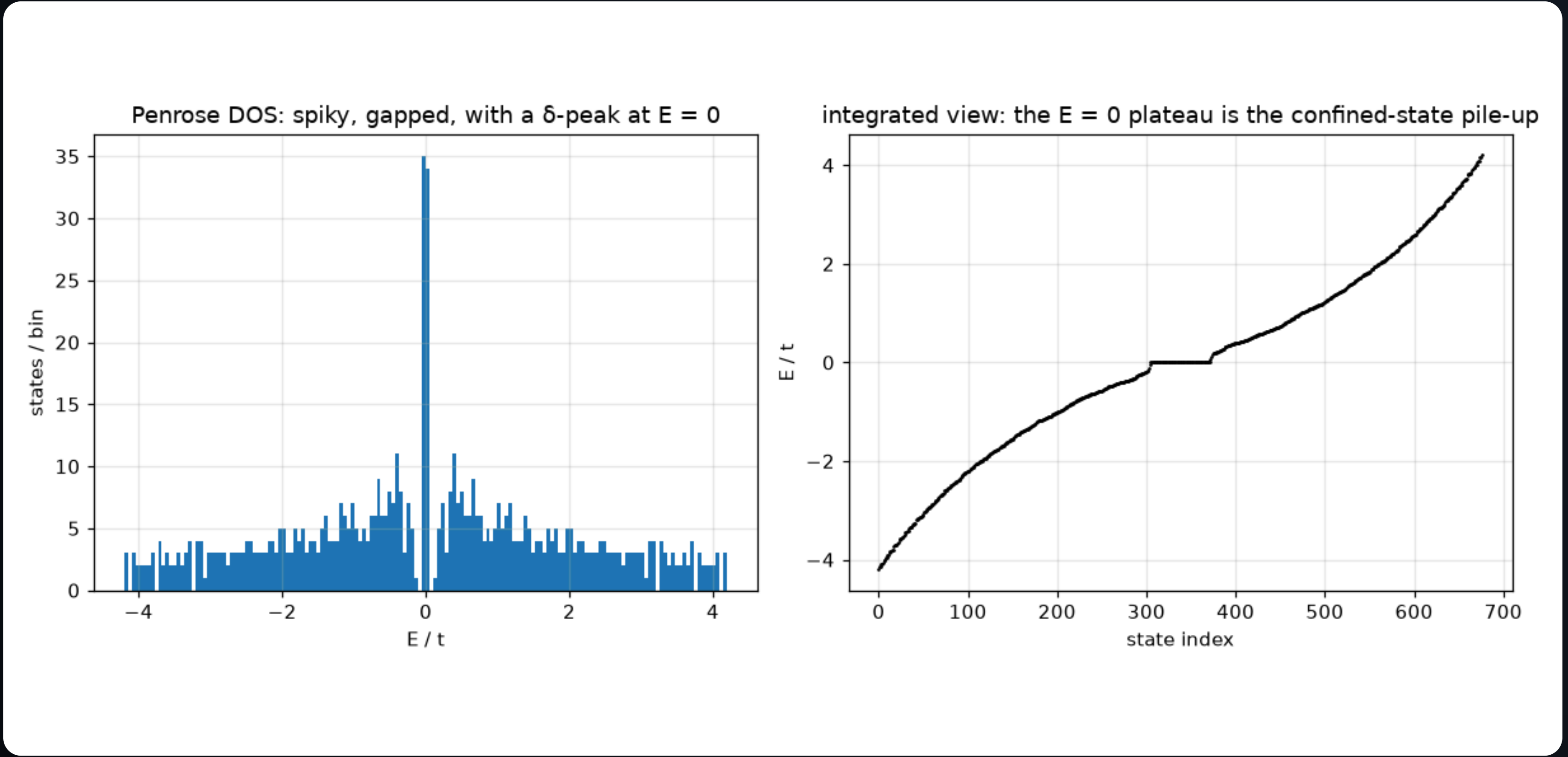


A Penrose tiling patch generated by de Bruijn's pentagrid construction: fat and thin rhombi, all edges of unit length, five-fold orientational order and no repeating cell anywhere. This is a legitimate PythTB model — a dim_k = 0 point cloud with hops on the rhombus edges — because nothing in the package's data model actually requires periodicity. — Figure 63 · notebook §23 · chapter 7, cell 9

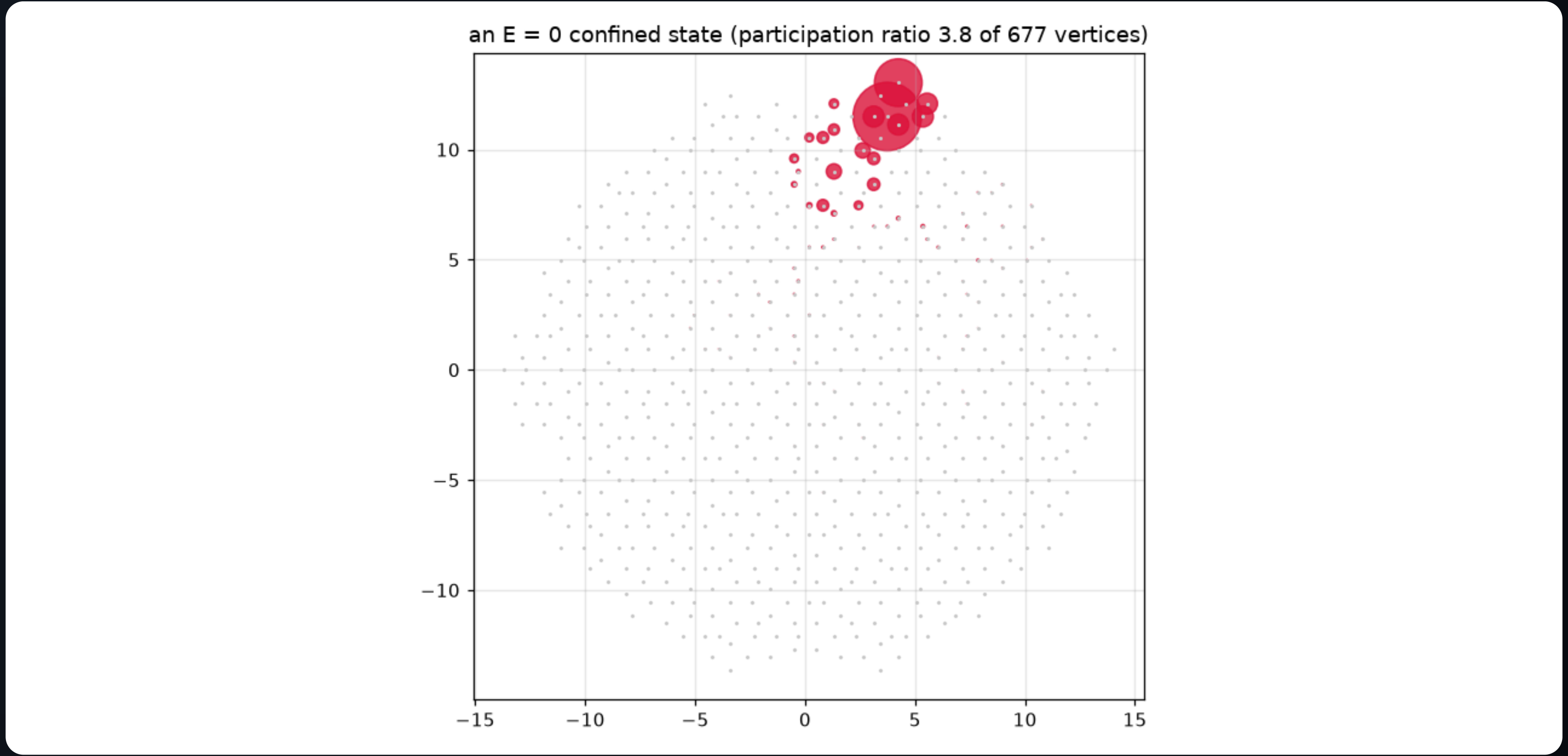
CORE

Spiky spectrum and confined states

The density of states is spiky, riddled with gaps, with a δ-peak at $E = 0$ that holds a finite fraction of all states (left). One such state (right) is strictly zero outside a few vertices: a **confined state**.



Left: the Penrose density of states — spiky, riddled with gaps, and carrying a δ-function at E = 0. Right: the sorted spectrum, where that δ-peak appears as a plateau of ~10% of all states at exactly zero energy: the strictly confined states, destructive-interference eigenstates that aperiodic order produces in macroscopic number (fraction 81 − 50φ in the infinite tiling). — Figure 64 · notebook §23 · chapter 7, cell 10



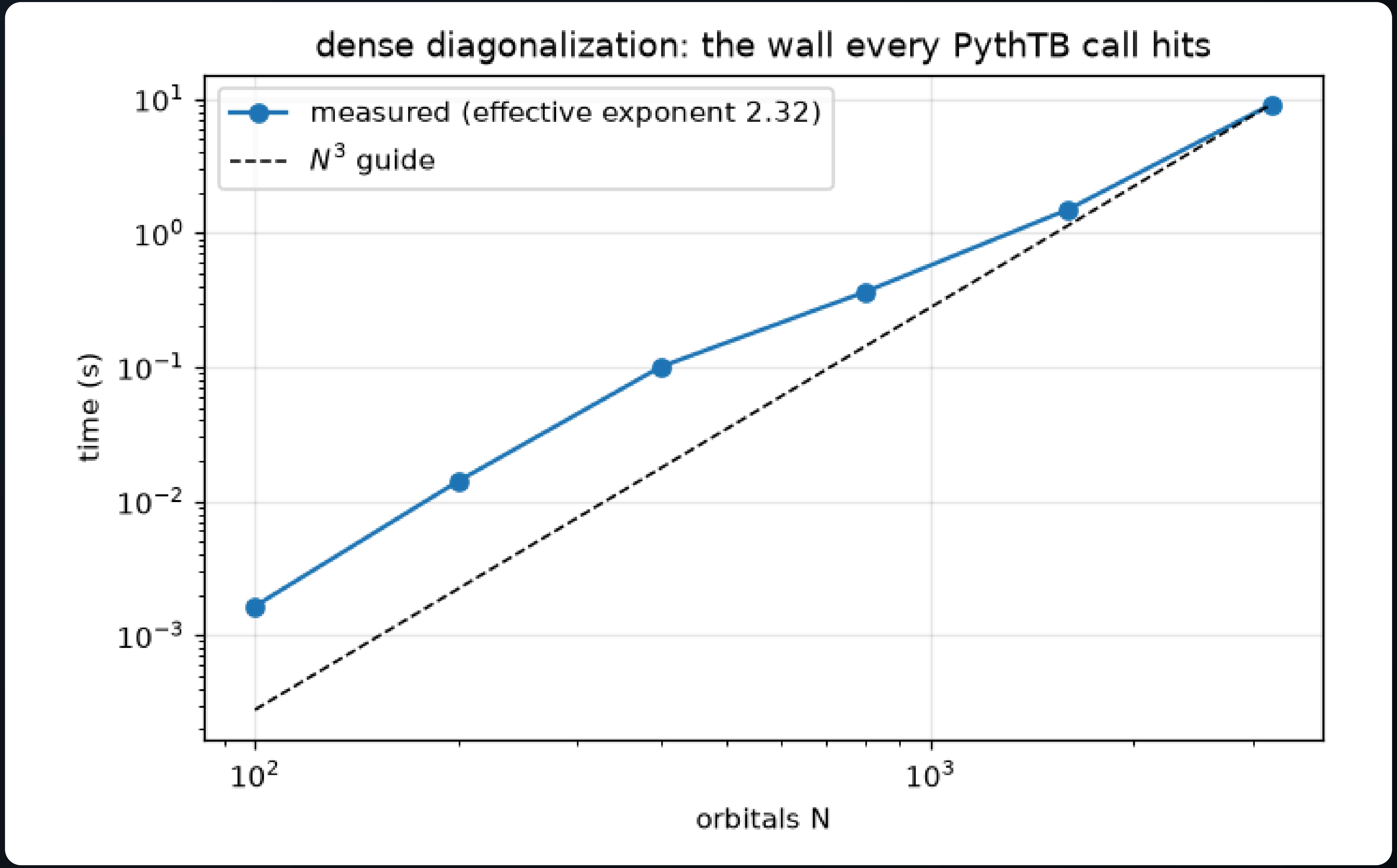
One confined state, extracted from the huge E = 0 degenerate subspace by projecting a single-site seed onto it: over 99% of its weight lives on a handful of vertices (red), strictly protected by interference — the amplitudes cancel on every site bordering the support. Unlike Anderson localization there is no exponential tail to speak of and no disorder: this is localization by geometry. — Figure 65 · notebook §23 · chapter 7, cell 11

- Confined states are the Lieb-lattice localized states of lecture 3 returning in a non-periodic setting: local geometry, not translation symmetry, produces them.
- Their fraction is exactly computable (Kohmoto–Sutherland): a macroscopic degeneracy in a system with no symmetry to explain it.
- The remaining states are critical: neither extended nor localized, with power-law scaling at every length.

The performance wall

Wall time of a full dense diagonalization versus matrix size, log-log. The slope is 3: $O(N^3)$. PythTB has no sparse path — every `solve_*` call pays this.

- 10^3 orbitals: fractions of a second. 10^4: minutes and gigabytes. 10^5: not with this tool.
- For large systems you want a few eigenvalues near a target (Lanczos, shift-invert) or none at all (kernel polynomial method).
- Those live in Kwant, scipy.sparse.linalg, and the KPM libraries — lecture 10.



Wall time of a full dense diagonalization versus matrix size, log-log, with the ideal N^3 slope as a guide. The measured effective exponent sits below 3 in this range only because threaded BLAS gains efficiency as matrices grow; the asymptote is inescapable. Every quantity PythTB computes — bands, Berry phases, markers, Wannier functions — stands on this staircase. — Figure 66 · notebook §24 · chapter 7, cell 13

MATH

Peierls phases, IPR, and the cost of dense algebra

PEIERLS SUBSTITUTION

$$t_{ij} \longrightarrow t_{ij}\,\exp\!\left(i\,(e/\hbar) \int_{r_i}^{r_j} A\cdot dl \right),\ \text{Landau gauge } A = (0, Bx, 0)\!:\ t_y(x) = t\, e^{2\pi i\,\varphi\, x/a},\quad \varphi = Ba^2e/h$$

HARPER EQUATION (Φ = P/Q)

$$\psi_{x+1} + \psi_{x-1} + 2\cos(\,2\pi\,\varphi\, x + k_y\,)\,\psi_x = (E/t)\,\psi_x,\ x = 1 \ldots q$$

INVERSE PARTICIPATION RATIO

$$IPR(\psi) = \Sigma_i\,|\psi_i|^4 \,/\, (\,\Sigma_i\,|\psi_i|^2\,)^2 \sim N^{-1}\ (\text{extended}),\ \xi^{-d}\ (\text{localized})$$

- The Harper equation is a $q\times q$ Bloch Hamiltonian for each (k_x, k_y): the magnetic supercell in PythTB is exactly this matrix.
- Dense Hermitian diagonalization: $\approx (4/3 + 2)\,N^3$ complex flops for values and vectors; memory $16N^2$ bytes.
- Sparse alternatives cost $O(N)$ per matrix–vector product; the number of products, not N, sets the time.

Limits & what next

What PythTB cannot do (transport, sparse, continuum, symmetry rails, interactions), where it goes wrong silently, and the road onward.

1. Five things PythTB cannot do
2. Silent wrongness
3. Interactions by hand: mean-field Hubbard on a zigzag ribbon
4. The capability matrix
5. Mean field, and the equation PythTB does not solve

Five things PythTB cannot do

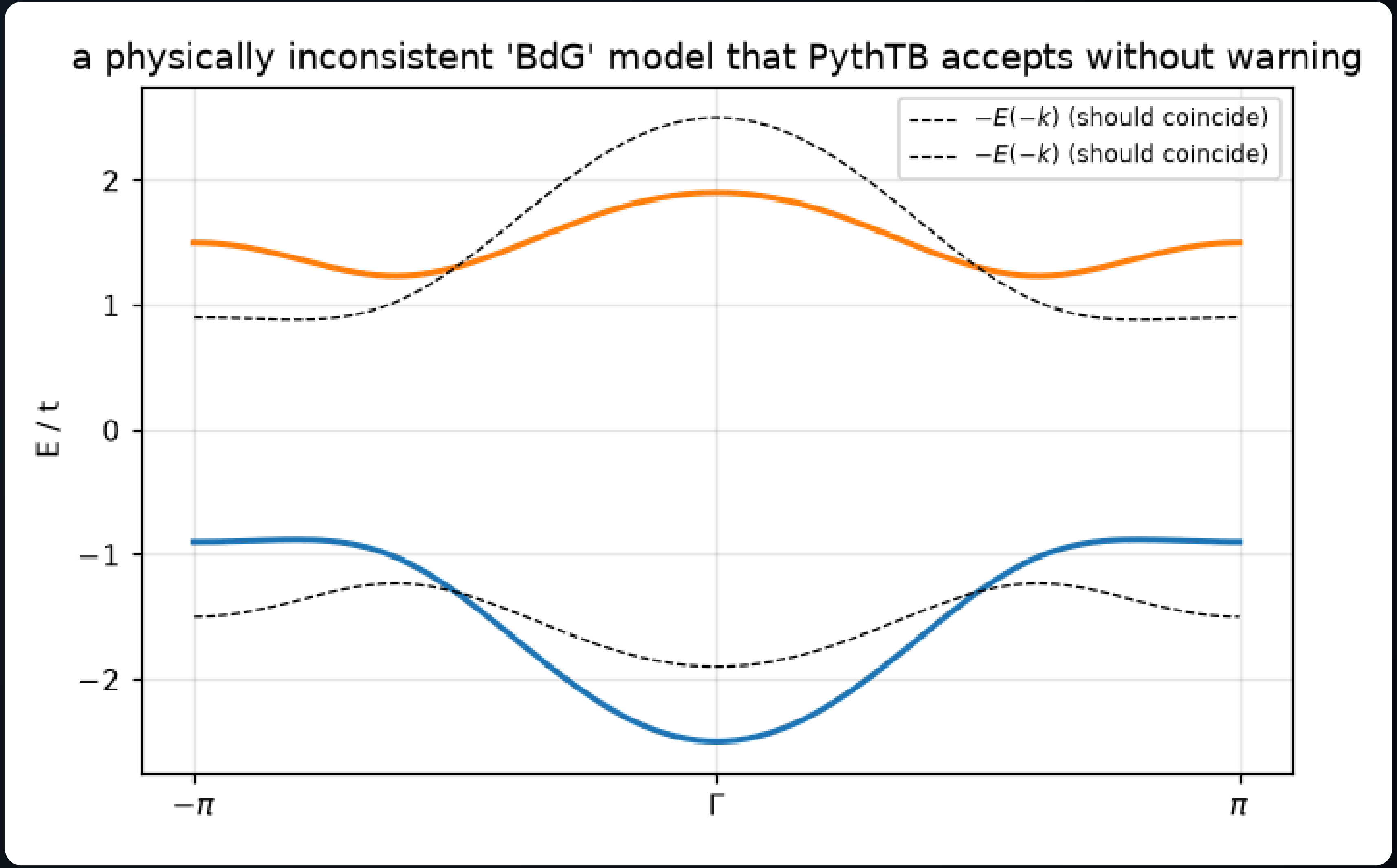
Knowing a tool's edge is part of knowing the tool. Each of these is a notebook section with a worked demonstration of the limit — and a pointer to what to use instead.

- **No transport**: no leads, no scattering matrix, no conductance. → Kwant (§25).
- **No sparse solvers, no KPM**: every call is dense and $O(N^3)$. → Kwant, scipy.sparse, KPM codes (§26).
- **No continuum models**: no $k{\cdot}p$ discretizer, no symbolic input. → Kwant's `continuum` (§27).
- **No symmetry rails**: a wrong BdG block runs without complaint (§28).
- **No interactions**: unless you write the self-consistent loop yourself (§29).

Silent wrongness

A 'BdG Hamiltonian' whose hole block was deliberately corrupted. PythTB builds it, solves it, plots it — and the spectrum is no longer particle–hole symmetric. No error, no warning.

- PythTB checks hermiticity and nothing else; every other symmetry is the user's responsibility.
- The cure is a test: assert $E \leftrightarrow -E$ symmetry on the spectrum, or check $\Xi H \Xi^{-1} = -H$ directly, every time you build a BdG model.
- The notebook's inline checks (`check(...)`) are this habit applied to every section.



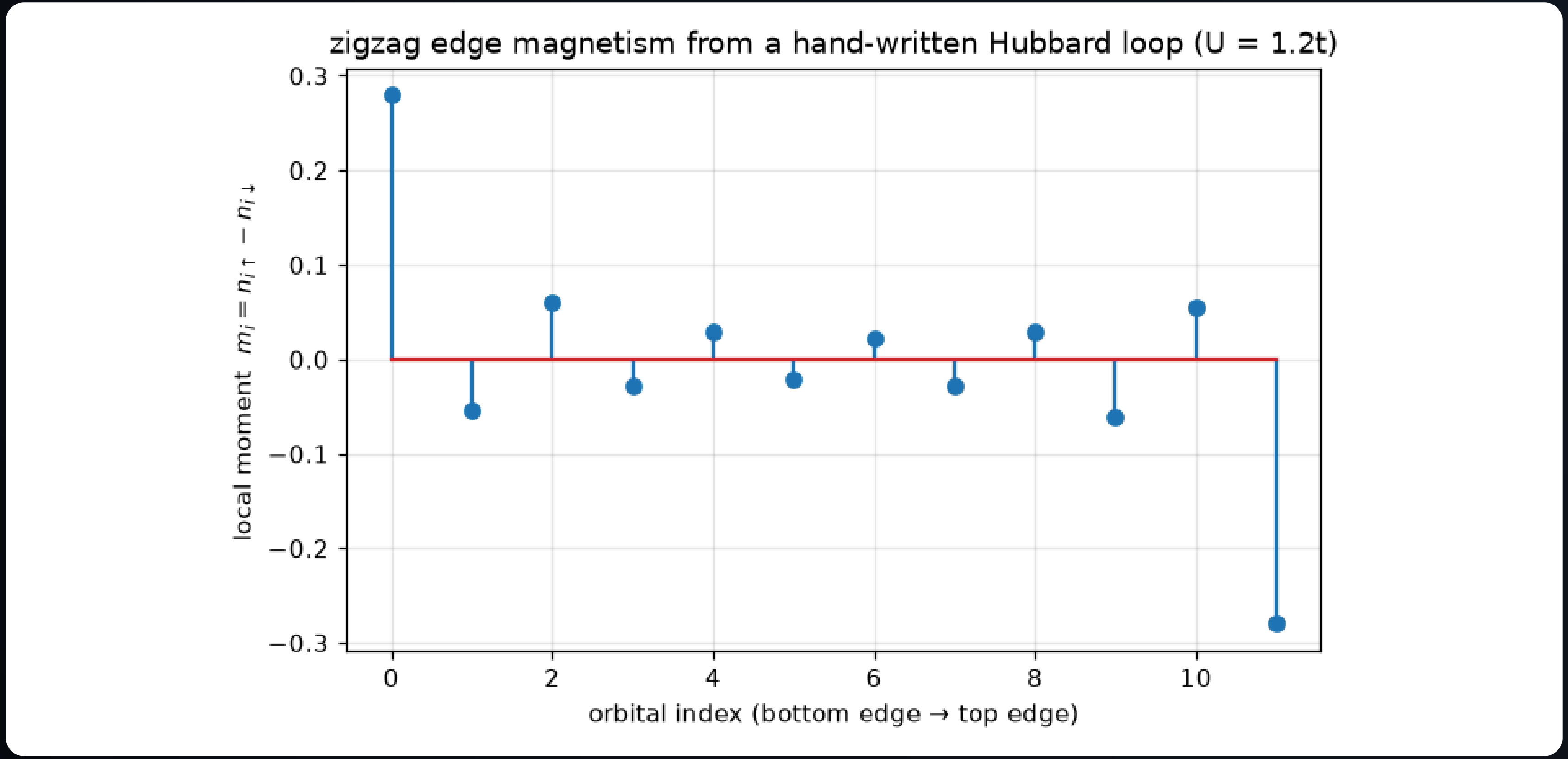
The demonstration of §28: a 'BdG Hamiltonian' whose hole block was deliberately corrupted (+0.7t instead of +t). Particle-hole symmetry demands the spectrum coincide with its negated mirror image (dashed) — it visibly does not — yet PythTB raised no objection at any point, because Hermiticity, which it does check, is intact. Symmetry correctness of a BdG problem is entirely the user's burden.

— Figure 67 · notebook §28 · chapter 8, cell 8

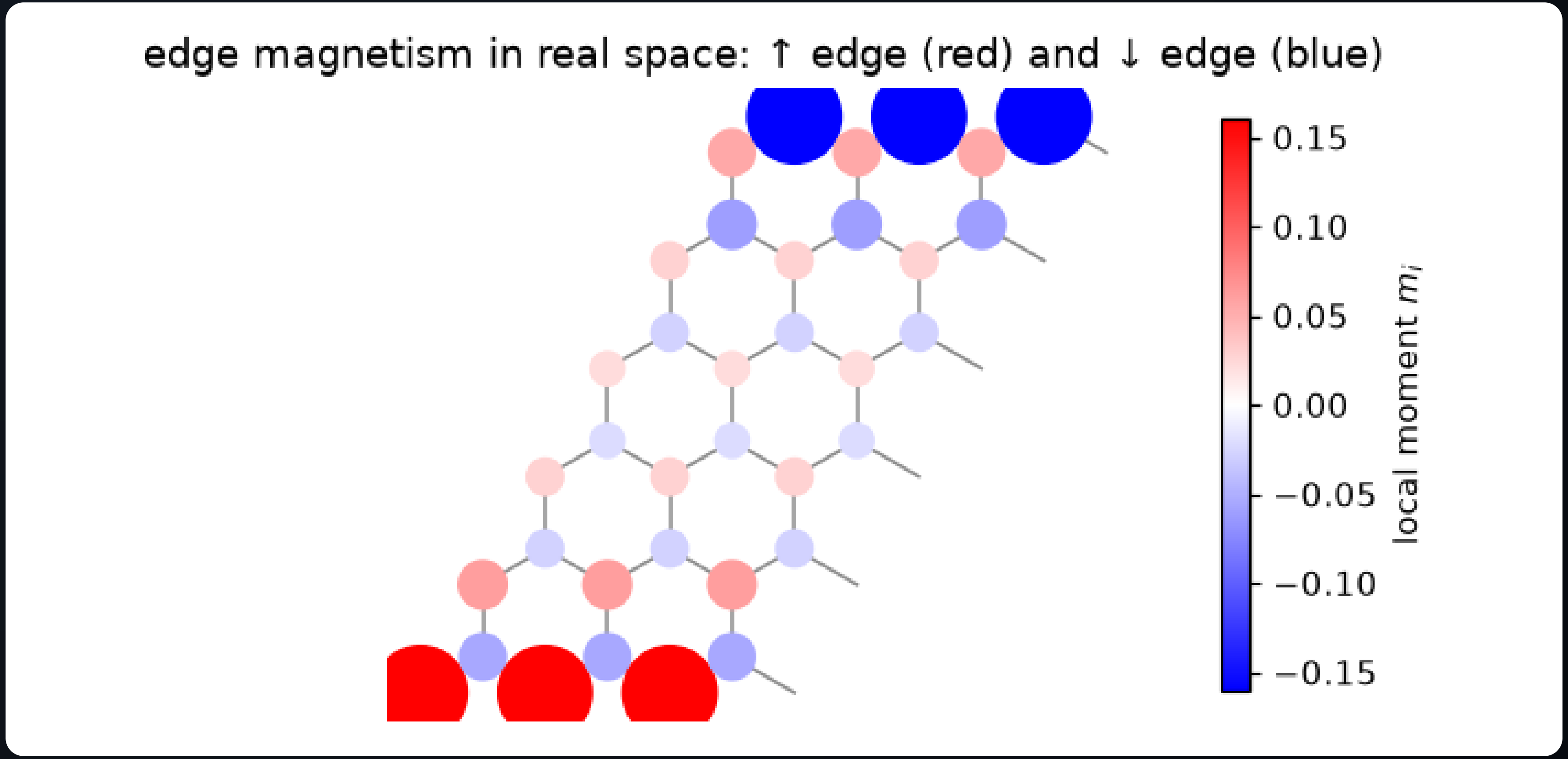
Interactions by hand: mean-field Hubbard on a zigzag ribbon

CORE

A self-consistent loop around PythTB: guess the local moments, build the spin-dependent onsite energies, solve, recompute the moments, repeat. The converged solution (left) shows opposite magnetization on the two zigzag edges, painted on the ribbon (right).



Self-consistent local moments $m_i = n\uparrow_i - n\downarrow_i$ across the zigzag ribbon, orbital by orbital from one edge to the other. The flat edge band of §6 polarizes under the Hubbard interaction: each edge orders ferromagnetically, the two edges antiferromagnetically (total Sz = 0, as Lieb's theorem demands on a balanced bipartite lattice), and the moments decay within a few rows into the bulk. — Figure 68 · notebook §29 · chapter 8, cell 10



The converged mean-field solution painted onto the ribbon itself (three periods shown; symbol size and color ∝ m_i): spin-up order along one zigzag edge, spin-down along the other, near-zero moment in the interior. The A-sublattice/B-sublattice character of the two edges (§6) is what selects the antiferromagnetic inter-edge alignment. — Figure 69 · notebook §29 · chapter 8, cell 11

- The flat edge band of lecture 3 has a huge density of states at the Fermi level: the Stoner criterion is met and the edges order.
- Antiferromagnetic across the ribbon, ferromagnetic along each edge: Fujita et al. (1996), later seen in STM.
- Thirty lines of Python; the physics is in the loop, PythTB is only the solver inside it.

CORE

The capability matrix

PythTB versus Kwant, honestly. Use both; know which one you are holding.

	PythTB 2.0	Kwant 1.5
Band structures, k-paths, meshes	yes, first-class	yes (wraparound / Bands)
Berry phase, curvature, Chern, Z_2 flow, axion	yes, built in	no (write it yourself)
Wannier functions, Wannier90 import	yes	no
Finite systems, ribbons, supercells, defects	yes (dense)	yes (sparse)
Transport: leads, S-matrix, conductance	no	yes, first-class
Sparse eigensolvers, KPM, 10^5+ sites	no	yes
Continuum k·p → lattice discretizer	no	yes (kwant.continuum)
Symmetry enforcement (BdG, conservation laws)	no	partial
Interactions / self-consistency	loop by hand	loop by hand
Learning curve	an afternoon	a week

Mean field, and the equation PythTB does not solve

MATH

HUBBARD MEAN FIELD

$$U\, n_{i\uparrow} n_{i\downarrow} \approx U(\, \langle n_{i\downarrow}\rangle\, n_{i\uparrow} + \langle n_{i\uparrow}\rangle\, n_{i\downarrow} - \langle n_{i\uparrow}\rangle\langle n_{i\downarrow}\rangle\,),\; m_i = \langle n_{i\uparrow}\rangle - \langle n_{i\downarrow}\rangle$$

SELF-CONSISTENCY

$$\varepsilon_{i\sigma}{}^{(n+1)} = U\, \langle n_{i,-\sigma}\rangle^{(n)},\; \langle n_{i\sigma}\rangle = \Sigma_{E_\alpha<E_F}\, |\psi_\alpha(i,\sigma)|^2,\; iterate\ until\ |m^{(n+1)} - m^{(n)}| < tol$$

LANDAUER–BÜTTIKER (KWANT'S PROBLEM)

$$G = (e^2/h)\, \mathrm{Tr}(\, t^\dagger t\,) = (e^2/h)\, \Sigma_n\, T_n,\; S = (\, r\ t'\ ;\ t\ r'\,)$$

- Mean field replaces a quartic operator by a quadratic one plus a constant: the result is again a tight-binding model, which is why the loop closes on PythTB.
- Convergence needs mixing ($m \leftarrow (1-\alpha)m_{old} + \alpha\, m_{new}$) and a symmetry-breaking seed, or the loop returns the paramagnet.
- The scattering matrix needs semi-infinite leads and a Green's function: a different mathematical object from a Hamiltonian's eigenvalues.

WHERE TO GO NEXT

You can now build, solve, and classify

Vanderbilt, Berry Phases in Electronic Structure Theory (the book PythTB was written for) · Asbóth, Oroszlány, Pályi, A Short Course on Topological Insulators · the PythTB documentation and examples · Kwant for transport · the companion notebook's exercises, with solutions